Mid-Term Report

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1 Current Status

As previously mentioned (2025/10/15) in my opening report, project-related coding is almost completed though there are certain aspects of *Tigris*[1] that could use a little bit more optional polishing. Specifically, we have completed

- the coding of *Tigris'* both as a language and a prover.
- passed the major modular feature test at `examples/`.

What's only left is

- the thesis.

Below, we present (non-definitive) **language specifications** and brief **internal design notes** of the language as **proof-of-progress**, which, shall include the following (based on that of OCaml):

- PL side (emphasized, in Lean):
  1. *Tigris*: The Programming language
     (a) Lexical conventions,
     (b) Values,
     (c) Names,
     (d) Type expressions,
     (e) Patterns,
     (f) Expressions, Declarations and Bindings
     (g) Type(class) definitions,
     (h) Implementation details i.e. Lexing, Parsing.
  2. Type System
     (a) Core calculus
     (b) Typing discipline
     (c) Exhaustiveness checking
     (d) Typeclass instance resolution
     (e) System F IR & elaboration
  3. The *Lambda* IR
     (a) Lowering
     (b) Optimizations

[1] Repository at https://github.com/notch1p/tigris

   (c) Incremental APIs
4. Codegen
   (a) Backend: Common Lisp (SBCL)
      i. Memory Representation
      ii. Runtime
5. FFI Utilities (SBCL)
6. CLI Interfaces (Interpreter, Compilers)
   (a) `Tigrisi`: IR-based REPL
   (b) `Tigrisl`: Compiler
   (c) `Tigris`: (Legacy, deprecated) Source-based REPL
7. Misc: Dependently-typed table printer

- Prover side (in Haskell):
  1. *Tigrisd*: The language itself
  2. Major features
  3. Type System, Semantics
     (a) …

Nomenclature In short: *TinyTheorem* = *Tigris* + *Tigrisd*

This document, while being a progress report only, is also an incomplete draft of the thesis' core.

Contents

Symbols				Keywords						
|	→	;;	⇒	mutual	infixl	infixr	match	then	let	fun
,	_	:	∀	extern	class	forall	data	prefix	postfix	in
				type	with	instance	else	and	rec	if

Table 1: Reserved symbols and keywords

2 The *Tigris* language

Tigris is a purely functional, flexible, static typed language with a focus on readability and syntactic uniformity that takes inspiration from Lean, OCaml and Haskell.

2.1 Lexical conventions

Below we use a augmented form of BNF (not to confused with ABNF) that accepts additional POSIX regex (specifically, `+*[^]`) on the RHS.

Encoding Tigris sources are expected to be UTF-8 encoded text.

Blanks Tabs, spaces, CR/LFs are considered blanks. Blanks are ignored and thus do not affect the semantic of the program.[2] Blanks separate adjacent identifiers, literals and keywords in order to prevent being parsed as one single identifier.

Comments One of `--` (Lean/Haskell/Lua style), `NB.` (J style) starts a *single-line* comment. Comments are treated as if they are blanks. Note that currently multiline comments are not supported.

Identifier Almost the same as OCaml/Haskell/Lean.

```
<ident>        ::= (<letter> | _) <rest>*
<upper-ident>  ::= <uppercase> <rest>*
<lower-ident>  ::= <lowercase> <rest>*
<letter>       ::= <upper-letter> | <lower-letter>
<rest>         ::= <letter> | 0..9 | _ | ' | ! | ?
<lower-letter> ::= a..z
<upper-letter> ::= A..Z
```

To simplify parsing, whether an identifier is capitalized or not carries additional information. In contrast, no distinction is made between `'!?`. This is permitted to facilitate the writing of idiomatic codes such as ending a nullable (`Option a`) or a non-local exiting function identifier in `?` or `!`, respectively.

Integer literals `<intlit> ::= ("-" | "+")? 0..9+`, in decimal.

String literals `<strlit> ::= "[^"]+"`

Reserved symbols and keywords See Table 1. Lexical ambiguities are resolved according to the *longest-match* rule.

[2] This is not the case on the prover side.

2.2 Value

Value refers to an expression that cannot be further reduced, a *normal form* so to speak. I.e. $\langle v, \sigma \rangle \Downarrow \langle v \rangle$ where v is one of

Integer numbers The range of an integer is implementation-dependent, that is, as of now, both SBCL backend and the interpreter (in Lean) support arbitrary-precision arithmetic.

Strings String values are finite sequences of characters, the size of which is implementation-dependent, that is, as of now, SBCL backend supports strings containing up to 2^{45} chars.

Tuples n-tuple is written $(v_1, \ldots, v_n)$.[3]

Records Records are syntactic sugars of inductive types, written $\{field_1 = v_1, \ldots, field_n = v_n\}$.[4] Currently, no type-directed parsing is implemented because of potential major refactors. Thus for records with same field labels, regardless of types, an ambiguous parsing warning is thrown but can be avoided with type ascription (same with patterns):[5]

```
type Point  = {x : Int, y : Int}
type Point' = {x : Int, y : Int}

-- [WARN] ambiguous record literal (using default 'Point'), consider adding type ascription;
-- as we currently do not have type-directed parsing.
-- candidates are [(Point, #[x, y]), (Point', #[x, y])]
let _ = {x = 1, y = 2}

let _ = ({x = 1, y = 2} : Point') -- success
```

For overlapping labels, a *longest-match* rule is used:

```
type Point   = {x : Int, y : Int}
type Point3D = {x : Int, y : Int, z : Int}
let (x, y) = ( {x = 1, y = 2}          -- x : Point
             , {x = 1, y = 2, z = 3})  -- y : Point3D
```

Constructed values refer to values that are constructed by a *value constructor* (or simply unqualified, *constructor*) of some inductive type. It is also called a *variant*. Constructors are like functions except they may not have parameters: for some constructor C_i, a variant is either C_i or $C_i\ fid_1\ \cdots\ fid_n$. The following variants are built-in:

Constructors	Types
`true`	Bool
`false`	Bool
()	Unit

Functions refers to the mappings from values to values. Details explained later.

2.3 Names

Identifiers are used to reference *language objects* by name: Table 2

[3] Tuples are specially treated instead of the usual Prod encoding.
[4] Typeclass instances are syntactic sugars of records.
[5] This annotation must be "next to" the record literal as it is processed by the parser rather than the typechecker.

Category	First letter
Values	any
Constructors	uppercase
Type constructors	uppercase
Record fields	any
Bound type variables	lowercase

Table 2: Case difference of identifier categories

Operator	Associativity
Type application	left
Product type ($*$)	right
Arrow type ($\rightarrow$)	right

Table 3: type operator associativity

2.4 Type expressions

```
<type-expr>    ::= <type-app> (-> <type-expr>)?                          (arrow type)
<type-atom>    ::= <ident>
                 | "(" <type-expr> ")"                                   (parenthesized)
                 | <type-ctor>
                 | sepBy(* | ×, <type-expr>)                             (product type)
                 | <type-args>
<type-app>     ::= <type-app>? <type-atom>                               (type application)
<type-ctor>    ::= <upper-ident>
<bound-ty>     ::= <lower-ident>
<type-binder>  ::= <bound-ty>                                            (type binder)
                 | "(" <bound-ty> : <intlit> ")"
<type-args>    ::= <bound-ty> | <type-expr>                              (type argument)
<qualified>    ::= "[" sepBy1(",", (<bound-ty> | <type-ctor>) <type-expr>+) "]"
<scheme>       ::= { FORALL <type-binder>+}? <qualified>? "," <type-expr>
FORALL         ::= forall | ∀
```

Note that, for juxtapositional type application[6], product type and arrow type, we have the following precedence & associativity table (higher comes first): Table 3

Type variables type variables (TVs) are identifiers with a lowercase beginning. The type (kind) of a TV is determined explicitly to aide the absent of kind inference. A TV denoted $(\alpha : n)$ is $\mathsf{Type} \rightarrow \mathsf{Type}$ which otherwise is Type.[7] In datatype definitions, type variables are always bounded, appearing at parameter position as well a subsequent type expression on the RHS. Note that validity checking of type expressions are mostly done at parsing time i.e. If some TV is unbound, a parsing error is immediately thrown. In type constraints a TV may be one of:

[6] similar to the a/b-type implementation in Haskell 2010 report.

[7] Note that however we do not allow curried HKT i.e. $(\alpha : 2)$ is instead, $\mathsf{Type} \times \mathsf{Type} \rightarrow \mathsf{Type}$. Because of this, we shall refer to it as *arity* rather than *kind* from now on

metavariables Also called *MVs*, denoted `?m.N` where `N` is a counter. Refers to a unspecified type that *must* be either instantiated[8] by a type or generalized when leaving the scope[9] MVs must not appear in the final scheme.

skolem-TVs Also called *skolems*, denoted `?sk.A` where `A` is the original TV. Skolem variables are TVs produced by the process of skolemization. These TVs are essentially type constructors[10] that may *not* be instantiated. This allows the typechecker to catch more errors:

```
instance : Functor Option =
  { map -- forall a b, (a -> b) -> Option a -> Option b
    | f, None ⇒ None
      -- [ERROR] Can't unify type ?sk.a with ?sk.b.
    | f, Some a ⇒ Some a
  }
```

Results in a typing error because both `a` and `b` are skolemized when entering the body of `map` and they may not be instantiate to each other. Here, we assume that the user forgot to apply `f`, which, should instead be `Some (f a)`. This the rigid nature of skolem-TVs allows the typechecker to catch this human error for us.

Note that, there is currently no implicit inference of type variables. They can only be introduced at a ∀ binder or datatype declarations.

Function types The type expression $ty_1 \to \cdots \to ty_n$ denotes the type of functions mapping of $ty_1, \ldots, ty_{n-1}$ to results of type ty_n.

Product types The type expression $ty_1 \times \cdots \times ty_n$ denotes the type of tuples whose elements are of $ty_1 \times \cdots \times ty_n$ respectively.

Constructed types Similar to TVs, type constructor has arity as well. *<type-ctor* is a type constructor with no parameter and a type expression. Otherwise, a type constructor must be fully applied.

arbitrary recursive types is rejected thus it is not possible to type `fun x -> x x`.

2.5 Patterns

Patterns, as in pattern matching, are templates of instructions used to *destructure* data structures of some shape. Patterns can be enriched by custom operators (described later) if they are associated with constructors however they are reduced to one of, or the composition of the primitive patterns below at parsing-time:

```
<pat>       ::= <pat-left> <infix-op> <pat>        infix operator enriched
              | <pat-left>
              | <prefix-op> <pat>                  prefix operator enriched
              | <pat> <postfix-op>                 postfix operator enriched
<pat-left>  ::= <pat-atom>
```

[8] The word *instantiate* is used throughout this text while its usage is not all well-defined: Sometimes, instantiate means *specialize*, where TVs are instantiate to concrete types; sometimes, it only indicates the substitution of a TV.

[9] that is, the whole toplevel phrase.

[10] that is, encoded as `TCons` to prevent illicit unification

Operator	Associativity
Constructor application	left
Prefix operator	
Postfix operator	
Infix operator	left
Infix operator	right

Table 4: Precedence of operators in patterns

```
                | <ctor> many1(<pat-atom>)            variant
<pat-atom>   ::= <lower-ident>                        variable
                | <ctor>                              constructor
                | "_"                                 wildcard
                | "{" sepBy1(",", <pat-field>) "}"    record pattern
                | "(" sepBy1(",", <pat>) ")"          product pattern
                | "(" <pat> : <type-expr> ")"         ascribed pattern
                | <literal>
<pat-field>  ::= <field> = <pat>
```

Note that the arity of a constructor must match the number of sub-patterns associated with it i.e. constructor in this context must not be partially applied. Mixfix (i.e. {pre,in,post}-fix) operators have the following precedence in patterns (higher comes first): Table 4

variable & wildcard patterns binds the name to the value, e.g.:

```
let fst (x, _) = x
```

Unlike other languages, pattern are not linear. A variable can be bound several times by a given pattern and overwrites previous bindings. Beware of the semantic confusion:

```
let prodEq : ∀a b, a × b → Bool
  | (x, x) ⇒ true
  | (x, _) ⇒ false
-- [WARN] Found redundant alternatives at
-- 2)  #[(x, _)]
```

It is not possible to decide the equality relation between two parts of a data structure through arbitrary patterns. Use the method (=) of typeclass Eq for that matter.

literal patterns test the equality of the constants.

```
let strToBool : String → Bool = fun
  | "true"  ⇒ true
  | "false" ⇒ false
-- [WARN] Partial pattern matching, an unmatched candidate is
-- [_]
```

parenthesized patterns come into effect when mixing with operator-enriched patterns:

```
type List a = Nil | Cons a (List a)
infixr 90 "::" ⇒ Cons
```

```
let f1 (x :: y :: _) = (x, y)
let f2 ((x :: y) :: _) = (x, y)
```

f1 and f2 are vastly different, according to *Tigris*' System F IR:

```
let (::) : ∀ α, α → List α → List α = (Λ α. Cons@α)

let f1 : ∀ α, List α → α × α =
  (Λ α. fun ?x₀ : List α ⇒ match ?x₀ with | Cons x (Cons y _) ⇒ (x , y))

let f2 : ∀ α, List (List α) → α × List α =
  (Λ α. fun ?x₀ : List (List α) ⇒ match ?x₀ with | Cons (Cons x y) _ ⇒ (x , y))
```

variant patterns are of the form $C_i\ p_1, \ldots, p_n$. It matches all variants whose constructor equals to C_i and whose subpatterns matches $p_1, \ldots, p_n$. An arity mismatch error is signaled if the number of subpatterns is unexpected by the constructor.

Record patterns are similar to record literals that will encounter ambiguous parsing if there exists records with same field labels.

```
type Point  = {x : Int, y : Int}
type Point' = {x : Int, y : Int}

let sum {x, y} = x + y
-- [WARN] ambiguous record pattern (using default 'Point'), consider adding type ascription;
-- as we currently do not have type-directed parsing.
-- candidates are [(Point, #[x, y]), (Point', #[x, y])]
let sum' ({x, y} : Point') = x + y
```

Since records are syntactic sugar of inductive types, same applies to record patterns. The denotational semantic is

$$[\![\textbf{type}\ T\ args* = \{(fid : \sigma)*\}]\!] := [\![\textbf{type}\ T\ args* = T\ \sigma*]\!]$$

where $tm*$ denotes the *splice* of tm, a sequential structure. Thus, a field can be extracted using the ordinary constructor application pattern:

```
class Functor (m : 1) =
  {map : forall a b, (a → b) → m a → m b}
let getMap (Functor f) = f
```

yields IR

```
let getMap : ∀ α, Functor α → ∀ a b, (a → b) → α a → α b =
  (Λ α. fun ?x₀ : Functor α ⇒ match ?x₀ with | Functor f ⇒ f)
```

2.6 Expressions

Expressions are the core of *Tigris*. Colloquially, we define the following metavariables:

- p_i denotes the i-th pattern in a pattern matching branch.
- C_i denotes the i-th constructor in a inductive type definition
- e, e', e_i denotes an arbitrary expression whose evaluation value depends on the context.

- v, v', v_i denotes a irreducible value.
- $a, b, c, d, e, x, y, z, w$ denotes a binder, or an identifier.
- f, g, h denotes an identifier and semantic-wise, a function.
- $[\![]\!]$ denotes blackboard style brackets, sometimes used as delimiters (if usable ordinary delimiters have run out) in metalanguage.
- greek letters mostly denotes types. Greek letters with a bar on them $\bar{\alpha}$ denotes a list of those letters, with implicit numbering $1, \ldots, i$.

A *Tigris* expression will, eventually, after exhaustive application of transformations, be translated into the core calculus `Expr` defined in `types.lean`.

```
<expr>         ::= <expr-infix>
                 | (<expr-infix> : <type-expr>)                          ascription
<expr-infix>   ::= <expr-left> <infix-op> <expr-infix>                   infix operator
                 | <prefix-op> <expr-left>                               prefix operator
                 | <postfix-op> <expr-left>                              postfix operator
                 | <expr-left>
<expr-left>    ::= fun (<expr-binder>+ ARROW <expr> | <expr-match>)      lambda abstraction
                 | let "rec"? <pat'> = expr in <expr>                    pattern matching let
                 | let "rec"? <let-binder> sepBy("and", <let-binder>)    let(rec) expression
                   in <expr>
                 | if <expr> then <expr> else <expr>                     conditional
                 | rec <expr>                                            fixpoint; deprecated
                 | match sepBy1(",", <expr>) with <expr-match>           pattern matching
                 | <expr-funapp>
<expr-funapp>  ::= <expr-funapp>? <expr-atom>                            function application
<expr-atom>    ::= <literal>
                 | <expr-ctor>
                 | <lower-ident>                                         variables
                 | "(" sepByN(",", <expr>) ")"          N = 0            unit
                                                        N = 1            parenthesized <expr>
                                                        N ≥ 2            product
                 | "{" sepBy1(",", <expr-field>) "}"                     record literal
<expr-ctor>    ::= <upper-ident>                                         constant constructors
<expr-match>   ::= ("|" sepBy1(",", <pat>) ARROW <expr>)+                pattern match body
<expr-field>   ::= <field> = <expr>
<expr-binder>  ::= <id> | <pat'>                                         binders
<pat'>         ::= "(" <pat> ")"                                         parenthesized <pat>
                 | <literal>                                             except literal patterns
<let-binder>   ::= <pat'> = <expr>                                       single-pattern binding
                 | <ident> <expr-binder>* <type-expr>? <let-rhs>         value/function binding
<let-rhs>      ::= "=" <expr> | <expr-match>                             let binding RHS
<literal>      ::= <intlit> | <strlit> | "true" | "false"
ARROW          ::= "⇒" | "->"
```

Note that, the `ARROW` lexer parses both lexeme ⇒ and →, only the first should be used, the alternative is to add some form of ML compatibility.

Basic expressions are all expressions list in `<expr-atom>` with the addition of function shorthand syntax.

Shorthand syntax is a Lean-like syntactic sugar for parenthesized expressions, within which, an underscore (_) or cdot (·) is transformed in preorder[11] against the expression to a function binder. This is similar to Haskell's operator sectioning. Respects precedence, e.g.

```
(_ + 1)

(_ + 1 * 2 / _)

(id _)
```

(a) shorthand syntax

```
fun x ⇒ x + 1

fun x y ⇒
  let tmp = 2 / y in
  let tmp = 1 * tmp in
  x * tmp

fun x ⇒ (id : forall a, a -> a) x
```

(b) desugared results

Parenthesized expressions can also contain type constraint, as in the example above.

Curried applications & definition Functional values are curried.[12] i.e. partial application is allowed and is in practice frequently used in idiomatic functional programming.

Function application associates to the left. It is denoted by juxtaposition of expressions. *Tigris* is call-by-value (CBV), thus the expression $e\ e'$ evaluates e to a functional value first, then e'.

Lambda abstraction takes the form of **fun** $p_1 \cdots p_n \Rightarrow e$

Function definition has two syntactic forms:

$$\begin{array}{l}\textbf{fun}\\ \mid p_{1,1}, \ldots, p_{1,m} \Rightarrow e_1\\ \vdots\\ \mid p_{n,1}, \ldots, p_{n,m} \Rightarrow e_n\end{array} \tag{1}$$

$$\textbf{fun}\ p_1 \cdots p_m \Rightarrow e \tag{2}$$

Both forms are syntactic sugar that are translated and, equivalent to

$$\begin{array}{l}\textbf{fun}\ x_1 \Rightarrow \cdots \textbf{fun}\ x_n \Rightarrow\\ \quad \textbf{match}\ x_1, \ldots, x_n\ \textbf{with}\\ \quad \mid p_{1,1}, \ldots, p_{1,m} \Rightarrow e_1\\ \quad \mid \overbrace{p_{2,1}, \ldots, p_{2,m} \Rightarrow e_2}^{\text{applies to form (1)}}\\ \quad \vdots\\ \quad \mid p_{n,1}, \ldots, p_{n,m} \Rightarrow e_n\end{array}$$

Syntactic intuition shows that canonically, a lambda abstraction only takes one (1) parameter.

[11] i.e. DFS.

[12] Constructors and functions form an *exhaustive partition* of functional value.

Let-bindings & expressions The `let` and `let rec` language constructs bind value names either locally or globally, together with `in <expr-body>`, they form expressions.

Nomenclature Here, a *let-binding* describes the $p_n = e_n$ portion, `and`-branches are implicitly assumed until mentioned otherwise; A *let-expression* includes the *body* portion as well.

Local non-recursive definitions the construct **let** $p_1 = e_1$ **and** $\cdots$ **and** $p_n = e_n$ **in** e evaluates[13] $e_1 \cdots e_n$ and matches the results against pattern $p_1 \cdots p_n$. If matching succeeds, e is evaluated in the lexical scope enriched by the bindings produced during matching, whose value is returned as the value of the whole let expression.[14] Pattern matching let-expression is translated to a `match .. with` construct.

The canonical let-binding described above can used to bind functional values though syntactic sugar exists: In a let expression, function may be defined as **let** $f\ p_1 \cdots p_n = e$, e.g.

```
let prodSum (x, y) z w = x + y + z + w
and x = 1
and y = 2
in prodSum (x, y) 3 4

let (x, y) = (1, 2) in x + y

let 0 = 1

let compose f g x = f (g x) in
let add20 = (_ + 20) in
let minus30 = fun x ⇒ x - 30 in
compose add20 minus30 50
```

In the third example, a warning is thrown;

```
[WARN] Partial pattern matching, possible cases such as [true] are ignored
```

At runtime, the nonrecoverable exception `MATCH-FAILURE` is thrown:

```
debugger invoked on a MATCH-FAILURE in thread
#<THREAD tid=145807 "main thread" RUNNING {1200030003}>:
  No branch can be matched against #[Toplevel]

Type HELP for debugger help, or (SB-EXT:EXIT) to exit from SBCL.

restarts (invokable by number or by possibly-abbreviated name):
  0: [ABORT] Exit debugger, returning to top level.

(|__start|)
   source: (FUNCALL #'|main| NIL #'IDENTITY)
0]
```

[13] virtually (in the context of a typechecker), in parallel; In practice, this is implementation-dependent. Cf. nested `let`.
[14] no sequential binding is performed. A name bounded to e_i is unbounded in e_{i+1}.

Local recursive definitions are introduced by the `let rec` construct:

$$\textbf{let rec } f_1\ p_1 \cdots p_n = e_1 \textbf{ and} \cdots \textbf{and } g \cdots = e_n \textbf{ in } e$$

To grossly simplify the feature of `let rec`: the names are considered already bound when the expressions e_1 to e_n are evaluated. In practice, the importance of this construct is obvious as this is the only way to facilitate (mutual) recursion in *Tigris*.

```
mutual
type Tree a = Empty | Node a (Forest a)
type Forest a = Nil | Cons (Tree a) (Forest a)
;;

let rec countTree
  | Empty ⇒ 0
  | Node _ f ⇒ 1 + countForest f
and countForest
  | Nil ⇒ 0
  | Cons t f ⇒ countTree t + countForest f
```

New users may be tempting to define recursive functions in a more "functional" way, but hesitate as they do not know whether it will work: **let rec** $f =$ **fun** $\cdots$ **and** $\cdots$ **and** $g =$ **fun** $\cdots$ **in** e. The answer is yes: Just the same as `let` binding, the canonical form of a letrec expression is precisely the above, what we introduced earlier is simply a syntactic sugar. It defines `f, g` as mutually recursive functions local to `e` This nature is reflected in the core calculus as whether recursive or not a let rec binding is, it is always transformed into a `Let` node, with the only difference being the RHS of a let rec binding is wrapped in `Fix`.

The semantic difference between `let rec` and `let` is minute, yet complicated. We largely follow OCaml's implementation but is even simpler and more strict.

Remark (Partition heuristic). the RHS of a letrec binding must have the form of `fun ..`; Otherwise, it is ruled as a let binding.

Using a simple[15] heuristic,[16] we partition a letrec group into one that is strictly *recursive*[17] and another one that becomes regular *let-binding*. Additionally, we require the LHS of a `letrec` binding to always be variables.[18] These assumptions are strict enough to rule out some valid programs.

```
let rec x = (1, x) and y = (2, x)

let rec x = (1, y) and y = 1
```

The first example is invalid in CBV semantics as it creates cyclic products whereas the second one is ruled out though valid. We find it appropriate as though example 2 is invented, it's coding style is inherently bad; Besides, the author also struggles to find a convincing example that justifies the preservation of the expressiveness lost here.

Global defintions are just like local ones, but is limited to the toplevel, where a let(rec) expression can be free of its body i.e. becomes a standalone let binding. Multiple toplevel let bindings can optionally be separated with `;;` for clarity.

[15] no additional free occurrence checking, or much of the condition checking in the OCaml manual ch. 12.1.
[16] one of the condition to be *statically constructive*.
[17] in the sense there is no recursive definition of non-functional values.
[18] Unlike Haskell, which translate it into a fixpoint application with pattern matching.

```
function k<T, U>(t: T, _: U): T {return t}
const id: <T>(t: T) ⇒ T = (t) ⇒ t
function map<T, U>(f: (t: T) ⇒ U, xs: T[]): U[] {
  let xss = []
  for (let x of xs) {
    xss.push(f(x))
  }
  return xss
}

let sns = map((n) ⇒ n.toString(), [1, 2, 3]);
```

(a) Gradual typing w/ limited local type inference

```
let k x _ = x
let id x = x
let rec map f
  | Nil ⇒ Nil
  | x :: xs ⇒ f x :: map f xs

let sns = map toString $ 1 :: 2 :: 3 :: Nil
```

(b) Static typing w/ global inference/generalization

Figure 2: comparison of TypeScript and *Tigris*

Type inference & generalization occurs in many places. Type inference provides the user with the ability to omit typing ascriptions while maintaining program type correctness (given the program itself is correct). Unlike other languages, *Tigris* has global type inference, enabling type-checkers to type programs with zero (taken literally) ascriptions, this is due to the type system having principal types i.e. the ability to infer a most general type. In other words, users can write like[19] Python/Ruby/LISP or any other dynamic language with full compile-time type safeguard.

Generalization applies to schemes and happens at let binding. It goes together with inference: Unlike other language, where users must write in some special language construct to define a polymorphic function, *Tigris* facilitates automatic generalization of expressions, which means, if an expression is polymorphic, then it *is*, without the need of supplying type ascriptions.

These two features free programmers from the verbose work of typing expressions while maintaining the type correctness of the program.

Control structures

Sequential operations are not builtin, one may substitute it with nested let expressions.

Conditional expressions are of the form **if** e **then** e_1 **else** e_2; which evaluates to e_1 if e yields the bool constructor `true` otherwise e_2.

Case analysis Also called pattern matching. The expression

$$
\begin{aligned}
&\textbf{match } e_1, \ldots, e_m \textbf{ with} \\
&\mid p_{1,1}, \ldots, p_{1,m} \Rightarrow e_1 \\
&\mid p_{2,1}, \ldots, p_{2,m} \Rightarrow e_2 \\
&\vdots \\
&\mid p_{n,1}, \ldots, p_{n,m} \Rightarrow e_n
\end{aligned}
$$

matches the row expression vector against the pattern matrix *in parallel*,[20] yielding the RHS of a row that first succeeds. At compile time, pattern matrices are checked of exhaustiveness and redundancy, using an efficient matric-modeled algorithm from Maranget:

[19] Only syntactically.

[20] Cf. product pattern, which is evaluated in preorder.

```
let (1, 2) = (1, 2) -- [WARN] Partial pattern matching, possible cases such as [(_, _)] are ignored

let f
  | (1, 2) ⇒ 1 + 2  -- [WARN] Found redundant alternatives at
                    --   3)  #[(_, y)]
  | (x, _) ⇒ x
  | (_, y) ⇒ y
```

If none matches, the nonrecoverable runtime exception `MATCH-FAILURE` is thrown. Pattern matching can be seen as a superset of the `switch` construct found elsewhere. It enables a declarative paradigm to elegantly destructure data structures and is at the core of modern functional programming.

Operators Operators are special strings bound to expressions. They are recognized by the parser and can be chained in expressions.

One feature that set *Tigris* apart from other languages is its ability to define custom operators from a infinite set of strings with associativity and precedence adjustable. A custom operator definition is a either a 4-tuple $(op, prec, assoc, impl)$ or a 3-tuple $(op, prec, impl)$, collected from one of the three operator declaration forms below:

```
<infixl-decl>  ::= infixl ":"? <op> <prec> ARROW <expr>    ; ⇒ (<op>, <prec>, L, <expr>)
<infixr-decl>  ::= infixr ":"? <op> <prec> ARROW <expr>    ; ⇒ (<op>, <prec>, R, <expr>)
<prefix-decl>  ::= prefix ":"? <op> <prec> ARROW <expr>    ; ⇒ (<op>, <prec>, <expr>)
<postfix-decl> ::= postfix ":"? <op> <prec> ARROW <expr>   ; ⇒ (<op>, <prec>, <expr>)
```

Where <op> denotes the operator string and <op> itself is a subset of <strlit>.

Definition (Operator candidate strings)**.** Unlike Haskell or OCaml, whose <op> comes from combinations of a predefined set of operator characters, *Tigris* follows a Lean-like approach and defines <op> to be *any string* except:

- those that begin with numbers,
- those that begin with or contain one of ⟦_,()[]{}[]⟧,
- those that, except the beginning, contain <blank>,
- those that equal to a reserved operator shown here: Table 1.

Blank characters *surrounding* <op> are automatically trimmed.

Precedence of the four operator classes (infixl, infixr, prefix, postfix) with respect to function application is the same as in patterns: Table 4. Although much[21] of the operators are left for users to define, some basic operators are defined (though they can be overridden) using the same mechanics.[22] [23] The addition of `$` and `@@` is a clever abuse of precedence and is to facilitate idiomatic function programming techniques. With the lowest precedence, it is possible to avoid parenthesization nested to the right, as in:

[21] that is, an (almost) infinite amount of potential operators.

[22] Though in practice it is more common to bind operators with methods from typeclass, we chose to keep it simple; Besides, It is also true that we currently do not have a *prelude*.

[23] As described in the expression grammar, following the usual practice, language constructs are hardcoded to have lower precedence than operators, thus `let n = 10 in n + x` parses as `let n = 10 in (n + x)` rather than `(let n = 10 in n) + x`.

Operator	Precedence	Class	Implementation
*	70	infixl	Primitive integer multiplication
/	70	infixl	Primitive integer division
+	65	infixl	Primitive integer addition
-	65	infixl	Primitive integer subtraction
=	50	infixl	Primitive integer equality
$	1	infixr	Function application
@@	1	infixr	Function application

Table 5: Predefined operators

```
let _ = f (g (h (x + y)))
let _ = f $ g $ h $ x + y
let _ = f @@ g @@ h @@ x + y
```

This becomes particularly handy if the argument of some function is a large expression and this technique is also widely used throughout the implementation of *Tigris* and *Tigrisd* themselves. It is also possible to bound a same operator to both prefix and postfix:

```
let not
  | true ⇒ false
  | false ⇒ true
let rec fact
  | 0 ⇒ 1
  | n ⇒ n * fact (n - 1)

postfix 65 "!" ⇒ fact
prefix 65 "!" ⇒ not
let main = (50!, !(true))
```

But currently it is not possible to do so between prefix and infix or postfix and infix. Common idiomatic use includes defining the `Cons` constructor of `List` as `(::)` and `append` as `(++)`:

```
type List a = Nil | Cons a (List a)
infixr 67 " :: " ⇒ Cons
let car (x :: _) = x
let cdr (_ :: xs) = xs
let rec map f
  | Nil ⇒ Nil
  | x :: xs ⇒ f x :: map f xs
let foldl1 f xs =
  let rec go xs =
    match xs with
    | x :: Nil ⇒ x
    | x :: y :: xs ⇒
      go (f x y :: xs)
  in go xs

infixl 65 " ++ " ⇒ append
let rec append
  | Nil, ys ⇒ ys
```

```
  | x :: xs, ys ⇒ x :: append xs ys
```

Multiple toplevel operator bindings can optionally be separated with ;; for clarity.

extern definition utilizes the FFI facility to declare *external*[24] functions.

```
<extern-decl> ::= extern <ident> <extern-symb> : <scheme>
<extern-symb> ::= <strlit>
```

The extern declaration introduces an *opaque*[25] name <ident> with scheme <scheme>, binding itself to an external function named <extern-symb>. Since *Tigris* is a pure language, useful function with side effects can be introduced with FFI. This is considered an advanced feature, interested users shall refer to subsection 4.5 for a detailed explanation.

2.7 Type, typeclass and instance definitions

datatype definition is used to introduce new types to the environment. It binds type constructors to datatypes in either of the forms: inductive types or record types. Though for the latter it is merely a syntactic sugar of the former as mentioned before.

```
<type-decl>    ::= mutual <type-decl1>+ ";;"                    (mutual rectype definition)
                 | <type-decl1>
<type-decl1>   ::= TYPE <type-ctor> <type-binder> = <type-constrs>
<type-field>   ::= <ident> : <type-expr>
<type-constrs> ::= <type-record>
                 | ("|" <type-ind>)+
<type-record>  ::= "{" sepBy1(",", <ident> : <type-expr>) "}"
<type-ind>     ::= <expr-ctor> <type-atom>+                     (inductive type branch)
TYPE           ::= type | data
```

Note that, the TYPE lexer parses both lexeme type and data, only the first should be used, the alternative is to add some form of Haskell compatibility.

An inductive type has the form

$$\textbf{type}\ T\ \bar{\alpha} = C_1\sigma_{1,1}\cdots\sigma_{1,c_1} \mid \cdots \mid C_n\sigma_{n,1}\cdots\sigma_{n,c_n}.$$

This declaration introduces a new type constructor T that takes n parameter *fully*, with one or more value constructors $C_1, \ldots, C_n$. The types of the constructors are given by

$$C_i : \sigma_{i,1} \to \cdots \to \sigma_{i,c_i} \to T\ \bar{\alpha}.$$

where the lowercase c_i represents the arity of C_i.[26]

```
type RGB = {r : Int, g : Int, b : Int}
type Color =
| Red | Green | Blue | Yellow | Black | White
| Custom RGB
```

[24] in the sense that it is not defined within the language.
[25] in the sense that it has no definition.
[26] The result type of a constructor must match the inductive type declaration as indexed families are not currently supported.

```
type Tree a = Lf | Br a (Tree a) a

mutual
type Tree a = Empty | Node a (Forest a)
type Forest a = Nil | Cons (Tree a) (Forest a)
;;
```

Type synonyms (equality constraints) and *contexts* (predicates) are currently not supported for type declarations. Note that, *Tigris* currently does enforce strict positiveness in inductive types.

Definition (Strict Positiveness)**.** A position is said to be strictly positive if

1. it is not in a function's argument type, and,
2. it is not an argument of any expression other than type constructors of inductive types.

Since *Tigris* is a programming language, we do not make this restriction as it may rule out some unproblematic type definition.

Typeclass definition Typeclass is the functional counterpart to what is called an *interface* in Object-oriented programming, except, it is more efficient and well-defined, as in methods are dispatched statically, as in enabling ad-hoc/parametric polymorphism in a principled fashion.

```
<class-decl>   ::= class <class-ident> <type-binder> = <type-record>
<class-ident>  ::= <type-ctor>
```

A class declaration introduces new class methods, whom are globally scoped. Classes are very similar to records because of their implementation[27], except it utilizes the typeclass facility. A class declaration takes the form of

$$\textbf{class}\ C\ \bar{\alpha} = \texttt{<type-record>}$$

and the type of the class method f_i is

$$f_i : \forall \bar{\alpha}, \bar{\beta}.\, \sigma_i$$

where $\bar{\beta}$ is the additional type variables bound at method level[28], and σ_i is the type signature of f_i modulo binder. Consider

```
class Eq a =
  {eq : a -> a -> Bool}
class Functor (f : 1) =
  {map : forall a b, (a -> b) -> f a -> f b}
class HEq a b =
  {heq : a -> b -> Bool}

let main = (map, eq, heq)
```

`eq` has type $\forall a\ [\mathsf{Eq}\ a].\, a \to a \to \mathsf{Bool}$, `map` has type $\forall f\ [\mathsf{Functor}\ f].\, \forall a\ b.\, (a \to b) \to fa \to fb$, `heq` has type $\forall a\ b\ [HEqab].\, a \to b \to \mathsf{Bool}$; Whereas `main`, i.e. some program that uses all the methods above without providing additional type ascription to *refine* the class, yields the verbose

$$\forall \alpha\ \beta\ \gamma\ \delta\ [\mathsf{Functor}\ \alpha, \mathsf{Eq}\ \beta, \mathsf{HEq}\ \gamma\ \delta]\ \forall a\ b.\, ((a \to b) \to \alpha\ a \to \alpha\ b) \times (\beta \to \beta \to \mathsf{Bool}) \times (\gamma \to \delta \to \mathsf{Bool}).$$

Default methods and superclass are not supported at this moment.

[27] dictionary passing.
[28] supported by polymorphic fields.

Instance definition Just as in object-oriented programming which interfaces are always associated with their implementations, so is typeclass. *Instances* are used to implement typeclasses.

```
<inst-decl> ::= instance ":"? <inst-scheme> = "{" <inst-method-decl> "}"
<inst-scheme> ::= {FORALL <type-binder>+}? <inst-context>? "," <class-ident> <type-expr>+
<inst-context> ::= "[" sepBy1(",", <qualified>) "]"
<inst-method-decl> ::= sepBy(",", <let-binder>)
```

For the sake of syntactic uniformity, the RHS of a instance declaration is similar to a record literal, with its field assignments being similar to a let-binding.

```
instance : Eq Int = {eq x y = __eqInt x y} -- __eqInt is primitive, requires η-conversion
instance : Functor Option =
  { map f
    | Some x ⇒ Some $ f x
    | None   ⇒ None
  }
instance HEq String Bool =
  { heq = fun
          | "true", true   ⇒ true
          | "false", false ⇒ true
          | _, _           ⇒ false
  }
instance forall a b [HEq a b], HEq b a = {heq y x = heq x y}
instance forall a [Eq a], HEq a a = {heq = eq}
```

Instance declarations may quantify type variables, Together with contexts, this can be used (as shown in example 4 below) to express the *implication* relations.[29] Here, we quantify a, b and an instance HEq a b and use it as the implementation for $HEq\ b\ a$ to express the assumed *commutative* nature of a heterogenous equality and to express the fact that the homogenous equality relation subsumes the hetero one. Instance resolution is described later. For now, users only need to remember that

1. a type may not be declared as an instance of some class more than once.
2. overlapping instance results in a compile-time ambiguous exception.
3. ambiguous type arising from the use of a method blocks typeclass elaboration and throws a compile-time ambiguous exception.

```
class HAdd a b c = {hadd : a -> b -> c}
instance HAdd Int Int Int = {hadd = add}

-- Ambiguous: HEq ?m6 Int: typeclass elaboration is stuck because of
-- metavariable(s)
--   [?m6]
-- induced by a call to heq. To resolve this, consider adding type ascriptions.
let f' = heq (hadd 2 3) 5
```

In this example, `hadd 2 3` is typed a metavariable, results in `HEq`'s instance lookup failure.

Although it is entirely possible to infer the context in a instance declaration just the same as in `main` in the above example, we nevertheless require users to provide explicit instance context.

Multiple toplevel type declarations can optionally be separated with `;;`[30] for clarity.

[29] That is, the $\rightarrow$ in first-order logic, or $\vdash$ in metalanguage.

[30] except `mutual` block where `;;` is required to close it.

2.8 Wrapping Up: Brief implementation notes

Below we briefly talk about some key points of *Tigris*' implementation, though readers shall expect a more thorough discussion in the thesis.

Lexing & Parsing Lexing stage and Parsing stage are fused. We employ the functional way of *monadic parsing*, which typically uses *parser combinators*, together with the flexible custom operators, an implementation of a fusion of combinatory parsing and Pratt parsing is ultimately used.

In practice, parser combinators are similar to a recursive descent parser or a hand-written one. The grammar is similar to a parsing expression grammar (PEG) which always selects the first production when encountering ambiguous rules. Though it is currently, not a packrat parser which is guaranteed to run in linear time. Backtracking is still extensively used in our case as a result of the our monad class implementation. A parser monad in *Tigris* is defined to be

```
abbrev TParser σ := SimpleParserT Substring Char
                    $ StateRefT (PEnv × String) (ST σ)
```

Where σ is a formal argument for the `ST`-backed parser monad, allowing efficient and proper mutable state to be stored in threadlocals rather than a simulated `StateT` that uses stack space. The monad transformer stack also has two transformers `SimpleParserT` and `StateRefT` to allow the fluent use of `State`-like methods in `ST` as well as the efficient use of `Substring` where we do not allocate new strings and instead uses a pair of start/end pointer as a *view* during parsing.

According to Wikipedia, the two major parser combinators are (given two parsers P and Q)

- the *alternative* combinator $p \oplus q\ (j) = \mathsf{P}\ j \cup \mathsf{Q}\ j$
- the *sequence* combinator $(p \circledast q)\ (j) = \bigcup\{Q\ k | k \in P\ j\}$

This is exactly, captured by the `MonadExcept`, `Alternative` class and the `Seq` class in monadic functional programming, and is implemented as

```
1  instance (σ ε τ m) [Parser.Stream σ τ] [Parser.Error ε σ τ] [Monad m]
2    : Monad (ParserT ε σ τ m) where
3    -- (...)
4    seq f x s := f s >>= fun
5      | .ok s f => x () s >>= fun
6        | .ok s x => return .ok s (f x)
7        | .error s e => return .error s e
8      | .error s e => return .error s e
9
10 instance (σ ε τ m) [Parser.Stream σ τ] [Parser.Error ε σ τ] [Monad m] :
11   MonadExceptOf ε (ParserT ε σ τ m) where
12   throw e s := return .error s e
13   tryCatch p c s := p s >>= fun
14     | .ok s v => return .ok s v
15     | .error s e => (c e).run s
16
17 instance (σ ε τ m) [Parser.Stream σ τ] [Parser.Error ε σ τ] [Monad m] :
18   OrElse (ParserT ε σ τ m α) where
19   orElse p q s :=
20     let savePos := Parser.Stream.getPosition s
21     p s >>= fun
```

```
    | .ok s v ⇒ return .ok s v
    | .error s _ ⇒ q () (Parser.Stream.setPosition s savePos)
```

Basic scaffolding and common combinators are already done by Dorais' lean4-parser and our implementation depends on this package. Additionally, we have defined several combinators are defined to aid the parsing of function application in `utils.lean` (especially, `chainl` and `chainr`):

```
/--
  `chainl1 p op` parses *one* or more occurrences of `p`, separated by `op`. Returns
  a value obtained by a **left** associative application of all functions returned by `op` to the
  values returned by `p`.
  - if there is exactly one occurrence of `p`, `p` itself is returned touched.
-/
partial def chainl1
  (p : ParserT ε σ τ m α)
  (op : ParserT ε σ τ m (α → α → α)) : ParserT ε σ τ m α := do
  let x ← p; rest x where
  rest x :=
    (do let f ← op; let y ← p; rest (f x y)) <|> pure x

partial def chainr1
  (p : ParserT ε σ τ m α)
  (op : ParserT ε σ τ m (α → α → α)) : ParserT ε σ τ m α := do
  let x ← p; rest x where
  rest x :=
    (do let f ← op; chainr1 p op <&> f x) <|> pure x
/- like `test`, but non-consuming -/
def test' (p : ParserT ε σ τ m α) : ParserT ε σ τ m Bool :=
  try lookAhead p $> true
  catch _ ⇒ return false
```

Typechecker There are two typecheckers available in the *Tigris* repository, a legacy, straight implementation of Algorithm W, as well as a constraint solver version that performs better with qualified types and is actually used. We shall only discuss the latter. Typecheckers are run inside the `InferC` Monad:

```
abbrev InferC σ := StateRefT CState (EST TypingError σ)

local instance : MonadLift (Except TypingError) (InferC σ) where
  monadLift
  | .error e ⇒ throw e
  | .ok res ⇒ return res
```

A tone-downed version of GHC's type variable leveling is used to deal with erroneous unification. Pattern matrices are checked at this stage of exhaustiveness and redundancy, using an efficient matric-modeled algorithm from Maranget.

The typechecker checks type, as well as producing a typed AST, which will then be *elaborated* to a System F-ish IR, defined very similar to the core calculus of *Tigris*, as:

```
inductive FExpr where
  | CI     (i : Int)    (ty : MLType)
  | CS     (s : String) (ty : MLType)
```

```
    | CB     (b : Bool)   (ty : MLType)
    | CUnit               (ty : MLType)
    | Var    (x : String) (ty : MLType)                -- monomorphic view at site
    | Fun    (param : String) (paramTy : MLType)
             (body : FExpr) (ty : MLType)              -- paramTy → body.ty = ty
    | App    (f : FExpr) (a : FExpr) (ty : MLType)     -- result type after application
    | TyLam  (a : TV) (body : FExpr)                   -- type abstraction
    | TyApp  (f : FExpr) (arg : MLType)                -- type application
    | Let    (binds : Array (String × Scheme × FExpr))  -- each binding elaborated & wrapped
             (body : FExpr) (ty : MLType)
    | Prod'  (l r : FExpr) (ty : MLType)
    | Cond   (c t e : FExpr) (ty : MLType)
    | Match  (discr : Array FExpr)
             (branches   : Array (Array Pattern × FExpr))
             (resTy      : MLType)
             (counterexample : Option (List Pattern))
             (redundantRows  : Array Nat)
    | Fix    (e : FExpr) (ty : MLType)                 -- rec tag
    | Proj   (src : FExpr) (mname : String)
             (idx : Nat) (ty : MLType)                 -- field projection
  deriving Repr, Inhabited
```

Using a pretty-printing algorithm similar to Wadler's, `FExpr` is pretty printed to have a syntax that is very much similar (except it always denotes signatures and type application) to a subset of *Tigris*. This algorithm is used for subsequent IRs.

From there, we begin the lowering process by lowering it to *Lambda* IR.

Type system, together with typechecking, is a major part of *Tigris*' implementation, we'll have a more thorough discussion below.

3 Type System of the *Tigris* language

Nomenclature Throughout this section, the word *constraint* is extensively used. Since we implement a constraint-solver based typechecker, broadly speaking, constraint can reference to

- the equality constraint, which is reflexive, commutative (though mapping is one-way in our case) and transitive.
- the predicate constraint, which is cumulative, as in this type of constraints can accumulate in the scheme's context.

Though, sometimes, it may also exclusively have the meaning 2). Readers should be able to deduce the meaning from the context as we stick to the broader sense in this text.

Tigris implements a constraint-based Hindley-Milner type system extended with

Predicates are of the form $C\bar{\alpha}$ where C is a class head. These are collected during inference and resolved via dictionary passing after constraint solving.

***Strict* Higher-Kinded Types** refers to the fact that higher-kinded TVs must be fully applied. This is done through the use of `KApp` which is just enough for abstractions like `Functor` and `Monad`, see footnote 7.

Limited Rank-N polymorphism refers to the fact that *schemes* (`TSch`) can appear inside monotypes `MLType`, but the system remains in rank-1 polymorphism through the addition of skolemization during type ascription checking.[31] This means when checking $e : \sigma$, we instantiate σ with skolem constants[32] and verify the inferred type unifies. Intuitively and informally this should be sound as

- skolem-TVs are rigid in the sense that they are unique and cannot be unified with other types.
- skolem-TVs are tracked through a rigid set that is used to reject substitutions which would otherwise escape scope.

Constraint generation & Solving Generation of constraints are handled by inference, which in turn, solves them in one pass. The solver returns a substitution (also called a *solution*) and residual predicates that become part of the generalized scheme.

Let-Polymorphism refers to let-bound variables are generalized over TVs such that $tv \notin \mathsf{ftv}\ \Gamma$, with predicate constraints becoming part of the scheme's context. Informally, it is correct to state that, program is still typeable even if users may not provide any signature due to the principality of HM systems.

3.1 Formal Grammar

The type system is built on its core calculus, an augmented version of the lambda calculus. To describe its typing discipline, we first present the formal grammar of the core calculus as well as the type expression, among other things. [33]

expr, e	::=		term
	|	*const*	constant
	|	*tname*	variable
	|	$\lambda\ x.\ e$	lambda abstraction
	|	$\textbf{let}\ x_1 = e_1\ \textbf{and}\ ..\ \textbf{and}\ x_n = e_n\ \textbf{in}\ e'$	letexp
	|	$e\ e'$	function application
	|	(e_1, e_2)	product
	|	$\textbf{if}\ e_1\ \textbf{then}\ e_2\ \textbf{else}\ e_3$	conditional
	|	$\textbf{fix}\ e$	fixpoint
	|	$\textbf{match}\ e_1, ..., e_n\ \textbf{with}\ Pmatrix$	pattern matching
	|	$e : sch$	type ascription

pattern, *pat*	::=		patterns
	|	*tname*	variable
	|	_	wildcard
	|	*const*	constant
	|	(pat_1, pat_2)	product
	|	$ctor\ pat_1\ ..\ pat_n$	constructor application

Pmatrix	::=		pattern matrix
	|	$\|pb_1\|\ ..\ \|pb_m$	

patbranch, *pb*	::=		pattern branch
	|	$pat_1, ..., pat_n \rightarrow expr$	

[31] as `gen` or `generalize` does not generate higher-rank binders.
[32] encoded by `TCon` i.e. a nullary constructor
[33] most grammar, rules, are generated from `report.ott` using Ott.

tyctor ::=
- | *upperident*
- | Int
- | Bool
- | String
- | Unit

tyvar, *tv*, *a*, *b* ::=
- | *lowerident*

tvset, $\bar{\alpha}$ ::= type variable set
- | $\varnothing$
- | $\{tv\}$
- | $\bar{\alpha}_1 \cup \bar{\alpha}_2$
- | $\bar{\alpha}_1 \cap \bar{\alpha}_2$
- | $\bar{\alpha}_1 \setminus \bar{\alpha}_2$
- | $\mathsf{ftv}(\Gamma)$
- | $\mathsf{ftv}(ty)$
- | $\mathsf{ftv}(\bar{\pi})$
- | $\mathsf{ftv}(pred)$

tyargs, *tys* ::= one or more type arguments
- | *ty*
- | *ty tys*

typeexpr, *ty*, τ ::= type expression

|	*tyctor*	nullary type constructor
|	*tv*	type variable
|	$ty \to ty'$	arrow type
|	$ty \times ty'$	product type
|	*tyctor tys*	nominal type application
|	$\kappa\ tys$	higher-kinded TV head application

predicate, *pred*, π ::= predicate
- | $ctor\ ty_1\ ..\ ty_n$

predicates, *preds*, $\bar{\pi}$ ::=
- | $\varnothing$
- | *pred*
- | $pred, \bar{\pi}$
- | $\bar{\pi}_1, \bar{\pi}_2$

scheme, *sch*, σ ::= scheme

|	$\forall\ tv_1\ ..\ tv_n.\bar{\pi} \Rightarrow ty$	
|	$(\bar{\alpha}.\bar{\pi} \Rightarrow ty)$	from tvset
|	*ty*	Monotype

context, Γ ::= typing environment
- | $\varnothing$
- | $\Gamma, x : sch$
- | $x : sch$

constraint, c	::=		constraint
	|	$ty_1 \sim ty_2$	equality constraint
	|	$pred_{evidx}$	predicate constraint with evidence

constraints, C	::=		constraint set
	|	$\varnothing$	
	|	c	
	|	C_1, C_2	set join

subst, θ	::=		substitution
	|	$\varnothing$	
	|	$tv \mapsto ty$	
	|	$\theta_1 \circ \theta_2$	composition

evidence, ev	::=		evidence term
	|	x	evidence variable
	|	$ctor\, ev_1 \,..\, ev_n$	dictionary constructor

evctx, Δ	::=		evidence context
	|	$\varnothing$	
	|	$\Delta, evidx : pred$	

3.2 Judgments & Inference rules

We now, present the judgments as well as the inference rule. Each judgement comes with a remark that informally explains the implementation.

Unification computes a most general unifier (MGU) for equality constraints. The unification engine

- checks occurrence to prevent infinite types as shown in section 2.4
- requires constructors to match exactly
- unifies `KApp` with concrete type applications `TApp`, provided arities match.
- α-renames schemes before unifying them structurally.

$\boxed{\theta = \mathsf{unify}\,(ty_1, ty_2)}$ *(Unification of types)*

U-Refl
$$\frac{}{\varnothing = \mathsf{unify}\,(ty, ty)}$$

U-VarL
$$\frac{tv \notin \mathsf{ftv}(ty)}{tv \mapsto ty = \mathsf{unify}\,(tv, ty)}$$

U-VarR
$$\frac{tv \notin \mathsf{ftv}(ty)}{tv \mapsto ty = \mathsf{unify}\,(ty, tv)}$$

U-Arrow
$$\frac{\theta_1 = \mathsf{unify}\,(a, a') \quad \theta_2 = \mathsf{unify}\,([\theta_1]b, [\theta_1]b')}{\theta_2 \circ \theta_1 = \mathsf{unify}\,(a \to b, a' \to b')}$$

U-Prod
$$\frac{\theta_1 = \mathsf{unify}\,(a, a') \quad \theta_2 = \mathsf{unify}\,([\theta_1]b, [\theta_1]b')}{\theta_2 \circ \theta_1 = \mathsf{unify}\,(a \times b, a' \times b')}$$

Constraint Solving are done left-to-right, threading the substitution through. Predicate constraints are collected and returned as residuals for instance resolution.

$\boxed{(\theta, \bar{\pi}) = \mathsf{solve}\,(C)}$ *(Constraint solving)*

$$\textsc{Slv-Empty}\ \frac{}{(\varnothing, \varnothing) = \mathsf{solve}\,(\varnothing)} \qquad \textsc{Slv-Eq}\ \frac{\theta = \mathsf{unify}\,(ty_1, ty_2) \quad (\theta', \bar{\pi}) = \mathsf{solve}\,([\theta]\,C)}{(\theta' \circ \theta, \bar{\pi}) = \mathsf{solve}\,(ty_1 \sim ty_2, C)} \qquad \textsc{Slv-Pred}\ \frac{(\theta, \bar{\pi}) = \mathsf{solve}\,(C)}{(\theta, [\theta]pred, \bar{\pi}) = \mathsf{solve}\,(pred_{evidx}, C)}$$

Instantiation replaces bound TVs with fresh TVs and returns the predicate constraints that must be satisfied.

$\boxed{(ty, \bar{\pi}) = \mathsf{inst}\,(sch)}$ *(Scheme instantiation)*

$$\textsc{Inst-Forall}\ \frac{tv'_1 = \mathsf{fresh} \quad .. \quad tv'_n = \mathsf{fresh} \quad \theta = \{tv_1 \mapsto tv'_1, .., tv_n \mapsto tv'_n\}}{([\theta]ty, [\theta]\bar{\pi}') = \mathsf{inst}\,(\forall\, tv_1 .. tv_n.\bar{\pi} \Rightarrow ty)}$$

Generalization abstract over TVs not free in the environment. Predicates mentioning such TVs become part of the context, that is,

A predicate is kept if and only if it mentions one or more qTVs[34] that also appears in the result type.

$\boxed{sch = \mathsf{gen}\,(\Gamma, ty, \bar{\pi})}$ *(Generalization)*

$$\textsc{Gen-Gen}\ \frac{\bar{\alpha} = (\mathsf{ftv}(ty) \cup \mathsf{ftv}(\bar{\pi})) \setminus \mathsf{ftv}(\Gamma)}{(\bar{\alpha}.\bar{\pi} \Rightarrow ty) = \mathsf{gen}\,(\Gamma, ty, \bar{\pi})}$$

Expression Typing generates constraints that are used in constraint solving to determine[35] the actual type. The judgment $\Gamma \vdash e \Rightarrow ty; C$ reads:

Under environment Γ, expression e has type ty generating constraints C.

Based on the core calculus, inference applies to:

Variables by looking up in Γ and instantiate the scheme,

Constants which already have concrete types,

Lambda abstraction of which parameter[36] gets fresh TVs,

Application by generating constraints that make the function type match,

Product by generating constraints that make the product type match,

[35] by applying all substitutions

[36] a lambda abstraction only takes one (1) parameter, as mentioned before.

Conditional by requiring then/else-branch to agree, condition to be Bool.

Let by inferring body *after* inferring/solving/generalizing RHS,

Let rec by extending Γ with fresh TVs for all binders *before* inferring each RHS, whose generated constraints equate these with inferred types,

Fix where **fix** $\lambda\ f.\ e$ has type of e,

Ascription by checking that inferred type is at least as general, which, takes different approach based on ranks where

- for rank-1 ascriptions, we simply unify.
- for rank-n (where $n \geq 2$, that is, if $TSch$ contains nested quantifiers), we have the following steps:

  1. **Instantiation** instantiate the scheme $\sigma = \forall \bar{\alpha}.\, \bar{\pi} \Rightarrow \tau$ with fresh type variables $\bar{\beta}$, yielding τ_{inst}.
  2. **Constraint solving** solve the current constraints augmented with $\tau_e \sim \tau_{\mathsf{inst}}$ to obtain a substitution θ'.[37]
  3. **Rigidity check** For each binding $\alpha \mapsto \tau'$ in the substitutions θ', if α is in the current rigid set $\mathcal{R}$, then τ' must equal α (from the property of rigids. i.e. it must *not* be specialized). Failing this check results in the ascription being rejected with a unification error.
  4. **Skolemization** As stated above and below, ensures the generality does match the declared (polymorphic) type.

Pattern matching where for

- each branch i: Patterns $p_{i,1}, \ldots, p_{i,m}$ are checked against the corresponding discriminant types. Γ is then enriched with pattern bindings. TVs arising from patterns are marked as *rigid* (within their branches) which prevents leaking[38] across branches. RHS is inferred after, requiring all branches to agree (through constraints).
- discriminants: Each one e_j is inferred independently, yielding types $\tau_1, \ldots, \tau_m$. whose types must agree with pattern types.

Additionally, pattern exhaustiveness and redundancy are checked separately (via `Exhaustive.exhaustWitness`) but not reflected in the typing judgment. Non-exhaustive matches or redundant branches generate warnings but are not type errors. Note that the latter is immediately removed from the matrix.

$\boxed{\Gamma \vdash e \Rightarrow ty;\, C}$ *(Constraint-based inference)*

I-Var
$$\frac{x : sch \in \Gamma \qquad (ty, \bar{\pi}) = \mathsf{inst}\,(sch)}{\Gamma \vdash x \Rightarrow ty;\, C}$$

I-Int
$$\frac{}{\Gamma \vdash intlit \Rightarrow \mathsf{Int};\, \varnothing}$$

I-String
$$\frac{}{\Gamma \vdash strlit \Rightarrow \mathsf{String};\, \varnothing}$$

I-True
$$\frac{}{\Gamma \vdash \mathbf{true} \Rightarrow \mathsf{Bool};\, \varnothing}$$

[37] This can be seen as a premature solving. We *tentatively* solve them only to use them in the next step that is rigidity check.

[38] this is mostly a precaution. The "leaking" described here is only likely to happen when pattern matching index families, which, is not supported, at all.

$$\frac{}{\Gamma \vdash \textbf{false} \Rightarrow \mathsf{Bool}; \varnothing}\ \text{I-False} \qquad \frac{}{\Gamma \vdash () \Rightarrow \mathsf{Unit}; \varnothing}\ \text{I-Unit} \qquad \frac{tv = \mathsf{fresh} \quad \Gamma, x : tv \vdash e \Rightarrow ty; C}{\Gamma \vdash \lambda\ x.\ e \Rightarrow tv \to ty; C}\ \text{I-Lam}$$

$$\frac{\Gamma \vdash e_1 \Rightarrow ty_1; C_1 \quad \Gamma \vdash e_2 \Rightarrow ty_2; C_2 \quad tv = \mathsf{fresh}}{\Gamma \vdash e_1\ e_2 \Rightarrow a; (C_1, C_2), ty_1 \sim ty_2 \to a}\ \text{I-App} \qquad \frac{\Gamma \vdash e_1 \Rightarrow ty_1; C_1 \quad \Gamma \vdash e_2 \Rightarrow ty_2; C_2}{\Gamma \vdash (e_1, e_2) \Rightarrow ty_1 \times ty_2; C_1, C_2}\ \text{I-Prod}$$

$$\frac{\Gamma \vdash e_1 \Rightarrow ty_1; C_1 \quad \Gamma \vdash e_2 \Rightarrow ty_2; C_2 \quad \Gamma \vdash e_3 \Rightarrow ty_3; C_3}{\Gamma \vdash \textbf{if}\ e_1\ \textbf{then}\ e_2\ \textbf{else}\ e_3 \Rightarrow ty_2; (((C_1, C_2), C_3), ty_1 \sim \mathsf{Bool}), ty_2 \sim ty_3}\ \text{I-Cond}$$

$$\frac{\Gamma \vdash e_1 \Rightarrow ty_1; C_1 \quad (\theta, \bar{\pi}) = \mathsf{solve}\,(C_1) \quad sch = \mathsf{gen}\,(\Gamma, [\theta]ty_1, \bar{\pi}) \quad \Gamma, x : sch \vdash e_2 \Rightarrow ty_2; C_2}{\Gamma \vdash \textbf{let}\ x = e_1\ \textbf{in}\ e_2 \Rightarrow ty_2; [\theta]C_2}\ \text{I-Let}$$

$$\frac{tv = \mathsf{fresh} \quad \Gamma, f : tv \vdash e \Rightarrow ty; C}{\Gamma \vdash \textbf{fix}\,(\lambda\ f.\ e) \Rightarrow ty; C, tv \sim ty}\ \text{I-Fix}$$

$$\frac{tv = \mathsf{fresh} \quad \Gamma, x : (tv) \vdash e_1 \Rightarrow ty_1; C_1 \quad (\theta, \bar{\pi}) = \mathsf{solve}\,(C_1) \quad sch = \mathsf{gen}\,(\Gamma, [\theta]ty_1, \bar{\pi}) \quad \Gamma, x : sch \vdash e \Rightarrow ty; C}{\Gamma \vdash \textbf{let}\ x = e_1\ \textbf{in}\ e \Rightarrow ty_1; [\theta]C}\ \text{I-LetRec}$$

Since match clause and type ascription are not easy to correctly type in OTT and too convoluted in terms of readability, we manually type them as follows:[39]

$$\frac{\begin{array}{c}\Gamma \vdash e_j \Rightarrow \tau_j\,;\, C_j \quad (\forall j \in 1..m) \\ \alpha = \mathsf{fresh} \\ \text{For each branch } i \in 1..k: \quad |(\bar{p}_i)| = m \quad \Gamma \vdash p_{i,j} \Leftarrow \tau_j \Rightarrow \Gamma_{i,j}\,;\, C_{p_{i,j}} \quad (\forall j \in 1..m) \\ \Gamma_i = \Gamma_{i,m} \quad (\text{final extended environment for branch } i) \\ \bar{\beta}_i = \mathsf{ftv}(\text{bound variables in } \bar{p}_i) \quad \mathsf{pushRigid}(\bar{\beta}_i) \quad \Gamma_i \vdash r_i \Rightarrow \tau_{r_i}\,;\, C_{r_i} \quad \mathsf{popRigid}() \\ C_{\mathsf{unify}} = \{\tau_{r_i} \sim \alpha \mid i \in 1..k\} \quad C = \bigcup_{j=1}^{m} C_j \cup \bigcup_{i=1}^{k} \left(\bigcup_{j=1}^{m} C_{p_{i,j}} \cup C_{r_i} \right) \cup C_{\mathsf{unify}}\end{array}}{\Gamma \vdash \textbf{match}\ e_1, \ldots, e_m\ \textbf{with}\ \overline{\mid p_{1,1}, \ldots, p_{1,m} \to r_1} \Rightarrow \alpha\,;\, C}\ \text{I-Match}$$

$$\frac{\Gamma \vdash e \Rightarrow \tau_e\,;\, C_e}{\Gamma \vdash (e : \tau) \Rightarrow \tau\,;\, C_e, \tau_e \sim \tau}\ \text{I-Ascribe-Mono}$$

$$\frac{\begin{array}{c}\Gamma \vdash e \Rightarrow \tau_e\,;\, C_e \\ \sigma = \forall \bar{\alpha}.\, \bar{\pi} \Rightarrow \tau \quad \bar{\beta} = \mathsf{fresh}^{|\bar{\alpha}|} \quad \theta = [\bar{\alpha} \mapsto \bar{\beta}] \quad \tau_{\mathsf{inst}} = \theta(\tau) \\ C_0 = \text{current constraints} \quad (\theta', \_) = \mathsf{solve}(C_0, \tau_e \sim \tau_{\mathsf{inst}}) \\ \mathcal{R} = \text{current rigid set} \quad \forall (\alpha \mapsto \tau') \in \theta'.\ \alpha \in \mathcal{R} \implies \tau' = \alpha \\ \tau_{\mathsf{sk}} = \mathsf{skolem}(\sigma) \quad \text{add constraints for } \bar{\pi}\end{array}}{\Gamma \vdash (e : \sigma) \Rightarrow \tau_{\mathsf{inst}}\,;\, C_e, \tau_e \sim \tau_{\mathsf{inst}}}\ \text{I-Ascribe-Poly}$$

[39] Readers shall note that because these rules are hand-typed, some symbols are not defined as a production rule above, or a formula in OTT, in this case, adequate natural language is used to define/describe them inline. The author has no one but himself to blame due to his unfamiliarity with OTT.

Patterns bind variables and generate constraints from the discriminants. The judgment $\Gamma \vdash pat \Rightarrow \Gamma'; C; ty$ reads:

Pattern *pat* against type *ty* extends Γ to Γ', generating constraints C.

$\boxed{\Gamma \vdash pat \Rightarrow \Gamma'; C; ty}$ *(Pattern inference)*

$$\text{P-Wild}\ \frac{}{\Gamma \vdash \_ \Rightarrow \Gamma; \varnothing; ty} \qquad \text{P-Var}\ \frac{x \notin \mathsf{dom}\,(\Gamma)}{\Gamma \vdash x \Rightarrow \Gamma, x : ty; \varnothing; ty} \qquad \text{P-ConstInt}\ \frac{}{\Gamma \vdash intlit \Rightarrow \Gamma; ty \sim \mathsf{Int}; ty}$$

$$\text{P-ConstStr}\ \frac{}{\Gamma \vdash strlit \Rightarrow \Gamma; ty \sim \mathsf{String}; ty} \qquad \text{P-ConstTrue}\ \frac{}{\Gamma \vdash \mathbf{true} \Rightarrow \Gamma; ty \sim \mathsf{Bool}; ty} \qquad \text{P-ConstFalse}\ \frac{}{\Gamma \vdash \mathbf{false} \Rightarrow \Gamma; ty \sim \mathsf{Bool}; ty}$$

$$\text{P-ConstUnit}\ \frac{}{\Gamma \vdash () \Rightarrow \Gamma; ty \sim \mathsf{Unit}; ty} \qquad \text{P-Prod}\ \frac{\begin{array}{c} a = \mathsf{fresh} \qquad b = \mathsf{fresh} \\ \Gamma \vdash pat_1 \Rightarrow \Gamma_1; C_1; a \\ \Gamma_1 \vdash pat_2 \Rightarrow \Gamma_2; C_2; b \end{array}}{\Gamma \vdash (pat_1, pat_2) \Rightarrow \Gamma_2; (C_1, C_2), ty \sim a \times b; ty}$$

Since constructor applications are not easy to correctly type in OTT and too convoluted in terms of readability, we manually type them as follows:

$$\frac{\begin{array}{c} ctor : \sigma \in \Gamma \qquad (\tau_{ctor}, \bar{\pi}) = \mathsf{inst}(\sigma) \qquad \mathsf{peel}(\tau_{ctor}) = (\tau_1, \ldots, \tau_n, \tau_{\mathsf{res}}) \qquad n = |\bar{p}| \\ \Gamma_0 = \Gamma \qquad \Gamma_{i-1} \vdash p_i \Leftarrow \tau_i \Rightarrow \Gamma_i \,;\, C_i \quad (\forall i \in 1..n) \qquad C_\pi = \bar{\pi} \end{array}}{\Gamma \vdash ctor\ p_1 \cdots p_n \Leftarrow \tau \Rightarrow \Gamma_n \,;\, C_1, \ldots, C_n, C_\pi, \tau \sim \tau_{\mathsf{res}}}\ \text{P-Ctor}$$

Where peel decomposes an arrow type into its argument types and result type:

$$\mathsf{peel}(\tau_1 \to \tau_2 \to \cdots \to \tau_n \to \tau_r) = (\tau_1, \tau_2, \ldots, \tau_n, \tau_r)$$

The constructor scheme σ is looked up in Γ and instantiated with fresh type variables. Predicate constraints from instantiation are added to ensure typeclass requirements are propagated.

Below, we list some remarks for concepts mentioned above that are not properly introduced.

Remark (Skolemization). replaces TVs with skolems that cannot unify with anything except themselves.

$$\mathsf{skolem}(\forall \bar{\alpha}.\, \bar{\pi} \Rightarrow \tau) = [\bar{\alpha} \mapsto \overline{\mathsf{sk}_\alpha}](\tau)$$

where each sk_α is a fresh skolem constant (implemented as a special type constructor, e.g., `?sk.`α in the implementation).

The word *rigid* is to capture the key nature of skolems: $\mathsf{sk}_\alpha \sim \tau$ succeeds iff $\tau = \mathsf{sk}_\alpha$.

Remark (Rigid Set/Stack). The type system maintains a stack of rigid type variable sets to handle nested scopes correctly.

$$\begin{array}{rl} \mathsf{pushRigid}(\bar{\alpha}): & \mathcal{R} := \mathcal{R} \cup \bar{\alpha}; \quad stack := \bar{\alpha} :: stack \\ \mathsf{popRigid}(): & \textbf{let}\ (S :: rest) = stack\ \textbf{in}\ \mathcal{R} := \mathcal{R} \setminus S; \quad stack := rest \end{array}$$

Invariant. At any point during inference, the rigid set maintains the fact that

$$\mathcal{R} \equiv \bigcup stack$$

(i.e. it is the union of all sets currently on the stack.) When solving constraints, any substitution $\alpha \mapsto \tau$ where $\alpha \in \mathcal{R}$ and $\tau \neq \alpha$ is rejected.

Note that the (:=) operation is implemented by the `ST` monad, through the `MonadStateOf` interface.

Remark (Solving Rigids). Though not shown above in the inference rule, the constraint solver respect rigid TVs:

$$\frac{\alpha \in \mathcal{R} \qquad \tau \neq \alpha}{\mathsf{solve}(\alpha \sim \tau) = \textbf{error}}\ \textsc{Solve-Rigid} \qquad \frac{\alpha \in \mathcal{R}}{\mathsf{solve}(\alpha \sim \alpha) = \varnothing}\ \textsc{Solve-Rigid-Ok}$$

Remark (Soundness). The type system should maintain soundness through several mechanisms:

Skolems During pattern matching and typing ascription, some TVs are marked as *rigid* and tracked in `rTV`. Substitutions that would specialize them to incompatible types are rejected by the solver. This is thought to prevent unsound generalizations.

Rank-N skolemization When checking $e : \forall\, tv.\pi \Rightarrow ty$, we replace tv with a fresh skolem constant (in order to make it truly general) to ensure expressions do work for all instantiations.

Occurs check is used to prevent users from constructing equirecursive types such as $\alpha \sim \alpha \to \beta$ or $\alpha \sim \alpha \times \beta$ arising from codes such as `fun x ⇒ x x` or `fun | (x, y) ⇒ x | z ⇒ z`, respectively.

Hygienic generalization Only TVs not free in Γ are generalized. Predicates are kept based on the condition described above that shall avoid vacuous constraints.

Syntax restrictions Lastly, if one looks closely at the BNF grammar of type expression or the formal grammar presented above, they will find that, it is actually not possible to write $(\forall a.a \to a) -> \mathsf{Int} \times \mathsf{Bool}$ as though `MLType` embeds scheme, the parsing facility does not admit schemes inside monotypes. Indeed, the only way to introduce embedded scheme is through polymorphic fields in datatype declarations, which, is not harmful. This is perhaps, the strongest restriction that is thought to guarantee the soundness of the type system.

Remark (Counterexamples: The need of rigids). Below we provide some counterexamples that emphasize the need of rigids:

- (**Wrong Specialization**) Consider the following

```
let id = (fun x ⇒ x : forall a, a -> a) 1

-- [ERROR] Can't unify type Int with ?3k.a.
let id'    : forall a, a -> a = fun x ⇒ (x : Int)
let id'' x : forall a, a -> a = x + 1
```

  The first example is acceptable: The ascription claims polymorphism, but the body is monomorphic, which is fine since we are instantiating the polymorphic function.

  In the second example however, It would be unsound for a, introduced as a rigid inside `fun x ⇒ ...` to be specialized to Int. Same applies to the third.

- (**Polymorphism recovery**) Typically, typeclass methods (which is very likely to be polymorphic themselves) are used directly which allows the special treatment of them like in Haskell '98. However, one may wish to write:

```
type Functor (f : 1) = Functor (forall a b, (a -> b) -> f a -> f b)

let prog (Functor f) =
  let l  = 1 :: 2 :: 3 :: Nil in
  and l' = "1" :: "2" :: "3" :: Nil in
  and const k _ = k in
  let x  = f (_ + 1) l in
  and y  = f (const ()) l'
  in (x, y)
```

This is perfectly acceptable, yet, do note the importance of rigids: a and b are rigid within each use, within each call. This allows the independent instantiation of them as f is specialized to different concrete type at x ($a \mapsto \mathsf{Int}, b \mapsto \mathsf{Int}$) and y ($a \mapsto \mathsf{String}, b \mapsto \mathsf{Unit}$).

3.3 Instance resolution

Merely associating instances with typeclasses is not enough at all: We, or, to be precise, the compiler, requires a general procedure (algorithm) to lookup the right instance for the right type of some class.[40] This process is called *instance resolution*. The algorithm described below searches the instance environment for a matching instance, recursively resolving any context constraints.

Instance metadata Each registered instance I's metadata is a structure

$$I = \langle \mathsf{iname}, \mathsf{cls}, \bar{\alpha}, \bar{\pi}_{\mathsf{ctx}} \rangle$$

where

- iname is the instance's unique identifier (naming convention: `i_Eq_0`),
- cls is the class head,
- $\bar{\alpha}$ is the list of type arguments in the instance head,
- $\bar{\pi}_{\mathsf{ctx}}$ is the list of context predicates.

e.g. an instance declaration `instance : forall α [Eq α], Eq (List α) = ...` produces

$$I = \langle \texttt{i\_Eq\_1}, \texttt{Eq}, [\texttt{List}\ \alpha], [\texttt{Eq}\ \alpha] \rangle$$

Head Unification refers to the procedure of checking whether a goal $C\ \bar{\tau}$ can match an instance head $C\ \bar{\alpha}$ before attempting full resolution.

[40] Or as in PL jargon: finding a *dictionary* (evidence term) that *witnesses* a predicate constraint $\pi = C\ \bar{\tau}$.

Algorithm 1 Head Unification

function UNIFYHEAD($\bar{\tau}_{\mathsf{goal}}, \bar{\alpha}_{\mathsf{inst}}$)
  if $|\bar{\tau}_{\mathsf{goal}}| \neq |\bar{\alpha}_{\mathsf{inst}}|$ **then**
    return error
  end if
  $\theta \leftarrow \varnothing$
  for $i \leftarrow 1$ **to** $|\bar{\tau}_{\mathsf{goal}}|$ **do**
    $\theta' \leftarrow \mathsf{unify}(\mathsf{apply}(\theta, \tau_i), \mathsf{apply}(\theta, \alpha_i))$
    $\theta \leftarrow \theta' \circ \theta$
  end for
  return θ
end function

Resolution Algorithm The main resolution algorithm maintains a *visited set* $\mathcal{V}$ to detect cycles (which indicates either ambiguous or divergent instance resolution).

Algorithm 2 Instance Resolution

function RESOLVE($\Gamma, \pi, \mathcal{V}$)
  Input: Environment Γ, predicate $\pi = C\ \bar{\tau}$, visited set $\mathcal{V}$
  Output: Evidence expression e s.t. $e : C\ \bar{\tau}$, or error

  if $\pi \in \mathcal{V}$ **then**
    return error: "cyclic instance resolution"
  end if
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{\pi\}$

  $insts \leftarrow \Gamma.\mathsf{instInfo}[C]$ ▷ All instances for class C
  if $insts = \varnothing$ **then**
    return error: "no instance for π"
  end if

  $found \leftarrow$ **none**
  for each $I = \langle \mathsf{iname}, C, \bar{\alpha}, \bar{\pi}_{\mathsf{ctx}} \rangle$ **in** $insts$ **do**
    try:
      $\theta \leftarrow$ UNIFYHEAD($\bar{\tau}, \bar{\alpha}$)
      $\bar{\pi}'_{\mathsf{ctx}} \leftarrow \mathsf{apply}(\theta, \bar{\pi}_{\mathsf{ctx}})$
      for each $\pi' \in \bar{\pi}'_{\mathsf{ctx}}$ **do**
        $\_ \leftarrow$ RESOLVE($\Gamma, \pi', \mathcal{V}$) ▷ Ensure solvable
      end for
      $e \leftarrow (\mathsf{iname} : C\ \bar{\tau})$ ▷ Ascribed instance reference
      if $found \neq$ **none then**
        return error: "ambiguous instance for π"
      end if
      $found \leftarrow$ **some**(e)
    catch: **continue** ▷ Try next instance
  end for

  if $found =$ **none then**
    return error: "no matching instance for π"
  end if
  return $found$
end function

Remark (ResolveM). The algorithm described above runs inside `ResolveM` monad. Just like any monad we previously mentioned, it is a `ST` backed state monad with error handling (hence `EST`), whose state is ResolveState, which is precisely the visited set $\mathcal{V}$.

```
abbrev ResolveState := Std.HashSet Pred
abbrev ResolveM σ := StateRefT ResolveState (EST TypingError σ)

variable {σ : Type}

instance : MonadLift (Except TypingError) (EST TypingError σ) where
  monadLift
  | .error e ⇒ throw e
  | .ok e ⇒ return e
```

Meanwhile, the `MonadLift` instance defines a natural transformation from `Except` to `EST`, lifting fail-able but pure actions to run in it.

Remark (Properties of instance resolution). The key design points of this algorithm are described below:

Termination has not been proved, as noted by the `partial` modifier in the source. Intuitively, It terminates if the dependency graph of a instance is well-founded i.e. does not contain infinite chains of instance dependencies. The set $\mathcal{V}$ (line 5) detects direct cycles, but may not (or does not) guarantee termination for all pathological instances.

Semantic coherence is enforced through a *single-match* policy: If multiple instances could match a goal (line 24), resolution fails with an ambiguity error, which, guarantees that the meaning of a program does not depend on which instance is arbitrarily chosen. However, users should note that, contrary to the usual practice, this restriction applies to all instances regardless of their generality, that is, the more specific one would not be chosen although it may make more sense doing so, and will result in a compile-time error. To allow the existence of multiple matched instances, one way is to introduce a precedence and a default rule like the above practice, but this currently is not supported.

Evidence generation refers to ascribed instance identifiers:

$$e = (\mathsf{iname} : C\ \bar{\tau})$$

Which is later *elaborated* into dictionary-passing style during the System F IR translation phase from which is then encoded in the field projection node Proj *src mname i ty*.

Context propagation refers to instantiating context predicates with the unifying substitution (line 19) and the recursively resolving of them (line 20), from instances with context predicates $\bar{\pi}_{\mathsf{ctx}}$ (e.g., `Eq a` in `instance : forall α [Eq α], Eq (List α) = ...`)

We shall simulate this algorithm with a simple example below: Consider resolving Eq (List Int) with instances:

$$I_0 = \langle \mathsf{i\_Eq\_0}, \mathsf{Eq}, [\mathsf{Int}], \varnothing \rangle,$$
$$I_1 = \langle \mathsf{i\_Eq\_1}, \mathsf{Eq}, [\mathsf{List}\ \alpha], [\mathsf{Eq}\ \alpha] \rangle.$$

1. Goal: Eq (List Int)
2. Try I_0: unify(List Int, Int) fails
3. Try I_1: unify(List Int, List α) succeeds with $\theta = [\alpha \mapsto \mathsf{Int}]$
4. Context: apply(θ, [Eq α]) = [Eq Int]
5. Recursively resolve Eq Int:
   (a) Try I_0: unify(Int, Int) succeeds with $\theta = \varnothing$
   (b) Context: $\varnothing$, no recursion needed
   (c) Return (i_Eq_0 : Eq Int)
6. Return (i_Eq_1 : Eq (List Int))

Remark (Comparison with GHC). GHC implements a more robust, sophisticated typeclass facility. Although *Tigris* has learned and taken many inspirations from it, its instance resolution strategy is much simpler than GHC's:

- Tigris does not allow overlapping instances, regardless of any ordering, whereas GHC provides precedence that controls the granularity of instance resolution;
- Tigris does not have instance chains as there is no mechanism for specifying fallback instances or instance priority

However, when compared to Haskell '98 or 2010, *Tigris*' instance declarations is somewhat more expressive as we have some form of `FlexibleContexts`[41] as it is possible to define `instance ∀a [Eq a], Eq (a * a) = ..;` Additionally, multi-parameter classes are also supported, such as the `HAdd` and `HEq` class mentioned above.

3.4 System F elaboration

The typed syntax `TExpr`, after instance resolution, is elaborated to `FExpr`. Below, we discuss an informal description of the elaboration.

Given a scheme

$$\forall \alpha_1 \cdots \alpha_n [P_1, \ldots, P_k], \tau$$

we elaborate a binding rhs to

$$\Lambda \alpha_1, \ldots, \alpha_n . \lambda d_1, \ldots, d_k .\ body$$

where $d_1, \ldots, d_k$ are the respective instances of preds $P_1, \ldots, P_k$. Except when the rhs already starts with the exact k dictionary lambdas, that is, wrappers auto-generated for class methods, in which case we only add `TyLams`.

At each `Var` occurrence:

$$\mathsf{Var}\ x : \sigma \text{ where } \sigma = \forall \bar{\alpha}\ \bar{P},\ tbody$$

is elaborated to

$$\begin{aligned}
&\mathsf{TyApp}\ \cdots\ (\mathsf{TyApp}\ (\mathsf{Var} xty_{\mathsf{mono}})\ \alpha_1)\ \cdots\ \alpha_n \\
&\mathsf{App}(\cdots) d_1 \\
&\vdots \\
&\mathsf{App}(\cdots) d_k
\end{aligned}$$

A dictionary argument[42] is either

- in scope, that is, matching one of the predicate templates introduced by lambdas, or,
- synthesized by instance resolution, producing an untyped expression. Inference and elaboration is then re-run on that expression to transform it to an `FExpr`. This procedure is recursive.

In case if a method field type is polymorphic (e.g. `Functor`), as in `TSch (Forall as [] τ)`, it is reified as a System F type abstractions by wrapping the projected field with `TyLam`, for each binder in `as`. This is thought to be canonical for Higher-ranked types. Do note, however, per-method predicate contexts inside **TSch** are currently unsupported and will be rejected.

[41] An extension of GHC that lifts the Haskell 98 restriction that the type-class constraints (anywhere they appear) must have the form `class type-variable` or `class (type-variable type1 type2 .. typen)`.

[42] named `d_P.cls_i` where P is the predicate, i is the instance counter for that class; a case it is memoized, the identifier is prepend with a `r`.

Dictionary memoization refers to the memoization of synthesized dictionaries. It is common for a single instance to be used multiple times, in which case it becomes rather inefficient, both compile-time and runtime as each encounter results in a newly synthesized dictionary, or a dictionary application of multiple dependent dictionaries (as is the case for instances with contexts). Memoization tracks the first usage of a dictionary at function-level, upon encountering multiple usage, the synthesized instance is reused, and the function body is prepended by a let-exp that binds this instance that *hoist* the inlined dictionary application. This is similar to common subexpression elimination, except in this case it is targeted and not general at all, which comes with the benefit of being more convenient to implement. Consider the hierarchy of `Functor`, `Applicative` and `Monad`, in a rather elaborated example:

```
class Applicative (f : 1) =
  { pure : ∀ a, a -> f a
    -- lazy to preserve eval semantics
  , seq : ∀a b, f (a -> b) -> (Unit -> f a) -> f b}
class Monad (m : 1) =
  { bind : ∀a b, m a -> (a -> m b) -> m b
  , return : ∀a, a -> m a}

infixl 55 ">>=" ⇒ bind
instance ∀(m : 1) [Monad m], Applicative m =
  { pure = return
  , seq f x = f >>= fun f ⇒ x () >>= fun x ⇒ return $ f x
  }
instance ∀(f : 1) [Applicative f], Functor f =
  {map f x = seq (pure f) fun _ ⇒ x}

instance Monad Option =
  { return = Some
  , bind
    | None, _ ⇒ None
    | Some x, f ⇒ f x}

let seqOp = seq (Some id)
```

With `i_Monad_0, i_Applicative_0` corresponding to the `Option` instance of `Monad` and the general `Applicative` instance, the System F IR result is

```
let seqOp : ∀ α, (Unit → Option α) → Option α =
  let rd_Applicative_1 : Applicative Option =
    let rd_Monad_0 : Monad Option = i_Monad_0
    in i_Applicative_0@Option rd_Monad_0
  in (Λ α. rd_Applicative_1[1, seq]@Option (Some@(α → α) id@α))
```

As shown above, memoization have come into effect as the application of dictionary is cached and is bound by `rd_Applicative_1`, which, has a counter of 1, this indicates that dictionary application creates (synthesizes) new dictionaries. Therefore it is much necessary to have memoization as it mainly boosts compile performance.

In practice, any elaboration runs inside the `F` monad, which, together with recursion state[43] , is defined to be

[43] It is worth noting that elaboration does not only depend on the state shown below, additional trivial environments are not shown here, please refer to `fsyntax.lean` and `fexpr.lean`.

```
structure FState where
  log    : Logger := ""
  memo   : Std.HashMap Pred (String × Scheme × FExpr) := ∅
  gensym : Nat := 0
deriving Inhabited

abbrev F := EStateM TypingError FState
```

A canonical state monad equipped with error reporting is used here, contrary to before: This is because of the many usage of local states in recursive call during elaboration. Stack-allocated (function parameters) states are much easier to work with in this case, than a *real* threadlocal mutable state.

Remark (SysF IR pretty-printed syntax). Here we have shown a more elaborated example of pretty-printed System F IR source. It should be obvious for anyone that is familiar with OCaml or Haskell, especially the type application syntax `<term>@<type-expr>`. But for reference, we present the relevant pretty-printer implementation below:

```
local instance : Std.ToFormat Pattern where
  format := .text ∘ Pattern.toStr
open Std Std.Format in
partial def FExpr.unexpand : FExpr → Format
  | .CI i _ | .CB i _ | .CS i _ ⇒ repr i
  | .CUnit _ ⇒ format ()
  | .App f a _ ⇒ parenL? f ++ group (indentD (parenR? a))
  | .Proj src mname i _ ⇒ parenL? src ++ sbracket (format i ++ ", " ++ format mname)
  | .Cond c t e _ ⇒
    "if" ◇ unexpand c ⟷ "then" ++ indentD (unexpand t)
    ⟷ "else" ++ indentD (unexpand e)
  | .Fix e _ ⇒ "rec" ◇ unexpand e
  | .Var x _ ⇒ format x
  | .Prod' p q _ ⇒ paren $ unexpand p ◇ "," ◇ unexpand q
  | .Fun param pTy body _ ⇒
    group $ "fun" ◇ param ◇ ":" ◇ pTy.toStr ◇ "⇒" ++ group (indentD (unexpand body))
  | .Match discr branches .. ⇒
    let discr := discr.map unexpand
    let br := branches.map fun (pats, e) ⇒
      "|" ◇ joinSep' pats ", " ◇ "⇒" ++ group (indentD (unexpand e))
    group $ "match" ◇ joinSep' discr ", " ◇ "with" ⟷ joinSep' br line
  | .Let binds body _ ⇒
    let (binds, recflag) := binds.foldl (init := (#[], false)) fun (acc, recflag) (id, sch, fe) ⇒
      match fe with
      | .Fix (.Fun _ _ body _) _ ⇒
        (acc.push $ id ◇ ":" ◇ toString sch ◇ group ("=" ++ indentD (unexpand body)), true)
      | _ ⇒
        (acc.push $ id ◇ ":" ◇ toString sch ◇ group ("=" ++ indentD (unexpand fe)), recflag)
    let recStr := if recflag then .text " rec " else .text " "
    "let" ++ recStr ++ joinSep' binds (line ++ "and ") ⟷ "in" ◇ unexpand body
  | .TyLam a body ⇒
    let (tvs, body) := Helper.peelTyLam [a.elimStr] body
    group $ paren $ "Λ" ◇ joinSep tvs " " ++ "." ++ indentD (unexpand body)
  | .TyApp f ty ⇒ parenL? f ++ "@" ++ parenT ty
where
```

```
parenR?
  | p@(.App ..) | p@(.Fun ..) | p@(.Cond ..) | p@(.Let ..) | p@(.Match ..) ⇒ paren (unexpand p)
  | p ⇒ unexpand p
parenL?
  | p@(.Fun ..) | p@(.Cond ..) | p@(.Let ..) | p@(.Match ..) ⇒ paren (unexpand p)
  | p ⇒ unexpand p
parenT
  | t@(TVar _) | t@(TCon _) | t@(TApp _ []) | t@(KApp _ []) ⇒ text t.toStr
  | t ⇒ paren t.toStr
```

Remark (On the usefulness of System F IR). Since the beginning of this project/thesis, the author himself has a very clear goal of compiling *Tigris* to Common Lisp (SBCL). Since LISP is a dynamically typed language with some form of type inference and propagation (SBCL-only, probably not gradual typing), There is not much need to do specialization of generics, or to preserve typing information for next-stage lowering.[44]

Because of this, we *did not* actually need a proper system F IR, despite we did, as typeclass elaboration is perfectly doable without it. This is where the author has displayed both his long-term thinking ability and his shortsightedness:

- The former, as in for later proper implementation of arbitrary rank polymorphism and/or inductive families, a System F IR is useful to have;
- The latter, as in after Lambda IR, typing information is gone since it does not preserve types, leaving codegen harder, and most importantly, *much less robust*. Where we have to record the *shapes* of symbols[45] and to do some form of pattern matching on IR to temporarily fix a bug. Moreover, it is much less feasible if we were to develop some other backends that need types (e.g. C).

  Although this does not have much to do with the usefulness of SysF IR, the insight is the same:

  > It is crucial to have a properly typed IR throughout compilation of programs.

4 The *Lambda* IR, CPS, Codegen & FFI

Elaboration to System F IR marks the end of "going up", whereas *Lambda* IR[46] serves as an actual intermediate language that marks the beginning of code lowering. *Lambda* IR should be classified as in A-normal form, or ANF for short, a type of IR frequently used in functional programming language compilers.

The essence of ANF is *names*. Object, or any (nontrivial) expression, is referred by name, by immediately capture them in a let-bound variable. This, to some form, mimics the assembly code of the same program. Although *Tigris* compiles to a high-level language, its form is not what normal programming styles looks like and thus SBCL may benefit from it, as the *Tigris* compiler is doing some of the jobs for it.

[44] though it has been painfully proven to be convenient, through coding practice.

[45] which itself, is not perfectly reliable.

[46] The name *Lambda* is a direct reference to the IR Appel used in his *Compiling with Continuations* and presumably used in the SML/NJ compiler. *Tigris* itself is a nod to Appel's *Modern Compiler Implementation in ML*, of which the cover features a tiger. *Tigris* is latin.

4.1 The *Lambda* IR

Definition of *Lambda* IR Design of *Lambda* IR focuses on five aspects: `Value`, `Rhs`, `Stmt`, `Tail` and `LExpr`. We shall present the definition of these parts individually, together with a brief explanation of their intents. Readers shall note that these definitions are mutually recursive.

`Value` Also called immediate values. They cannot be further reduced.

```
inductive Value where
  | var    (x : Name)
  | cst    (k : Const)
  | constr (tag : Name) (fields : Array Name)
  | lam    (param : Name) (body : LExpr)   -- single-argument lambdas; multi-arg via tuples
deriving Repr, Inhabited
```

Note that *closures* (lambdas) at this stage are not yet converted and capture free variables.

`Rhs` denotes the right-hand sides of ANF where all arguments are variables

```
inductive Rhs where -- x : Rhs
  | prim     (op : PrimOp) (args : Array Name)          -- x := op(args_1, ..., args_n)
  | proj     (src : Name) (idx : Nat)                   -- x := src_idx
  | mkPair   (a b : Name)                               -- x := ⟨a, b⟩
  | mkConstr (tag : Name) (fields : Array Name)         -- x := tag⟦fields_1, ..., fields_n⟧
  | isConstr (src : Name) (tag : Name) (arity : Nat)    -- x := src matches ⟪tag/arity⟫
  | call     (f : Name) (arg : Name)                    -- x := f(arg)
deriving Repr, Inhabited
```

`Stmt` right now only has single let binding, which, in practice, is no difference from a letexp.

```
inductive Stmt where
  | let1 (x : Name) (rhs : Rhs)
deriving Repr, Inhabited
```

`Tail` denotes the tail positions, which includes return, application, conditional and pattern matching branches. `Tail` is important in marking tail positions, which is utilized in later tailcall transformations.

```
inductive Tail where -- tl : Tail
  | ret  (x : Name)                                  -- tl := RET x
  | app  (f : Name) (arg : Name)                     -- tl := f(arg)^T
  | cond (condVar : Name) (tBranch eBranch : LExpr)  -- tl := if cond then t else e
  | switchConst (scrut : Name)                       -- tl := case^C of cases_i
      (cases : Array (Const × LExpr))
      (default? : Option LExpr)
  | switchCtor (scrut : Name)                        -- tl := case of cases_i
      (cases : Array (Name × Nat × LExpr))
      (default? : Option LExpr)
  | matchFail (pats : Array Name)                    -- tl := MATCHFAIL(pats)
deriving Repr, Inhabited
```

Note that constant switches and constructor switches are treated differently – This is to facilitate later constant folding more convenient to implement.

MATCHFAIL is made a special token to denote the *fallback* branch of a partial match. As said before, entering such branch results in a nonrecoverable runtime error. The reason that it takes `pats`, an array of `Names` i.e. strings, is to provide the users with the symbols of the match discriminants, which, is directly passed onto codegen, printing from the LISP debugger.

`LExpr` is the main body of an program in *Lambda* IR.

```
inductive LExpr where
  /-- always ends in tails -/
  | seq    (binds : Array Stmt) (tail : Tail)
  | letVal (x : Name) (v : Value) (body : LExpr) -- letι x = v in body
  /-- Nested let Rhs bindings. -/
  | letRhs (x : Name) (rhs : Rhs) (body : LExpr) -- let x = rhs in body
  /-- Recursive function bindings. -/           -- letω
  | letRec (funs : Array LFun) (body : LExpr)    --   label fun₁ : ⋯
                                                 --   label fun₂ : ⋯
                                                 --   ⋮
                                                 -- in body
deriving Repr, Inhabited, BEq

/-- A top-level function in the Lambda IR. -/
structure LFun where
  fid    : Name
  param  : Name
  body   : LExpr
deriving Repr, Inhabited
```

Here, normally what would be one letexp is divided into 3 sub-lets based on their RHS: a **let**ι that binds immediate value, a **let** that binds regular `Rhs`, and finally a **let**ω that binds a group of mutually recursive functions.

In addition to these mutually recursive definitions, we additionally have `LFun` that represents a top-level function. Readers with a keen heart will notice that `letRec` carries an array of `LFuns` – `LFun` at this position doesn't mean that `funs` have to be toplevel. We use `LFun` instead of `Name × Name × LExpr` is purely out of performance reasons: they have compatible representation, which, could save some potentially unnecessary conversions.

Sequence operations – `seq` – although *Tigris* is a pure language, we use `seq` to capture tails as it is the only constructor in `LExpr` that takes `Tails`. Also, someday may turn *Tigris* into a impure functional language like OCaml which, can be convenient if `seq` is present in `LExpr`.

Lambda IR is pretty-printed. In order for readers to get acquainted with the syntax, we'll omit the implementation of pretty printer, and instead, provide a concrete example of a simple evaluator:

```
type Expr a =
  | Atom a
  | Add (Expr a) (Expr a)
  | Sub (Expr a) (Expr a)
  | Mul (Expr a) (Expr a)
  | Div (Expr a) (Expr a)

let rec eval
```

```
  | Atom a ⇒ a
  | Add e1 e2 ⇒ eval e1 + eval e2
  | Sub e1 e2 ⇒ eval e1 - eval e2
  | Mul e1 e2 ⇒ eval e1 * eval e2
  | Div e1 e2 ⇒ eval e1 / eval e2

let prog =
  Mul (Atom 20) $
    Sub (Mul (Atom 10)
             (Atom 20)) $
        Div (Atom 2400) $
          Add (Atom 120) $
            Add (Mul (Atom 10)
                     (Atom 20)) $
                (Atom 0)
;;
let 3860 = eval prog
```

yields, in *Lambda* IR

```
  main (arg) {
  letω
    label eval(args0):
      let _pL#?x₀ = args0[0]
      case _pL#?x₀ of
        «Add/2» →
          let p1 = _pL#?x₀[0]
          let p2 = _pL#?x₀[1]
          letι u3 = ()
          let pair4 = ⟨p1, u3⟩
          let call5 = eval(pair4)
          letι u6 = ()
          let pair7 = ⟨p2, u6⟩
          let call8 = eval(pair7)
          let p9 = ADD(call5, call8)
          RET p9
        (...)
        «Atom/1» →
          let p28 = _pL#?x₀[0]
          RET p28
        (...)
  in
  letι c38 = 20
  let con39 = Atom⟦c38⟧
  (...)
  letι c42 = 20
  let con43 = Atom⟦c42⟧
  let con44 = Mul⟦con41, con43⟧
  (...)
  letι c51 = 20
  let con52 = Atom⟦c51⟧
```

Primitive operation	Description
`add`	Integer addition
`sub`	Integer subtraction
`mul`	Integer multiplication
`div`	Integer division
`eqInt`	Integer equality
`eqStr`	String equality
`eqBool`	Bool equality

Table 6: Primitive operations of *Lambda* IR

```
  let con53 = Mul⟦con50, con52⟧
  (...)
  letι u61 = ()
  let pair62 = ⟨con60, u61⟩
  let call63 = eval(pair62)
  letι c64 = 3860
  let cmp65 = EQi(call63, c64)
  if cmp65 then RET con60 else MATCHFAILURE
}
```

we shall mention some peculiar details during lowering, which itself, is implemented in CPS style: As stated above, having a system F is somewhat overkill since the compilation target being a dynamically typed language. To reduce noises, a system F IR is stripped of TyLam, TyApp.

Shortpath: Constructors & Primitive operations First, we list the seven primitive operations and nullary constructors that is not impacted by the additional overhead introduced by the function calling convention: Table 6

Detection of this operations is through pattern matching of the application of these primitives, hence it is sometimes necessary to η-expand them in, e.g., method registration which is seen before. In practice, primitives and normal functions are only differentiated by their names. That is, applications for primitives can also be done through funapp branch – However, it is rather inefficient to do so, which is why the `lowerF` function handles theses primitives (and nullary constructors) first:

```
partial def lowerF
  (e : FExpr) (ρ : Env)
  (ctors : Std.HashMap String Nat)
  (k : Name -> M σ LExpr) : M σ LExpr := do
  match matchBinaryPrim e with
  | some (op, x, y) ⇒
    lowerF x ρ ctors fun vx ⇒
      lowerF y ρ ctors fun vy ⇒ do
        let r <- fresh "p"
        let cont <- k r
        return .letRhs r (.prim op #[vx, vy]) cont
  | none ⇒
    match matchCtorApp ctors e with
    | some (cname, args, ar) ⇒
      if ar == 0 && args.isEmpty then
        let r <- fresh "con"
```

```
      let cont <- k r
      return (.letRhs r (.mkConstr cname #[]) cont)
    else lowerCtorApp cname args ar ρ ctors k
  | none ⇒ (...) -- normal lowering branches.
```

Dictionary projections are nodes that are specialized for typeclass methods. Normally, methods are projected through the use of pattern matching, according to canonical dictionary passing implementations. This compilation overhead is reduced by the use of `Proj`s, which, is threaded through *Lambda* IR lowering.

Function, function applicaton & uncurrying Proper functional languages enforces currying. However, it is inefficient to generate n functions for a n-arity function, which, is also harder for target languages to do tailrec optimizations, which, is why most functional language compilers do *uncurrying* – the action of *squashing* curried parameters into a single canonical products[47] that can be encoded by the `lam` constructor. The lowering of functions have several steps below:

1. **`decomposeLamChain`** is used to squash *consecutive* function[48] heads i.e. `fun ..` [49] which returns a 3-tuple $(p0, rest, core)$ where $p0$ refers to the first parameter, *rest* refers to the rest and *core* refers to the function body.

2. **η-expansion** not only refers to the practice in lambda calculus, in our case, it refers the corner case of applying a projected[50] function not properly covered by the `App` branch: Consider this example:

   ```
   type BoxedAdd = Boxed (Int -> Int -> Int)
   let boxedAdd = Boxed (_ + _)

   let unwrapBoxed (Boxed f) = f
   let (Boxed unboxedAdd') = boxedAdd
   let main =
     ( unwrapBoxed boxedAdd 20 20
     , unboxedAdd' 30 30)
   ```

   where `unwrapBoxed boxedAdd 20 20` is called a *higher-order application*, where, the application of `boxedAdd` at `main`. This, however, cannot be caught by `lowerF` as it only recognizes the application head `unwrapBoxed` and it sees no application inside `unwrapBoxed`, thus, this application would not be translated correctly by a naive interpreter. In this case, `unwrapBoxed` is implicitly η expanded (in spirit) as `let unwrapBoxed (Boxed f) x y = f x y`. As shown in the LISP object code below:

   ```
   (defun |fn1001| (|payload| |k|) ; unwrapBoxed
     (declare (optimize (speed 3) (safety 0) (debug 0)) (ignorable |payload|))
     (let ((|α| (car |payload|)))
       (let ((|_pL#?x₀| (car |α|)))
         (let ((|_pR#?x₀| (cdr |α|)))
           (let ((|_pL#η0| (car |_pR#?x₀|)))
             (let ((|_pR#η0| (cdr |_pR#?x₀|)))
   ```

[47] in our implementation, specifically.
[48] i.e. `fun x ⇒ fun y ⇒ ...`
[49] squashing is unconditional – even though several `fun` heads might be bounded by different names, they are squashed as well.
[50] from pattern matching

```
(cond
  ((eq (car |_pL#?x₀|) '|Boxed|)
    (let ((|p5| (svref (cdr |_pL#?x₀|) 0)))
      (let ((|pair6| (cons |_pL#η0| |_pR#η0|)))
        (let ((|_code1002| (svref (cdr |p5|) 0)))
          (let ((|Γc1003| (svref (cdr |p5|) 1)))
            (let ((|pc1004| (cons |pair6| |Γc1003|)))
              (funcall (the function |_code1002|) |pc1004| |k|))))))))))))
```

where ..ηN like `_pR#η0` denotes a parameter produced by η-expansion.
One may wonder, how can we keep semantic consistency of CBV if we do η-expansion – Since some programs (e.g. Y combinator) would otherwise diverge. The answer is we (sort of) *can't*, which is why, η-expansion is only applied to the condition described above.
Besides, Lean does it as well and is also based on similar condition (type signature), See this proofexchange post [51] by me and this github issue@leanprover/lean4#5908 [52].

3. **Syntactic arity** refers to the user-defined function arity, which can introduce ambiguity when mixed with higher-order function/application as seen above. Normally, the lowering facility determines a function's arity based on its type signature, this, is proved in practice to be not adequate. Consider `unwrapBoxed` : `BoxedAdd` → `Int` → `Int` → `Int` from the example above[53]: `unwrapBoxed` has an arity of 3, however, it should only have 1, otherwise if it were passed the additional two parameters, it would have no way to deal with it (as the body does not reference them.). In this case, only after η-expansion can it be allowed to have an arity of 3. Syntactic arity is much used in uncurried application.

4. **`destructArgsPrelude`** is used to destruct uncurried parameters – a single products. It produces several bindings that prepend the function body that represent the original parameters, as shown above with a consecutive `car`/`cdr`s. We shall discuss this topic later in closure conversion.

5. **`lowerFCore`** A call to which can be though as living in the bottom of a continuation call stack. `lowerFCore` is *homoiconic* in the sense that it represents a tail in the lowering process, as well as lowers an expression that is the tail in the object language that is *Lambda* IR. It is defined as

```
@[inline] partial def lowerFCore
  (e : FExpr) (ρ : Env) (ctors : Std.HashMap String Nat) : M σ LExpr :=
  lowerF e ρ ctors $ pure ∘ .seq #[] ∘ .ret
```

   Which, does not take a continuation to transfer control.

Implementing uncurrying is a joint effort on both function definition and application. We shall define the parameter layout: for a function of arity n, parameter $a_1, \ldots, a_n$, we construct a tuple pl (payload) where

$$pl = \begin{cases} ((), ()) & n = 1, a_1 = () \\ (a_1, ()) & n = 1 \\ (a_1, \ldots, a_n) & n \geq 2 \end{cases}$$

The author must admit that this is not exactly the best design in hindsight as it is `List`-like, but isn't exactly a list, especially for $n \geq 2$. Without typing information, it is harder to identify the shapes, a difficulty encountered later in codegen. Also, we should have encoded arity information

[51] https://proofassistants.stackexchange.com/a/4424
[52] https://github.com/leanprover/lean4/issues/5908
[53] Although tedious, we shall remind the readers of arrow types nest to the right.

inside function heads before *Lambda* IR, instead of collecting on-demand with less information and robustness, cluttering the code.

Since functions are now uncurried, we shall make a distinction between full applications and unsaturated (partial) ones: full applications are divided into several cases whereas partial application generates wrapper lambdas on demand. Interested readers shall refer to `lowerFunApp` in `ftransform.lean` that deal with various different scenarios of function application on a case-by-case basis: unary, partial, broken type/syntactic arity mismatch (called "broken" in code, the addition of dictionary parameters).

Constructor applications are done in a similar (as in partial application) way, but it is, at its core, not a function and does not have a body, which is why it is more simple to deal with.

Pattern matching is compiled using a *decision tree* approach[54] from Maranget. Based on this approach, we define three types that encapsulate the state of decision tree compilation and the tree itself: a `Sel` (similar to a `Proj` that encodes projection information), a `RowState` which encodes a row vector a in pattern matrix, and `DTree`, the tree itself.

```
inductive Sel where
  | base (idx : Nat)
  | field (s : Sel) (idx : Nat)
deriving Repr, Inhabited

structure RowState where
  pats : Array Pattern
  rhs  : FExpr
  binds: Array (String × Sel) := #[]
deriving Repr, Inhabited

inductive DTree where
  | fail
  | leaf (row : RowState)
  | testConst (j : Nat) (branches : Array (TConst × DTree)) (default? : Option DTree)
  | testCtor  (j : Nat) (branches : Array (String × Nat × DTree)) (default? : Option DTree)
  | splitProd (j : Nat) (next : DTree)
deriving Repr, Inhabited
```

In PL jargon, the target of a pattern matching compilation is always called *matching automata*, which is either *decision trees* or *backtracking automata*. The latter being the simplest one, similar to one's thinking process: from the top-leftmost of a matrix, test every pattern, then move on to the next row, until it reaches bottom-rightmost, which, has the advantage of maintaining linear space complexity in terms of *code size*. But, as stated in its name, it may backtrack, being potentially inefficient at runtime.

In contrast, we implement a decision tree compiler that makes sure no duplicate matching of the same value against the respective pattern. An example is taken from Maranget's work to illustrate this issue: Consider

```
let prog
  | _, false, true ⇒ 1
  | false, true, _ ⇒ 2
  | _, _, false    ⇒ 3
  | _, _, true     ⇒ 4
```

[54] Exhaustive/Redundancy checking can be seen as a much lightweight algorithm derived from this approach

which is compiled to (only the relative parts are shown here; adjusted for better readability; ∅ denotes a default branch)

```
case x of
  false → case y of
            true  → case z of true → RET 2 | false → RET 2 | ∅ → RET 2
            false → case z of true → RET 1 | false → RET 3
            ∅     → case z of true → RET 4 | false → RET 3
  ∅     → case y of
            false → case z of true → RET 1 | false → RET 3
            ∅     → case z of true → RET 4 | false → RET 3
```

We can see that, at worst, only 3 comparisons need to be made, much efficient compared to a backtracking automata.[55] It is worth noting that a lighter version of the algorithm described in the paper above is used in practice because some information has already been obtained during checking: the `Match` node of `TExpr` has a slot specifically to store the exhaustiveness of the matrix, sparing recalculations here. The wrapper type `Subarray` is used to minimize intermediate allocation of arrays.

4.2 Closure conversion of the *Lambda* IR

Closure conversion refers to hoisting locally bound functions to the toplevel, transforming it into a globally bound function. The main challenge is to eliminate free variables captured by local lambda abstractions.

While some text suggest that closure conversion is different from lambda lifting as the former does not add free variables to parameter list and does not adjust call sites, we use this word to describe the implementation *Tigris* used that takes the middle ground between lambda lifting and closure conversion, hence, these two words can be used interchangeably though we largely stick to the latter.

Some might suggest, since we are targeting Common Lisp, which is already a functional language itself with lambda abstractions, why the hassle – The reason is mostly a virtuous one – We reinvent the wheels to demonstrate the efforts put into the project, as a training in getting acquainted with practical implementations of functional languages as well as serving as an example for educational purposes.

The middle ground refers to the approach of lifting every `Lam` as well as adding an environment which stores all captured variables, this environment itself, is then passed alongside the function pointer, which, as a whole, is called a function object. When applying function objects, this environment (referenced as Γ) is assembled with uncurried parameters (referenced as *payload*) which is then passed to the function pointer.[56] This is called a middle ground because, although all free variables are eliminated, a function object is also created, though, the function itself is lifted to global. It is only when referencing[57] such functions, function objects are used instead of a bare function pointer. Besides, `Lam`s are eliminated during this process, therefore, in spirit and in the author's opinion, this is an implementation of lambda lifting.

Function calling convention , or encoding of function objects, is described informally as follows

1. *payload* refers to the single parameter taken by lifted functions, $payload = \langle arg, \Gamma \rangle$ where arg refers to the product constructed from parameters

[55] although this matrix can be further simplified e.g. simply return 2 when $(x, y, z) = (\mathsf{F}, \mathsf{T}, \cdot)$, but such optimization has not be implemented.

[56] e.g. This pointer, in the SBCL backend, is simply a symbol that reference functions.

[57] i.e. calling/passing.

2. Γ, or closure environment refers to explicit values holding captured free variables. It is encoded as a constructed value $\mathbf{E}[\![fv_1, \ldots, fv_n]\!]$. In practice, to easily maintain its uniqueness, a non-parsable `U+1D404 MATHEMATICAL BOLD CAPITAL E` is used instead.
   Γs are treated as if they are variants, encoded with `mkConstr`.

3. *Closure*, or *function object*, is a 2-field constructed value $\mathbf{C}[\![ptr, \Gamma]\!]$ where *ptr* refers to a code pointer (function pointer), which is a string internally. In practice, to easily maintain its uniqueness, a non-parsable `U+1D402 MATHEMATICAL BOLD CAPITAL C` is used instead.
   Closures are treated as if they are variants, encoded with `mkConstr`.

4. *Application* of a closure project ptr, Γ and pass (arg, Γ) to *ptr*.

5. *Mutual recursion*, or recursion in general from `let rec` groups are converted so that each function body projects (arg, Γ) from payload. Calls from within the body reference the code pointer directly with the same env.[58] In case of mutual recursion, all functions in a `let rec` group shall share the same Γ layout.

Note that, since *Tigris* does not distinguish bindings of functional value from regular value, canonically speaking, every function definition in the form of a let-binding is no difference from binding names to lambda abstractions, which means, all functions i.e. all `Lam`s are converted. In this sense, what would be called a "global" function collects free variables from the toplevel i.e. global definitions (if it is referenced). Consider

```
let x = 1;; let prog y = x + y + 1;; let main = prog 2
```

Though `x` and `prog` are both toplevel bindings, since the latter references the former, it is captured by Γ and projected inside `prog`'s body nonetheless, as shown in CC'd IR:

```
fn1000 (payload) {
  let α = payload[0]
  let Γ = payload[1]
  let c0 = Γ[0]
  let _pL#y = α[0]
  let p2 = ADD(c0, _pL#y)
  letι c3 = 1
  let p4 = ADD(p2, c3)
  RET p4
}

main (payload) {
  letι c0 = 1
  let Γ = E⟦c0⟧
  let lam5 = C⟦fn1000, Γ⟧
  letι c6 = 2
  letι u7 = ()
  let pair8 = ⟨c6, u7⟩
  _code1001 := lam5[0];
  Γc1002 := lam5[1];
  pc1003 := ⟨pair8, Γc1002⟩;
  _code1001(pc1003)ᵀ
}
```

[58] This ensures efficient recursive calls as if they are native functions in the object language.

Remark (Performance concerns). Some might suggest that, for "globally defined functions" (in the traditional sense) that do not contain free variables (i.e. $\Gamma = \varnothing$), it is rather inefficient for them to be packaged inside a closure, then unwrapping them again for no reasons, instead, why not reference them directly[59] – The author wholeheartedly agree as *Tigris did*[60] have this optimization but it was considered experimental and issues did arise after enabling it, which is why it is temporarily removed, not that it matters a lot since most function calls are recursive calls and calls from within or from functions within a mutual group are already handled in this way.

Remark (Peephole: fuse immediate tailcall). A simple pattern matching on IR is used to detect: if an expression has the pattern $f^{\mathsf{T}}(a)$ where f is the closure just bound for codeptr fid and env Γ, we rewrite this tail to $fid^{\mathsf{T}}(\langle a, \Gamma \rangle)$; Old closure is handled and removed by DCE pass in the optimization module (which shall be discussed below).

```
def fuseImmediateTailCall (envName x fid : Name) : LExpr -> LExpr
  | .seq binds (.app f a) ⇒
    if f == x then
      let pl := "ρ"
      .seq (binds.push (.let1 pl (.mkPair a envName))) (.app fid pl)
    else .seq binds (.app f a)
  | .letVal y v b ⇒ .letVal y v (fuseImmediateTailCall envName x fid b)
  | .letRhs y r b ⇒ .letRhs y r (fuseImmediateTailCall envName x fid b)
  | .letRec fs b ⇒ .letRec fs (fuseImmediateTailCall envName x fid b)
  | e ⇒ e
```

4.3 Optimizations of the *Lambda* IR

Optimizations are mostly done surrounding closure conversion: before CC and re-runs after to handle noises produced during CC. Currently, there are 3 passes: constant folding/propagation, copy propagation/dead code elimination (DCE) and tailcall-ify.

Constant folding & propagation refers to propagation of values that can be computed at compile time. Currently, the compiler handles $(+, -, *, \div)$ on integers and equality is handled for all primitive types: integer, bool, string. Although typically these operations come with laws such as associativity, commutativity, and distributivity, it is *not* respected by the optimizers i.e. $10+10+x+10$ will be simplified to $20 + x + 10$, but not $30 + x$.

It would not be as interesting if the optimizer can't interpret very basic control flow clauses such as conditional and pattern matching. e.g. `let x = true in if x then 1 else 2` is simplified to 1 and a more complex one:

```
type X = {x : Int, y : Int}
let s = {x = 1, y = 2}
let main =
  let {x, y} = s in x + y
```

Which gets simplified to 3. Constant folding is mostly useful when paired with typeclasses: consider

```
class Eq a = {eq : a -> a -> Bool}
type Option a = None | Some a
infixl 50 "=" ⇒ eq
```

[59] by bare pointers i.e. names
[60] See older commits in the repository.

```
let eqOpt = fun
  | Some x, Some y ⇒ x = y
  | None, None     ⇒ true
  | _, _           ⇒ false
instance forall a [Eq a], Eq (Option a) = {eq = eqOpt}

let main x y = Some x = Some y
```

if a method is registered by an accessible[61] variable, constant folding will be able to eliminate the dictionary projection of said method, replacing it with that variable, as both the instance and the method implementation `eqOpt` is visible to the optimizer. In this case, (=), or `eq`, is replaced by `eqOpt` in `main`, making dictionary passing truly zero-abstraction cost.[62]

Copy propagation & dead code elimination is used in conjunction with other passes such as constant folding and closure conversion, as noted above. Copy propagation uses the transitivity of (=) in let-bindings, e.g. `let x = 1 in let y1 = x .. yn = y_n_1 and z = 2 in y_n + z` simplifies to 3. The optimizer knows that y_n alias y_{n-1} and *chases* this alias based on the law above, together with constant folding, yields 3.

Meanwhile, dead code elimination is as simple as it sounds: it eliminates unused bindings/expressions. Such as any unused global binding in `main` is removed, provided that there is not a transitive dependency. Although, it is worth noting that since *Tigris* is purely functional, DCE was used to be radical, removing any unused let-binding. This behavior is now changed if the LHS is a wildcard pattern as though *Tigris* is without `Seq`,[63] let expression can be abused as in `let _ = act in ...` to run `act` with actual side effects such as IO, introduced by (the abuse of) the FFI system (described later).

Tailcall optimization is crucial for functional programming languages and is also guaranteed by many. Although readers are usually familiar with the topic if they are experience using such languages, we shall spare no trouble in explaining its importance.

Functional languages are differentiated from others by their extensive usage of functions (subroutines) in programs, which, when enter, pushes a new frame that serves as the context[64] of said call onto the call stack. The stack is unwinded as the control *returns* from the subroutine. When the call stack runs out of space, a *stack overflow* exception is thrown and the program is terminated. However, it turns out that calling a function in the *middle* of some code is different from calling it at the *end* since the latter can be seen as a transfer of control[65] that can be implemented by the machine instruction `j` instead of `call` (assuming RISC-V ISA), which, allows the stack to unwind once control is transferred, freeing up stack spaces.

Obviously, stack spaces are usually consumed by recursive calls. A recursive function is said to be *tail-recursive* if it ends with the call to itself.[66] *Tailcall optimization* refers to making sure functions like this (including non-recursive) are guaranteed to have stack space complexity $\mathcal{O}(1)$ rather than $\mathcal{O}(n)$. See examples: Figure 3.

[61] that is, it must be accessible at callsite, not shadowed by local binds.
[62] though it is not in this case. Since the `Option` instance for `Eq` is a function.
[63] in source language, that is.
[64] including callsites (return points) and stack-allocated values.
[65] which, of course includes control structures such as pattern matching, conditional and non-local exit i.e. a transfer of control (and sometimes values) to an exit point for reasons other than a normal return, such as those produced by `goto`, `throw` and early returns.
[66] in this case, frames are pushed and popped *consecutively*, between recursive calls.

```
let rec countTree
  | Empty ⇒ 0
  | Node _ f ⇒ 1 + countForest f
and countForest
  | Nil ⇒ 0
  | Cons t f ⇒ countTree t + countForest f
```

(a) non-tail recursive

```
let rec even
  | 0 ⇒ true
  | n ⇒ odd $ n - 1
and odd
  | 0 ⇒ false
  | n ⇒ even $ n - 1
```

(b) tail-recursive

Figure 3: Non-tailcall vs. tailcall

Tailcall can be transformed, either by hand, or by mechanical CPS conversions (discussed later). Intuitively, tail recursive functions have similar though process and semantics to *iterations*.[67]

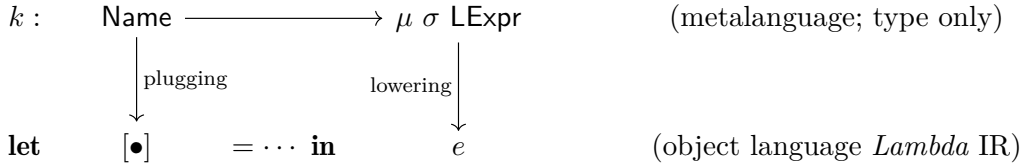
Some might ask, from the description above, it seems that tailcall optimization is a rather low-level optimization. What does it mean when it is done at IR level, specifically, the *Lambda* IR?

The answer is that we *mark* tail call positions (encoded as `app`, from `Tail`) and pass down this information to the backend whom is responsible for making sure it stays in tail position in the object code that would otherwise be difficult to do so without the marks. A tailcall is represented, as previously mentioned, as $f(arg)^{\mathsf{T}}$ where f is a function name or code pointer depending on the stage (before CC or after).

Optimizations are summed up by this method definition:

```
def optimizeLam (e : LExpr) : LExpr :=
  let e1 := constantFold e
  let e2 := copyPropDCE e1
  let e3 := tailcallify e2
  let e4 := copyPropDCE e3
  e4
```

We sum up this section by talking about the *ubiquitous* usage of CPS style in the lowering implementation, technical details are omitted as we shall only discuss the intuition here. The continuation function k, by homoiconicity, represent a *pluggable* expression (of the rest program) in the object language:



where the hole (•) may or may not (but should, if not useless) be bound in e. Pluggable refers to the fact that applying such continuation to[68] some name (e.g. m) yields the rest of program with m bounded in e. Internally, it just runs through whatever monadic actions inside μ, in the explicit CPS stack, that, is maintained by a large closure whose body is the composition of continuation functions of the lowering actions, where, μ is defined by[69]

```
abbrev M σ := StateRefT (Nat × AMap) (ST σ)
```

with AMap being the syntactic arity map from Name to Nat. In practice, this trick is used to *propagate* variable names generated by the metalanguage to the object language.

[67] which, itself, is $\mathcal{O}(1)$ in stack space.
[68] "applying to" can also be substituted for "plugging up".
[69] another use of the ST monad, reason same as before.

Incremental APIs and the *Tigrisi* IR interpreter is a experimental extension. While the *Tigrisl* compiler compiles programs in one go, it is rather inefficient for REPLs as they have different use logic: users of a REPL session usually enter a program *piece by piece*, while expecting immediate result after entering them. A naive REPL might track everything they entered since the start of REPL then *re-compiles* it per user entrance, which, for multi-stages language implementations with several passes is horribly inefficient. Thus, the incremental APIs are born, which lives in the `Incremental` namespace.

When implementing the incremental APIs, we not only have to track every state [70], but also the use of it inside `{ftransform,lift,opt}.lean` (is it intertwined, etc):

```
structure LoweringState where
  gensym : Nat
  env    : Env
  ctors  : Std.HashMap String Nat
  arity  : Std.HashMap String Nat
deriving Inhabited
```

Interested users shall refer to them in the bottoms of `ftransform.lean` and `lift.lean`. This is considered highly experimental and unstable since it is not thoroughly tested.

the *Tigrisi* IR interpreter targets the *Lambda* IR and can run independently regardless of with incremental APIs or not. Though in the other case it runs much slower due to worse compilation performance. The interpreter itself is stable, but due to its partial dependency on the incremental APIs, we still consider it experimental.

Interpreters, just like virtual machines, have `states`, such as the OCaml VM that runs the bytecode produced by `ocamlc`. Same is applied to *Tigris*. Largely, these states are introduced by bindings, whether local or global, or functional. In this case, the state is a *mapping*, or *table* (e.g. a Hashmap or Treemap). An important design aspect to consider is to make the state as lightweight as possible, since allocating them, projecting/modifying them consumes time and space. In our implementation, we define immediate values and environments as:

```
inductive Val where
  | unit
  | int (i : Int)
  | bool (b : Bool)
  | str (s : String)
  | pair (p q : Val)
  | ctor (tag : Name) (fields : Array Val)
  | code (pointer : Name)
deriving Repr, BEq, Inhabited

abbrev Env     := Std.HashMap Name Val       -- local binding table
abbrev FunTab  := Std.HashMap Name LFun      -- Function table
abbrev GlobalEnv := Std.HashMap Name Val     -- global binding table
```

On encountering a `let*`[71], relevant bindings are inserted to one of the three tables defined above.

Tigrisi runs in `EIO`[72] and is *multi-threaded*: evaluation runs in a separate thread. This way it can respond to the keystroke `<C-c>` from user, on encountering which the REPL thread is terminated.

[70] and temporary states that is dropped after the compilation of a program, but should be saved in this case.

[71] `*` is wildcard, matching empty string, ι or ω, as in the pretty-printed syntax of *Lambda* IR. It has nothing to do with sequential application from LISP or OCaml.

[72] that is, `IO` with exception handling.

Termination of threads is *cooperative*: A joint effort of the interpreter, through `checkInterrupt` up front at crucial branch (`eval*` functions):

```
/-! leval.lean -/
private def checkInterrupt : EIO TypingError Unit :=
  IO.checkCanceled >>= fun
  | true  => throw .Interrupted
  | false => return ()

/-! ffidecl.lean -/
@[extern "lean_sigint_pipe"]
opaque installSigintPipe : IO Int32
@[extern "lean_read_fd_byte"]
opaque readFdByte : @&Int32 → IO Int32

/-! Tigrisi.lean -/
-- (.. within main ..)
let stdin ← getStdin
let replST ← mkRef LApp.initREPL
let EVS : IO.Ref EvalState ← mkRef none

let fd ← installSigintPipe

if fd ≥ 0 then
  discard $ asTask (prio := .dedicated) do
    repeat do
      let r ← readFdByte fd
      if r ≤ 0 then pure ()
      else
        if let some t ← EVS.get then
          IO.cancel t
        else pure ()
-- (..)
```

with the help of CLI frontend (above) and Lean's C FFI interface (we omit the win32 implementation):[73]

```
typedef lean_obj_res OBJRES;
/* (.. win32 impl ..) */
static int32_t sig_pipe[2] = {-1, -1};

static void sigint_handler() {
  // write is async-signal-safe
  char b = 1;
  if (sig_pipe[1] >= 0) {
    (void)!write(sig_pipe[1], &b, 1);
  }
}

LEAN_EXPORT OBJRES lean_sigint_pipe() {
```

[73] this approach is adopted from the deprecated REPL `tigris` (which is a subset of the current *Tigris*).

```
14      if (sig_pipe[0] != -1)
15        return lean_io_result_mk_ok(lean_box(sig_pipe[0])); // already installed
16      if (pipe(sig_pipe) != 0)
17        return lean_mk_io_user_error(
18            lean_mk_string("sigint pipe installation failed"));
19      struct sigaction sa;
20      memset(&sa, 0, sizeof(sa));
21      sa.sa_handler = sigint_handler;
22      sigemptyset(&sa.sa_mask);
23      sa.sa_flags = SA_RESTART; // restart interrupted syscalls
24      if (sigaction(SIGINT, &sa, NULL) != 0) {
25        close(sig_pipe[0]);
26        close(sig_pipe[1]);
27        sig_pipe[0] = sig_pipe[1] = -1;
28        return lean_mk_io_user_error(lean_mk_string("call to sigaction() failed"));
29      }
30      return lean_io_result_mk_ok(lean_box(sig_pipe[0]));
31    }
32
33    LEAN_EXPORT lean_obj_res lean_read_fd_byte(int32_t fd) {
34      char b;
35      for (;;) {
36        ssize_t r = read(fd, &b, 1);
37        if (r == 1)
38          return lean_io_result_mk_ok(lean_box(1));
39        if (r == 0)
40          return lean_io_result_mk_ok(lean_box(0)); // EOF
41        if (errno == EINTR)
42          continue; // retry
43        return lean_mk_io_user_error(lean_mk_string("read() failed"));
44      }
45    }
```

The CLI frontend is responsible for launching evaluation threads `EVS` as well as a *sentinel thread* that keeps reading the sigint pipe, on detecting cancel signals from the user, signals `EVS` to exit, which is then responded by `checkInterrupted` within the thread, who exits *gracefully* i.e. through normal returns. `EVS` is a threadlocal mutable state `IO.Ref EvalState` where `EvalState` is defined as

```
abbrev EvalState := Option
                  $ Task
                  $ Except Error
                  $ LApp.REPLState × LInterpreter.Val × Scheme
```

where, `Task` is a Lean primitive representing asynchronous computation, similar to *coroutines*, such as Scala's `Future`, Javascript's `Promise` and Rust's `JoinHandle`. The POSIX sigint pipe can only be read if it is installed. Implementation of this pipe, its handler and reader is in C, through the FFI to be called in Lean, as shown above.

Now, if some computation diverges, e.g., `let rec loop x = loop x in loop ()`, users can press `<C-c>` to interrupt a thread: Figure 4. Though, users should note that there is a caveat: an tigris REPL launched through Lake[74] has `broken sigint pipe` where `<C-c>` just terminates the program,

[74] Lean's build system

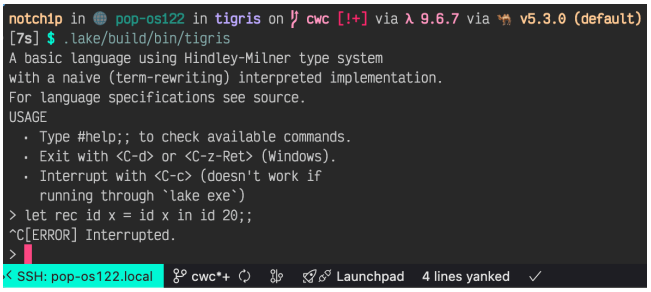


Figure 4: A user interrupting a computation

instead, users should launch the program directly, which, should be the default behavior if not building from source.

The *Tigrisi* interpreter can be invoked with `interpretL` (w/o incr. APIs) or `interpretI`, of which the former is stable and is used to quickly test the lowering stage to see if there is any issue.

4.4 CPS transformation

CPS transformation refers to the transformation of *Lambda* IR into its CPS counterparts where

- every function takes an explicit continuation parameter k,
- function calls pass results to continuations rather than returning,
- continuations are *not* first-class: they are either locally-bound labels or the special parameter k,
- ANF is maintained.

The use of CPS, as explained in the section about tailcall, is to transform non-tailrec functions. This is rather specific as in general, mechanized CPS is used to encode complex control flow, similar to a `longjmp` and CPS transformation is to make control flow explicit, at the expense of increased heap usage, as continuations are usually lambda abstraction[75] them selves, being passed around together with function parameters. The intuition is that, CPS trades heap space for stack space, by allocating and *compositing* continuations, which, can be thought as

> an explicit call stack (that replaces the actual call stack) which is unwinded when applied.

The CPS IR is similar to *Lambda* IR, defined as (`CName` is `Name`)

```
1  inductive CRhs where
2    | prim    (op : PrimOp) (args : Array CName)
3    | proj    (src : CName) (idx : Nat)
4    | mkPair  (a b : CName)
5    | mkConstr (tag : Name) (fields : Array CName)
6    | isConstr (src : CName) (tag : Name) (arity : Nat)
7    | const   (k : Const)
8    | alias   (y : CName)   -- x = y
9  deriving Repr, Inhabited, BEq
10
11 mutual
```

[75] which usually is allocated in heap. Though most compilers may force stack allocation if there is an optimization window, that defeats the purpose of using CPS to save stack space if it is too naive.

```
12  inductive CTail where
13    | appFun      (f : CName) (payload : CName) (k : CName)
14    | appKont     (k : CName) (value : CName)
15    | ite         (condVar : CName) (tBranch eBranch : CExpr)
16    | switchConst (scrut : CName)
17                  (cases : Array (Const × CExpr))
18                  (default? : Option CExpr)
19    | switchCtor  (scrut : CName)
20                  (cases : Array (Name × Nat × CExpr))
21                  (default? : Option CExpr)
22    | halt        (value : CName)
23    | matchFail   (pat : Array Name)
24  deriving Repr, Inhabited, BEq
25
26  /-- CPS expression in ANF: pure let1 / letKont / local fun groups, ending with a tail. -/
27  inductive CExpr where
28    | let1    (x : CName) (rhs : CRhs) (body : CExpr)
29    | letKont (kid : CName) (param : CName) (kBody : CExpr) (body : CExpr)
30    | letRec  (funs : Array CFun) (body : CExpr)
31    | tail    (t : CTail)
32  deriving Repr, Inhabited, BEq
33
34  structure CFun where
35    fid          : CName
36    payloadParam : CName
37    kontParam    : CName
38    body         : CExpr
39  deriving Repr, Inhabited, BEq
40  end
```

The CPS transformation is the last step of the IR lowering process, which, is further lowered by the SBCL backend to Common Lisp. Semantics of the actual program should be similar to that of CPS, provided if there is not any backend inconsistency[76]

First, we present the formal grammar of *Lambda* IR, and its CPS target:

$$
\begin{aligned}
v \in \mathsf{Val} &::= x \mid k \mid \mathsf{ctor}(t, \bar{x}) \mid \lambda p.\, e \\
r \in \mathsf{Rhs} &::= \mathsf{prim}(op, \bar{x}) \mid \mathsf{proj}(x, i) \mid \mathsf{mkPair}(x, y) \\
&\quad \mid \mathsf{mkCtor}(t, \bar{x}) \mid \mathsf{isCtor}(x, t, n) \mid f(a) \\
s \in \mathsf{Stmt} &::= \mathbf{let}_1 x = r \\
\tau \in \mathsf{Tail} &::= \mathbf{ret}\, x \mid f(a)^{\mathsf{T}} \\
&\quad \mid \mathbf{if}\, c\, \mathbf{then}\, e_1\, \mathbf{else}\, e_2 \\
&\quad \mid \mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to e_i}, e_?) \\
&\quad \mid \mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to e_i}, e_?) \\
&\quad \mid \mathsf{matchFail}
\end{aligned}
$$

[76] which, unfortunately, remains an assumption because *Tigris* is not formally verified. Note that, the `M` monad is continued to use for this transformation.

$$
\begin{aligned}
e \in \mathsf{LExpr} ::= {} & \overline{s};\ \tau \\
\mid {} & \mathbf{let}_{\iota}\, x = v\, \mathbf{in}\, e \\
\mid {} & \mathbf{let}_{1} x = r\, \mathbf{in}\, e \\
\mid {} & \mathbf{let\ rec}\ \overline{f_i(p_i) = e_i}\, \mathbf{in}\, e \\
r_c \in \mathsf{CRhs} ::= {} & \mathsf{prim}(op, \bar{x}) \mid \mathsf{proj}(x, i) \mid \mathsf{mkPair}(x, y) \\
\mid {} & \mathsf{mkCtor}(t, \bar{x}) \mid \mathsf{isCtor}(x, t, n) \mid k \mid x \\
\tau_c \in \mathsf{CTail} ::= {} & f(p, k) \mid k(v) \mid \mathsf{halt}(v) \\
\mid {} & \mathsf{ite}(c, e_1, e_2) \\
\mid {} & \mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to e_i}, e_?) \\
\mid {} & \mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to e_i}, e_?) \\
\mid {} & \mathsf{matchFail} \\
e_c \in \mathsf{CExpr} ::= {} & \mathbf{let}_{1} x = r_c\, \mathbf{in}\, e_c \\
\mid {} & \mathbf{letK}\, \kappa(p) = e_c\, \mathbf{in}\, e_c \\
\mid {} & \mathbf{let\ rec}\ \overline{f_i(p_i, k_i) = e_i}\, \mathbf{in}\, e_c \\
\mid {} & \mathsf{tail}\, \tau_c
\end{aligned}
$$

We define the *domain of semantics* (i.e. datatypes) as:[77]

$$
\begin{aligned}
\mathsf{Val} &= () + \mathbb{Z} + \mathbb{B} + \mathsf{String} + (\mathsf{Val} \times \mathsf{Val}) + (\mathsf{Name} \times \mathsf{Val}^*) + \mathsf{Name} \\
\rho &= \mathsf{Name} \rightharpoonup \mathsf{Val} \\
\Phi &= \mathsf{Name} \rightharpoonup (\mathsf{Name} \times \mathsf{Name} \times \mathsf{CExpr}) \\
\Phi_\kappa &= \mathsf{Name} \rightharpoonup (\mathsf{Name} \times \mathsf{CExpr}) \\
\mathsf{Ans} &= \mathsf{Val} + \mathsf{Error}
\end{aligned}
$$

The transformation $\mathcal{C} : \mathsf{LExpr} \to \mathsf{CExpr}$ is defined with respect to a *current continuation* k, which is either a continuation variable or should be the identity continuation $\mathrm{id}_{\mathsf{Ans}}$[78] for top-level expressions.

Below, we present the denotational semantics for this transformation. Note that $C^k_{[\bullet]}$ represents a a transformation with bound continuation k, with a hole denoting an element of the input domain $[\bullet]$, which is one of v, r, s, τ, e; And $C^k_s[\![\bullet]\!](e_c)$ threads e_c. We simplify `let*` and `switch*` to just one binding/one discriminant.

$$
\begin{aligned}
\mathcal{C}_v[\![x]\!] &= x && \text{(Values)} \\
\mathcal{C}_v[\![k]\!] &= \mathsf{const}(k) \\
\mathcal{C}_v[\![\mathsf{ctor}(t, \bar{x})]\!] &= \mathsf{mkCtor}(t, \bar{x}) \\
\mathcal{C}_r[\![\mathsf{prim}(op, \bar{x})]\!] &= \mathsf{prim}(op, \bar{x}) && \text{(Rhs)} \\
\mathcal{C}_r[\![\mathsf{proj}(x, i)]\!] &= \mathsf{proj}(x, i) \\
\mathcal{C}_r[\![\mathsf{mkPair}(x, y)]\!] &= \mathsf{mkPair}(x, y) \\
\mathcal{C}_r[\![\mathsf{mkCtor}(t, \bar{x})]\!] &= \mathsf{mkCtor}(t, \bar{x}) \\
\mathcal{C}_r[\![\mathsf{isCtor}(x, t, n)]\!] &= \mathsf{isCtor}(x, t, n)
\end{aligned}
$$

[77] think of $(+)$ and $(\times)$ as coproduct and product, respectively.
[78] The standard *driver* in SBCL backend is this.

$$
\begin{aligned}
\mathcal{C}_r[\![f(a)]\!] &= \text{(see call transformation below)} \\
\mathcal{C}^k_\tau[\![\mathbf{ret}\, x]\!] &= \mathsf{tail}\, k(x) && \text{(Tails)} \\
\mathcal{C}^k_\tau[\![f(a)^{\mathsf{T}}]\!] &= \mathsf{tail}\, f(a,k) \\
\mathcal{C}^k_\tau[\![\mathbf{if}\, c\, \mathbf{then}\, e_1\, \mathbf{else}\, e_2]\!] &= \mathsf{tail}\, \mathsf{ite}(c, \mathcal{C}^k[\![e_1]\!], \mathcal{C}^k[\![e_2]\!]) \\
\mathcal{C}^k_\tau[\![\mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to e_i}, e_?)]\!] &= \mathsf{tail}\, \mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to \mathcal{C}^k[\![e_i]\!]}, \mathcal{C}^k[\![e_?]\!]) \\
\mathcal{C}^k_\tau[\![\mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to e_i}, e_?)]\!] &= \mathsf{tail}\, \mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to \mathcal{C}^k[\![e_i]\!]}, \mathcal{C}^k[\![e_?]\!]) \\
\mathcal{C}^k_\tau[\![\mathsf{matchFail}]\!] &= \mathsf{tail}\, \mathsf{matchFail} \\
\mathcal{C}^k_s[\![\bar{s};\, \tau]\!] &= \mathcal{C}^k_s[\![\bar{s}]\!](\mathcal{C}^k_\tau[\![\tau]\!]) && \text{(Stmt)} \\
\mathcal{C}^k_s[\![\epsilon]\!](e_c) &= e_c \\
\mathcal{C}^k_s[\![(\mathbf{let}_1 x = r); \bar{s}]\!](e_c) &= \begin{cases} \mathbf{letK}\, \kappa(x) = \mathcal{C}^k_s[\![\bar{s}]\!](e_c)\, \mathbf{in}\, \mathsf{tail}\, f(a,\kappa) & r = f(a) \\ \mathbf{let}_1 x = \mathcal{C}_r[\![r]\!]\, \mathbf{in}\, \mathcal{C}^k_s[\![\bar{s}]\!](e_c) & \text{otherwise} \end{cases} \\
\mathcal{C}^k[\![\mathbf{let}_\iota\, x = v\, \mathbf{in}\, e]\!] &= \mathbf{let}_1 x = \mathcal{C}_v[\![v]\!]\, \mathbf{in}\, \mathcal{C}^k[\![e]\!] && \text{(letVal)} \\
\mathcal{C}^k[\![\mathbf{let}_1 x = f(a)\, \mathbf{in}\, e]\!] &= \mathbf{letK}\, \kappa(x) = \mathcal{C}^k[\![e]\!]\, \mathbf{in}\, \mathsf{tail}\, f(a,\kappa) && \text{(letRhs)} \\
\mathcal{C}^k[\![\mathbf{let}_1 x = r\, \mathbf{in}\, e]\!] &= \mathbf{let}_1 x = \mathcal{C}_r[\![r]\!]\, \mathbf{in}\, \mathcal{C}^k[\![e]\!] \quad (r \neq f(a)) \\
\mathcal{C}^k[\![\mathbf{let\ rec}\, f(p) = e_f\, \mathbf{in}\, e]\!] &= \mathbf{let\ rec}\, f(p, k_f) = \mathcal{C}^{k_f}[\![e_f]\!]\, \mathbf{in}\, \mathcal{C}^k[\![e]\!] && \text{(letRhs)}
\end{aligned}
$$

Finally, we present the denotational semantics of CPS IR *itself*. Just as above, the semantics is given by

$$
\begin{aligned}
\mathcal{E} &: (\mathsf{CExpr} \oplus \mathsf{CTail}) \to \rho \to \Phi \to \Phi_\kappa \to (\mathsf{Val} \to \mathsf{Ans}) \to \mathsf{Ans} \\
\mathcal{R} &: \mathsf{CExpr} \to \rho \to \mathsf{Val}
\end{aligned}
$$

where the fifth parameter i.e. of type $\mathsf{Val} \to \mathsf{Ans}$ is the *outermost continuation*, or the *meta-continuation*, denoted μ, which represents the toplevel continuation that is $\mathrm{id}_{\mathsf{Ans}}$ or halt. We write $\rho[n \mapsto v]$ to denote the insertion[79] of mapping $n \mapsto v$ to ρ; Readers should acknowledge that ρ, n, v are metavariables. Note that $\mathcal{E}$ takes both tails and expressions.

$$
\begin{aligned}
{}^{80}\mathcal{R}[\![\mathsf{prim}(op, \bar{x})]\!]\rho &= \delta(op, \rho(\bar{x})) && \text{(CRhs)} \\
\mathcal{R}[\![\mathsf{proj}(x,i)]\!]\rho &= \pi_i(\rho(x)) \\
\mathcal{R}[\![\mathsf{mkPair}(x,y)]\!]\rho &= (\rho(x), \rho(y)) \\
\mathcal{R}[\![\mathsf{mkCtor}(t,\bar{x})]\!]\rho &= (t, \rho(\bar{x})) \\
\mathcal{R}[\![\mathsf{isCtor}(x,t,n)]\!]\rho &= \begin{cases} \mathsf{true} & \rho(x) = (t, \bar{v}) \wedge |\bar{v}| = n \\ \mathsf{false} & \text{otherwise} \end{cases} \\
\mathcal{R}[\![\mathsf{const}(k)]\!]\rho &= k \\
\mathcal{R}[\![\mathsf{alias}(y)]\!]\rho &= \rho(y) \\
\mathcal{E}[\![f(p,k)]\!]\rho\Phi\Phi_\kappa\mu &= \begin{cases} \mathcal{E}[\![e_f]\!]\rho'\Phi'\Phi'_\kappa\mu & \Phi(f) = (p_f, k_f, e_f) \\ \mathsf{Error} & \text{otherwise} \end{cases} && \text{(funapp)}
\end{aligned}
$$

[79] insertion follows the standard behavior: if the key m exists, its value is replaced by v.

$$\begin{aligned}
\text{where} \quad \rho' &= \{p_f \mapsto \rho(p),\ k_f \mapsto \rho(k)\} \\
\Phi' &= \Phi \\
\Phi'_\kappa &= \varnothing \quad \text{(local continuations are scoped)}
\end{aligned}$$

$$[81]\mathcal{E}[\![\kappa(v)]\!]\rho\Phi\Phi_\kappa\mu = \begin{cases} \mathcal{E}[\![e_\kappa]\!]\rho[p_\kappa \mapsto \rho(v)]\Phi\Phi_\kappa\mu & \Phi_\kappa(\kappa) = (p_\kappa, e_\kappa) \\ \mu(\rho(v)) & \kappa = k_{\mathsf{param}} \\ \mathsf{Error} & \text{otherwise} \end{cases} \quad \text{(kontapp)}$$

$$\mathcal{E}[\![\mathsf{halt}(v)]\!]\rho\Phi\Phi_\kappa\mu = \rho(v) \quad \text{(halt)}$$

$$\mathcal{E}[\![\mathsf{ite}(c, e_1, e_2)]\!]\rho\Phi\Phi_\kappa\mu = \begin{cases} \mathcal{E}[\![e_1]\!]\rho\Phi\Phi_\kappa\mu & \rho(c) = \mathsf{true} \\ \mathcal{E}[\![e_2]\!]\rho\Phi\Phi_\kappa\mu & \text{otherwise} \end{cases} \quad \text{(conditional)}$$

$$\mathcal{E}[\![\mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to e_i}, e_?)]\!]\rho\Phi\Phi_\kappa\mu = \begin{cases} \mathcal{E}[\![e_j]\!]\rho\Phi\Phi_\kappa\mu & \exists j.\ \rho(x) = k_j \\ \mathcal{E}[\![e_?]\!]\rho\Phi\Phi_\kappa\mu & \forall i.\ \rho(x) \neq k_i \wedge e_? \text{ exists} \\ \mathsf{Error} & \text{otherwise} \end{cases} \quad \text{(const switch)}$$

$$\mathcal{E}[\![\mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to e_i}, e_?)]\!]\rho\Phi\Phi_\kappa\mu = \begin{cases} \mathcal{E}[\![e_j]\!]\rho\Phi\Phi_\kappa\mu & \rho(x) = (t_j, \bar{v}) \wedge |\bar{v}| = n_j \\ \mathcal{E}[\![e_?]\!]\rho\Phi\Phi_\kappa\mu & \text{no match}; e_? \text{ exists} \\ \mathsf{Error} & \text{otherwise} \end{cases} \quad \text{(ctor switch)}$$

$$\mathcal{E}[\![\mathsf{matchFail}]\!]\rho\Phi\Phi_\kappa\mu = \texttt{MATCH-FAILURE} \quad \text{(matchfail)}$$

$$\mathcal{E}[\![\mathbf{let}_1 x = r\ \mathbf{in}\ e]\!]\rho\Phi\Phi_\kappa\mu = \mathcal{E}[\![e]\!]\rho[x \mapsto \mathcal{R}[\![r]\!]\rho]\Phi\Phi_\kappa\mu \quad \text{(let1)}$$

$$[82]\mathcal{E}[\![\mathbf{letK}\ \kappa(p) = e_\kappa\ \mathbf{in}\ e]\!]\rho\Phi\Phi_\kappa\mu = \mathcal{E}[\![e]\!]\rho\Phi\Phi_\kappa[\kappa \mapsto (p, e_\kappa)]\mu \quad \text{(letKont)}$$

$$\mathcal{E}[\![\mathbf{let\ rec}\ f(p, k_f) = e_f\ \mathbf{in}\ e]\!]\rho\Phi\Phi_\kappa\mu = \mathcal{E}[\![e]\!]\rho\Phi[f \mapsto (p, k_f, e_f)]\Phi_\kappa\mu \quad \text{(letRec)}$$

$$\mathcal{E}[\![\mathsf{tail}\ \tau]\!]\rho\Phi\Phi_\kappa\mu = \mathcal{E}[\![\tau]\!]\rho\Phi\Phi_\kappa\mu \quad \text{(tail)}$$

Experienced readers should immediately notice some key properties based on the definition of this CPS IR, nevertheless, we list them as remarks below:

Remark (Non-first class continuations). refer to continuations appear only in specific syntactic positions, as listed below:

1. As the second argument to function calls: $f(p, k)$
2. As the target of continuation application: $\kappa(v)$
3. Bound by **letK**: $\mathbf{letK}\ \kappa(p) = e\ \mathbf{in}\ \ldots$

This is enforced by the definition (abstract syntax) of CPS IR, where there is no `CRhs` that captures a continuation as a value. Some users might be disappointed as such IR could not support advanced control primitives such as call/cc or shift/reset from delimited continuation. Although such primitives can be more efficiently treated by the compiler, this form of CPS really is not widely used and can be substituted for more boilerplate code. Even If one must use it, we suggest an *internalized* version `Cont` which is more local with good maintainability[83] through the `Monad` interface, similar to the one from `Control.Monad.Cont` in GHC. Refer to testcase `examples/cont.tig` for an example.

[80] where δ is the interpretation of primitive operations and π_i is projection.

[81] The second case handles the distinguished continuation parameter: applying it invokes the toplevel continuation *mu*, effectively returning from the current function.

[82] Note that continuations are *not* values in the environment; they are stored in a separate continuation table Φ_κ, reflecting that they are not first-class.

[83] since CPS itself can be convoluted and counterintuitive of which the incautious use is just as bad as `goto`.

Remark (Guaranteed tailcall). refers to the fact that all function calls in CPS are in tail position.

$$\mathcal{E}[\![f(p,k)]\!] \ldots \text{ directly invokes } f \text{ without building stack frames}$$

Non-tail calls in the source are transformed via **letK**:

$$\textbf{let}_1 x = f(a) \textbf{ in } e \quad \rightsquigarrow \quad \textbf{letK } \kappa(x) = \mathcal{C}^k[\![e]\!] \textbf{ in } \mathsf{tail}\, f(a,\kappa)$$

where the continuation κ captures the rest of the computation e.

Remark (Closure representation after CPS). Previously, we discussed the encoding and calling convention of function objects, this, is largely unchanged after CPS:

$$\mathbf{C}[\![ptr, \Gamma]\!]$$

where Γ is the captured environment. In CPS, function calls become:

$$\mathcal{E}[\![f(p,k)]\!] = \ldots \text{ where } f \text{ is a code pointer}$$

The payload p is a pair (α, Γ) where α contains the user arguments.

Remark (CPS Module semantics). A CPS module consists of:

$$M = (\bar{F}, F_{main})$$
$$F = (f, p, k, e)$$

The semantics of a module is:

$$\mathcal{M}[\![(\bar{F}, F_{main})]\!] = \mathcal{E}[\![e_{main}]\!]\rho_0\Phi_0\varnothing\,\mathrm{id}_{\mathsf{Ans}}$$

where:

$$\Phi_0 = \{f_i \mapsto (p_i, k_i, e_i) \mid F_i \textbf{ of } (f_i, p_i, k_i, \cdot) \in \bar{F}\} \cup \{f_{main} \mapsto (p_m, k_m, e_m)\}$$
$$\rho_0 = \{p_m \mapsto ((), \mathbf{E}[\![]\!])\}$$

The initial payload is $(((), ()), \mathbf{E}[\![]\!])$[84]. Though, this behavior can be changed in the backend by modifying the driver wrapper code to pass whatever value to `main` in Common Lisp.

4.5 Backend: Common Lisp (SBCL) & FFI

Before proceeding further, let us recap all the stages in the implementation of Tigris, through datatypes:

Tigris source $\xrightarrow{①}$ Expr $\xrightarrow{②}$ TExpr $\xrightarrow{③}$ FExpr $\xrightarrow{④}$ LExpr $\xrightarrow{⑤}$ CExpr $\xrightarrow{⑥}$ LISP source

where ① : lexing, parsing ② : typechecking ③ : instance resolution, SysF elaboration

④ : *Lambda* IR lowering ⑤ : CPS conversion ⑥ : SBCL codegen

The SBCL backend simply translate CPS IR to Common Lisp source. Since SBCL is functional and dynamically typed, specialization or most type information is not needed, continuations are created as-is and *defunctionalization* is not attempted. The reason for choosing Common Lisp instead of other languages such as JS or Go is because the author considers himself a LISPer, who enjoys using macros.

Informally, we describe the translation as follows:

[84] Recall, for function with parameter a of arity 1, we construct $(a, ())$ even if $a = ()$.

toplevel functions refer to all functions, of shape $fid(pl, k)$, as `(defun fid (pl k))`;

let-binding i.e. `let1` as the standard `(let ((n v)) ..)`;

kont-binding i.e. `letKont k v = ..` is translated to `(labels ((k (v) ..)) body)`;

kont-app i.e. `APPLY k v` as `(funcall (the function k) v)`

closure i.e. $\mathbf{C}[\![ptr, \Gamma]\!]$ as `#'ptr` if ptr is toplevel/label-bound, or otherwise, store as-is;

Γ i.e. $\mathbf{E}[\![\bar{f}]\!]$, for each field f,

- store as-is if f is a local functional value,
- store `#'f` if f is toplevel/label-bound,
- store as-is, otherwise.

funcall i.e. $f(a, k)$ as `(funcall (the function f) ..)` where f is local, or `(funcall #'f ..)` where f is toplevel/label-bound

conditionals i.e. $\mathsf{ite}(c, t, e)$ as `(if c t e)`

switch where

- $\mathsf{switch}^{\mathsf{Const}}(x, \overline{k_i \to e_i}, e_?)$ as `(cond (((equal x k_i) e_i) .. (t e_?)) ..)`
- $\mathsf{switch}^{\mathsf{ctor}}(x, \overline{(t_i, n_i) \to e_i}, e_?)$ `(cond (((eq x t_i) e_i) .. (t e_?)) ..)`

product i.e. $pr = (p, q)$ to `(cons p q)` and projects with the standard `(car pp)` or `(cdr pp)`

variant i.e. $ctag[\![\overline{fld}]\!]$ as `(cons tag (vector @fld))`[85] with projection x_i as `(svref (cdr x) i)`. **E** and **C** are encoded this way as well.

test-ctor i.e. $\mathsf{isCtor}(x, tag, \dot{)}$ as `(eq (car x) tag)`. Note that arity is not checked at this time. *Symbols* appear inside CPS IR are escaped with bars `||` to preserve the case, as in

```
def sym (s : Name) : String :=
  let esc := s.foldl (init := "|") fun acc ch ⇒
    match ch with
    | '|' ⇒ acc ++ "\\|"
    | '\\' ⇒ acc ++ "\\\\"
    | _ ⇒ acc.push ch
  esc.push '|'
```

The caveats, as described in section 4.1 (uncurrying), because type information is erased, we collect *shapes* on demand to *alleviate* this problem, where `Shapes` can be thought as degenerate types

[85] as in `'simple-vector`, the fastest stock array variant. Note that `@x` denotes the splice of x.

```
  inductive Shape where
    | unknown
    | pair
    | ctor (tag : Name) (arity : Nat)
    | fn -- function object inside value cell
  deriving Inhabited, BEq, Repr
abbrev ShapeEnv := Std.HashMap Name Shape
```

where ShapeEnv associates a name with a shape. The initial shape of a name is always unknown, but will be *refined* as the code is being compiled. However we use the word *alleviate* because this method is not robust as it it local to functions and does not deal with the root cause: design flaws. Major refactor of adding type information to Lambda IR, threading through CPS transformation and codegen must be made to address the issue.

The runtime system refers to the primitive operator implementations, driver code and the definitions of low-level exceptions, e.g. the string equality %string= , the integer addition %int+ and the MATCH-FAILURE condition +NOMATCH+[86] is implemented as

```
(declaim (inline %string=))
(defun %string= (s1 s2)
  (declare (type simple-string s1 s2))
  (the boolean (sb-kernel:%sp-string= s1 s2 0 nil 0 nil)))

(declaim (ftype (function (integer integer) integer) %int+) (inline %int+))
(defun %int+ (a b)
  (declare (optimize (speed 3) (debug 0) (safety 0)))
  (sb-kernel:two-arg-+ a b))

(define-condition match-failure (error)
  ((discriminant :initarg :discr :reader discriminant :type string))
  (:report
   (lambda (condition stream)
     (format stream "No branch can be matched against ~A" (discriminant condition)))))
(defconstant +NOMATCH+ 'match-failure)
```

Interested readers should refer to runtime.lisp.

Common Lisp have all kinds of equality relations. To minimize overhead on dynamically typed languages, we largely follow Table 7. EQ is the most efficient as it compares memory address (i.e. physical equality rather than structural equality), which, is used most, in our case, such as comparing constructors as they are always unique, which has the same memory address. Compared to the usual encoding of constructors as integers, we achieve same performance here with better readability since the name of the constructor is the symbol itself.

Remark (Inlining runtime). The content of runtime.lisp is *inlined* in the compiler, through a trick used in the build script (also in Lean.) which detects whether there is a change in the file, and writes to runtime.lean if so:

```
input_file runtime.lisp where
  path := __dir__ / "runtime.lisp"
```

[86] Common Lisp have *conditions*, which, is arguably more expressive and general then other languages' exception system as it encodes all kinds of control flow, not just exceptions. In fact, it can be used to implement an internalized continuation facility.

To compare against..	Use..
Objects/Structs	`EQ`
Symbols	`EQ`
NIL	`EQ/NULL`
T	`EQ`
Precise numbers	`EQL`
Floats	`=`
Char	`EQL/CHAR=`
List/Cons/Seq	`EQUAL`
String	`EQUAL/EQUALP/STRING=` (symbols as well)
Tree	`TREE-EQUAL`

Table 7: LISP equalities

```
    text := true
  target runtime.lean : Unit := do -- NOTE: see also `meta` (modifier)
    let runtimep ← runtime.lisp.fetch
    let rstrBegin := "r###\"\n".toUTF8
    let rstrEnd   := "\"###".toUTF8
    runtimep.mapM fun path ⇒
      IO.FS.withFile path .read fun rt ⇒
        IO.FS.withFile (__dir__ / "runtime.lean") .write fun h ⇒ do
          let runtime ← rt.readBinToEndInto rstrBegin
          h.putStrLn s!"/-- generated from {path} -/"
          h.putStrLn "def runtime : String :="
          h.write $ runtime.pushMany rstrEnd
```

FFI or *foreign function interface*, refers to calling functions external to the language. Since *Tigris* compiles to SBCL, a rather high and low-level[87] language, FFI is obtained for *free* if one is familiar with the calling convention described before as we do not have to deal with memory management or boxing/unboxing. Metaprogramming is used extensively to make the use of FFI as convenient as possible. Since *Tigris* is not able to see an externally defined function, it puts every trust it has in the user whom is responsible for

making sure the function is correctly typed, both in LISP and in the `extern` declaration.

However, schemes declared this way can not be checked for their *sanity*, i.e., it is possible to inhabit the types that should otherwise be empty, or types that can only be proved (inhabited) with circular argument such as $\forall\alpha.\ \alpha$ or $\forall\alpha,\beta.\ \alpha \to \beta$. Though, the former already exists because of the deprecated fixpoint combinator `rec`, as in `rec` id_α,[88] whereas the latter could be useful, though unsafely, when specialized to $\forall\alpha.\ \mathsf{Unit} \to \alpha$ (as seen in an example later).

We illustrate the use of FFI through a somewhat elaborate example:

```
extern println "%println" : forall a, a -> Unit
extern append "%string-append" : String -> String -> String
extern toString "%to-string" : forall a, a -> String
```

[87] Common Lisp is high-level in having GC and being a dynamically typed functional language; and is low-level in having basic special operators and a bunch of low-level features such as `go`, `block`, `progn` that dates back to 1950s.

[88] while it damages the soundness of the type system, we keep it nonetheless for educational purposes. Besides, users can avoid it, since normal programming do not use `rec` at all.

```
extern read "%read" : forall a, Unit -> a
infixl 65 "++" ⇒ append

let fact n =
  let rec go acc
    | 0 ⇒ acc
    | m ⇒
      let _ = println $ "fact "
                    ++ toString (n - m) ++ " = "
                    ++ toString (acc * m)
      in go an @@ m - 1
  in go 1 n
postfix 50 "!" ⇒ fact

let main =
  let _ = println "Enter a number:" in
  let x = read () in
  x ! -- x! is also a valid identifier. So we parenthesize it/use juxtaposition.
```

`read` is an unsafe function whose return type can be specialized to any type, in this case, `Int`. Internally, a name associated to a FFI declaration is transformed, during parsing, to its FFI name, e.g., `read` becomes `%read`. Where the implementations of these external functions are

```
(defforeign |%to-string| (x _) ;; e.g. toString : ∀a, a -> String
  (declare (ignore gamma _))
  (funcall k (princ-to-string x)))
(defforeign "%string-append" (x y) ;; e.g. string-append : String -> String -> String
  (declare (ignore gamma) (type string x y))
  (funcall k (concatenate 'string x y)))
(defforeign |%read| (_ __) ;; e.g. read (unsafe) : ∀a, Unit -> a
  (declare (ignore gamma _ __))
  (funcall k (read)))
(defforeign |%println| (x _) ;; e.g. println : ∀a, a -> Unit
  (declare (ignore gamma _))
  (princ x) (terpri)
  (funcall k nil))
```

Compiling and running the program results in[89]

```
$ tigrisl examples/fact.tig --link ffi.lisp -o dist-fact.fasl && chmod +x dist-fact.fasl
; wrote (...)/dist-fact.fasl
; compilation finished in 0:00:00.021
$ ./dist-fact.fasl
Enter a number:
> 5
fact 0 = 5
fact 1 = 20
fact 2 = 60
fact 3 = 120
```

[89] though `ffi.lisp` in this case in automatically linked. Linking it again with `--link` would make SBCL complain about duplicate definitions. `--link` here is only to demonstrate its use.

```
fact 4 = 120
120
```

For users with LISP experience, they will quickly notice the unfamiliar notation `defforeign`. This, together with other gadgets, are LISP macros to make the definition of foreign functions easier, which lives inside `ffi.lisp`. Amongst them, the most important are

- `WITH-ARGS` destructures `payload = (cons α Γ)` and binds each user argument to the symbols provided and a special variable `GAMMA` denoting the captured environment. Basically, it generates the prelude part of a function where payload are destructured according to the provided *lambda list*[90]. This macro is not supposed to be used along; It is intended to be called by `defforeign` and users should use that instead. `with-args` is defined as

```
(defmacro with-args (payload arg-list &body body)
  (let* ((alpha (gensym "ALPHA"))
         (tmp   (gensym "TMP")))
    (labels ((build-binds (args)
                          (cond
                           ((null args) '())
                           ((null (cdr args)) `((,(car args) (car ,tmp))))
                           (t
                             (let ((head (car args))
                                   (rest (cdr args)))
                               (append `((,head (car ,tmp))
                                         (,tmp  (cdr ,tmp)))
                                 (if (null (cdr rest))
                                     `((,(car rest) ,tmp))
                                     (build-binds rest))))))))
      `(let* ((,alpha (car ,payload))
              (gamma  (cdr ,payload))
              (,tmp   ,alpha)
              ,@(build-binds arg-list))
         (declare (ignorable gamma))
         ,@body))))
```

- `DEFFOREIGN` defines a tigris-callable foreign function. It is modeled after `defun` and combines `defparameter`/`with-args`. A foreign function definition takes the form of[91]

  defforeign *function-name ffi-lambda-list* $\{form\}^* \longrightarrow$ *closure-object*

  Aside from `GAMMA`, it additionally binds `k` i.e. the continuation, applying `k` can be thought as returning the value of the function as this is the canonical way to do so. `function-name`, or `name` in the implementation below, is the name of the foreign function, can be a string `"%println"` (preserves case), or a symbol `|%println|`. Users are highly recommended to escape the symbol (to preserve cases) as LISP by default does not distinguish cases but *Tigris* does. `defforeign` is implemented as

[90] LISP jargon. Refers to s list that specifies a set of parameters (sometimes called lambda variables) and a protocol for receiving values for those parameters.

[91] Here we use the same BNF syntax from X3J13/CLtL2. Unexperienced users may interpret it as: `(defforeign function-name (x y) forms)` that returns a closure object, where `(x y)` is `ffi-lambda-list` representing function parameters that may vary in length; `forms` refers to a list of forms that is the function body.

```
(defmacro defforeign (name arg-list &body body)
  (flet ((name-to-string (x)
                         (etypecase x
                           (string x)
                           (symbol (symbol-name x))))
         (escape-bar (s)
                     (with-output-to-string (out)
                       (loop for c across s do
                             (case c
                               (#\| (write-string "\\|" out))
                               (#\\ (write-string "\\\\" out))
                               (t (write-char c out))))))
         (make-closure (f)
                       `(cons C (vector
                                 (function ,f)
                                 (cons E (vector))))))
    (let* ((ns     (name-to-string name))
           (fn-sym (intern (escape-bar ns)))
           (cell   (intern ns))
           (clos   (make-closure fn-sym)))
      `(eval-when (:compile-toplevel :load-toplevel :execute)
         (defun ,fn-sym (payload k)
           (with-args payload ,arg-list ,@body))
         (defparameter ,cell ,clos)))))
```

Note that the use of special operator `eval-when` is to make sure that any foreign functions defined in this way gets compiled and bound when `(load ..)`.

With these macros, defining foreign functions should be as easy as defining normal LISP functions.

caution! Any function output by the SBCL backend has the following compiler instructions:

```
(declare (optimize (speed 3) (safety 0) (debug 0)))
```

which removes most safety features of LISP, maximizing performance. Any error can result in a compromised LISP image (memory environment) with unreadable low-level error messages. It is best to remove this instruction when debugging.

`tigrisl` is the CLI interface to interact with the compiler. `tigrisl` uses a simple hand-written recursive descent parser to parse command line arguments; Multiple input files are compiled in parallel with the same amount of threads launched. An invocation takes the form of

`tigrisl` [*flags*] *ifiles* [`-o` *ofiles*]

where *ifiles* and *ofiles* refers to a list of input and output files that should either

- matches one to one. As in, for all i, input file name f_i^{in}, there always exists a output file name f_i^{out}, vice versa.
- matches one to one, with residual *ifiles*. As in, for *ifiles* $f_1^{\mathsf{in}}, \ldots, f_i^{\mathsf{in}}, f_{i+1}^{\mathsf{in}}, \ldots, f_n^{\mathsf{in}}$ and *ofiles* $f_1^{\mathsf{out}}, \ldots, f_i^{\mathsf{out}}$, we have $f_1^{\mathsf{in}}, \ldots, f_i^{\mathsf{in}}$ matching $f_1^{\mathsf{out}}, \ldots, f_i^{\mathsf{out}}$, with residual *ifiles* $f_{i+1}^{\mathsf{in}}, f_n^{\mathsf{in}}$ using default output naming convention i.e. appending with `.lisp`.

- any other situation results in a exception.

...and notable *flags* are

- `--lam`, emit raw *Lambda* IR with optimizations;
- `--cc`, emit CC'd *Lambda* IR;
- `--cps`, emit CPS IR;
- `--sysf`, emit System F IR; These four flags can be used together.
- `-nl, --no-lisp`, do not emit Common Lisp source.
- `-ne, --no-entry`, do not generate entrypoint, where entrypoint is also called driver code, that is the following:

```
(defun |__start| ()
  (format nil "~A"
    (funcall #'|main| nil #'identity)))
```

- `--legacy`, use deprecated typechecker/IR/backend. Not recommended;
- `-l, --link <path>`, link additional lisp source from <path>, can be used multiple times. Default behavior: link `ffi.lean`. Note that, internally, linking a file f adds a `(load f)` to the top of the object code. The order of multiple `--link`s is preserved here.
- `-o <ofiles>`, *positional vararg*, anything beyond this point is considered a part of `-o`. `<ofiles>` has file extension-directed output where `*.fasl` generate FASL executables and anything else for LISP sources. FASL generation is done by concatenating every linked LISP files together with the generated object code, then feed them into the launched SBCL with *block compilation* on to generate a monolithic binary. This binary can usually run directly in shell. If that is not the case, users may also use `sbcl --script <fasl>` to run fasl files.
- `--fasl`, emit FASL, applies to all input files.

5 PP: The *dependently-typed* table printer

What's the fun when there is *Tigrisd*, the theorem prover counterpart of *Tigris*, but the implementation of them doesn't involve some dependent type practices – Actually, both CLI interfaces and REPLs use a dependently-typed table printer that lives inside the `PP` module, not to mention the termination proofs of some recursive functions.

The author, and this project advocates for the use of dependent types in programming and below, we'll discuss the design and implementation of this simple printer, hopefully, to provide some insights into how dependent types can help users catch more unexpected errors at compile-time, rather than runtime.

We wish the types of table are distinguished by their headers, say, the header (name, gender, age) should be different from (product name, manufacturer, serial number, price). To achieve this at compile-time, we shall rely on the typechecker. Suppose the header is a list, we may turn it into a *dynamically-sized*[92] product *type* whose length and component type must match for the product types to be unified (Text.SString is stylized string):

```
abbrev ToProd (as : List α) (motive : Type -> Type) : Type :=
  match as with
  | [] ⇒ PUnit
```

[92] In the sense that its size depends on the length of the header.

```
  | [_] ⇒ motive α
  | _ :: x :: xs ⇒
    motive α × ToProd (x :: xs) motive

variable (header : List Text.SString)
abbrev Row := ToProd header id
```

The above reads: ToProd (as : List α) f can be eliminated to one of 1) PUnit[93], 2) $f\ \alpha$, or 3) $f\ \alpha \times$ ToProd (cdr as) f, based on the *shape* of as:

1. where as is empty;
2. where as is a *singleton*;
3. where as contains *two or more* elements.

This effectively tells us that ToProd can *eliminate* to one of the three[94] types depending on the value of as, specifically, its shape. Readers should pay great attention to this approach, or thought process, as the pattern matching used here is interpreted differently[95] (more of a *recursor*-based interpretation),[96] which is mirrored later.

Note that *motive* is type theory jargon. It represents the target type (or a part of it) we want it to be eliminated to. Informally, it can be best compared to the function used when mapping some data structures. Also, the use of `abbrev` rather than `def` here guarantees the definition to be *reducible*, not *semireducible* or *irreducible*, allowing the automatic unfolding of the definition when using tactics such as `rfl`, short for reflexivity.

Now, we may encode an entry (a `Row`) of a table as `ToProd header id`. This way, we require the entry of some table to match the shape of its header, otherwise it is a type error; And the table type itself, obviously, can be defined as

```
abbrev TableOf := Array $ Row header
```

We use `Array` here because it is overridden in the runtime with *real continuous data structures* rather than a linked list `List`. Besides, it also makes sense as we want *the shape* of the entry to match its header, but there should not be any size restriction when it comes to the size of a table, i.e. the numbers of entry a table has.

Furthermore, since we are making a *printer*, we want the output to look good, or at least, looks like a table. Here we define

```
inductive Alignment | left | center | right
inductive PadsBy | perCell | perCol
abbrev Align := ToProd header λ _ ⇒ Alignment
abbrev OverrideWidth := ToProd header λ _ ⇒ Option Nat
```

where

- `Align` is a tuple of the same shape as a header, denoting each column's alignment;
- `PadsBy` is the padding strategy. `perCell` makes every cell to have the same size as the largest one in them. This is rarely useful, but may look good if the table isn't too large; `perCol` is the one that is actually useful and more familiar to users since it applies the same strategy but local to each column.

[93] *universe-polymorphic* version of Unit. Though since the result lives in Type ≡ Type 0 ≡ Sort 1, it is effectively Unit.
[94] To be precise, it is $2 + n$ for as of length n.
[95] different than a *operational* interpretation, in the sense of regular programming.
[96] And it is indeed in the form of recursor invocations inside the Lean *kernel*.

```
<ofiles>          a list of output files written to
  file extension-directed output:
   - '*.fasl' for FASL output, regardless of --fasl
   - anything else for LISP output
```

```
(.str "<ofiles>", .str "a list of output files written to"
(.str "", .byl "file extension-directed output:")
(.str "", .str " - '*.fasl' for FASL output, regardless of
(.str "", .str " - anything else for LISP output")
```

Figure 5: output of the table printer and its equivalent table entry

- `OverrideWidth`, despite its name, stores the calculated width of each column. It is named `OverrideWidth` because it can be overridden (*before* the calculation) to have more manual control over the output, providing better granularity.

Based on this, we sketch the configuration of this printer as

```
structure PPSpec (header : List Text.SString) where
  align    : Align header
  width    : OverrideWidth header := 0
  header?  : Bool := true
  margin   : Nat := 3
  padsBy   : PadsBy := .perCol
  truncate : Bool := false
```

where `margin` is *sum* of a cell's left and right margin. And `truncate`, which is a flag that toggles the skip of a cell if it is empty i.e. gets replaced by the cell to its right (also respects `margin`); Despite its simple nature, this is actually more interesting than one might imagine, as this could be used to create indentation, together with margin, to achieve some interesting effects, such as simulating tool tips, which is most useful when printing command helppages as shown in Figure 5.

Now, we can actually implement this printer by first defining some helper functions that calculates the width of every column: Figure 6. This is where the aforementioned thought process come into use. The type `ToProd`, can somewhat be interpreted as having multiple values (though this "value" are *types*), as it *depends* on its `List` argument, though, it remains as one type at a moment because it is statically typed. We illustrate the usage of this thought process by introducing a concept called *discriminant refinement*.

Discriminant refinement is, a jargon, though fairly straightforward, stating that the type information of a match discriminant can be *refined* based on the pattern matching against it. An example being `match x with | 0 ⇒ ...`[97], where Lean knows that within this branch, `x` must be 0, or as a *hypothesis* that becomes part of the *ambient types*: $h : x = 0$. – A very reasonable, and true take. This technique, is used fairly a lot in programming with dependent types, as is in this case: we *guide* Lean the underlying type of `ToProd` through patterns against `header`:

1. a `[]`, stating *header* $=$ [], refines `ToProd` to `PUnit`, as stated in its definition;
2. a `[_]`, stating *header* $= [\cdot]$, a singleton, or, a `List` with one element of type α, refines `ToProd` to $f\alpha$, as stated in its definition;
3. an Erlang-like `%[_, _ | _]`, which is syntactic sugar for `_ :: _ :: _`, stating *header* $= \cdot :: (\cdot :: \cdot)$, a list with two or more elements, refines `ToProd` to a dynamically-sized product.

Where f is the motive, in this case, the constant function that maps all types to `Option Nat`. The point of these seemingly useless patterns differs from the regular purpose of providing templates

[97] Though, in non-dependent type context, this feature may need to be enabled manually, in the form of `match h : x with ..`, by prefixing `x` with a name `h`, for binding the hypotheses in each branch in order to for them be picked up by the typechecker.

```
def ovTostr (ov : OverrideWidth header)
  : String := match header with
  | [] ⇒ "" | [_] ⇒ toString ov
  | %[_, _ | _] ⇒ toString ov.1 ++ ovTostr ov.2

def zeroOverride (header : List Text.SString)
  : OverrideWidth header := match header with
  | [] ⇒ () | [_] ⇒ none
  | %[_, y | ys] ⇒ (none, zeroOverride (y :: ys))

def maxN (h₁ h₂ : OverrideWidth header)
  : OverrideWidth header := match header with
  | [] ⇒ () | [_] ⇒ max h₁ h₂
  | %[_, _ | _] ⇒ (max h₁.1 h₂.1, maxN h₁.2 h₂.2)

def merge (h₁ h₂ : OverrideWidth header)
  : OverrideWidth header := match header with
  | [] ⇒ () | [_] ⇒ max h₁ h₂
  | %[_, _ | _] ⇒ (h₂.1 <▷ h₁.1, merge h₁.2 h₂.2)

instance : ToString $ OverrideWidth header := ⟨ovTostr⟩
instance : Union    $ OverrideWidth header := ⟨merge⟩
instance : Max      $ OverrideWidth header := ⟨maxN⟩
instance : Zero     $ OverrideWidth header := ⟨zeroOverride header⟩
```

Figure 6: Helper functions used to calculate column width

for destructuring; Instead, it is to provide additional information to the typechecker, which, if not omitted, results in a type error as Lean cannot deduce the type of `header` on its own. It is the user's responsibility to do case-by-case analysis just as in proofs. From the outsider's perspective, it is as simple as to mirror the cases in the type definition.

Remark (Common usage of discriminant refinement). Furthermore, discriminant refinement deals with surrounding environment. `foldr1` on `List` usually does not permit empty lists, applying to such is a runtime error in Haskell, for example. In Lean, we simply make this requirement an assumption and defines the function as:

```
def foldr1 (f : α -> α -> α) (as : List α) (h : as ≠ []) : α :=
  match as with
  | [] ⇒ h rfl ▷.elim -- or simply remove this branch.
  | [x] ⇒ x
  | x :: y :: xs ⇒ f x (foldr1 f (y :: xs) nofun)
```

where h is refined to $[] \neq []$ in the empty branch where we manually *proved false* and used *ex falso* to match the expected type. Though, this branch can simply be removed as Lean can automatically proves it `false`. Similiarly `nofun` is a shorthand used here to prove $y :: xs \neq []$, but the proof itself automatically handled by Lean. Users calling this function must prove that as is nonempty through actual proofs or simply introducing a hypothesis using `if h : as ≠ [] then ...` Though, we shall stop here as this is beyond the scope of this text, which is not a tutorial.

Getting back to implementing table printer, we found ourself having several polymorphic helpers in the sense that they are independent of `header` and we may use it to implement some operations like `toString`, $(\cdot \cup \cdot)$, max and 0 where it denotes the zero element of some algebraic structure. From there, implementing printers becomes trivial.

This concludes the description of *Tigris*.

6 *Tigrisd*: a lightweight theorem prover

Tigrisd is a experimental dependently-typed theorem prover that demonstrates clear, modular and exemplary theorem prover design. *Tigrisd*, together with *Tigris*, is a bundled solution serving educational purposes that hopes to become useful in programming language/theorem proving education. *Tigrisd* is deeply influenced and takes much inspiration in terms of design/implementation from pi-forall[98]

While currently *Tigrisd* is written in Haskell, because of the latter's strong and vibrant ecosystem with libraries like `Unbound` providing useful methods to manipulate the lambda calculus and the AST so that we do not have to reinvent the wheel. [99] Though, we may someday rewrite it in Lean. The point being that at least there is a working prototype.

Nomenclature The word *term*, when used along, includes what would be considered a *type* in regular programming languages. When *term* and *type* are used together, sometimes the former includes latter, sometimes it does not, readers should be able to deduce the exact meaning of it; Unconditionally, *type* is used exclusively to reference types, no matter dependent or not.

6.1 The basics

The syntax is of *Tigrisd* is very similar to *Tigris*, with a few exceptions, especially in relation to its dependently-typed nature. The implementation of this syntax also uses monadic parsing, in the form of parser combinators, specifically, the *layout* combinators, which use off-side rules (i.e. indentation aware). This decision is made this way because proofs are generally more verbose than programs. Some of the connectives can be omit if we use indentation-aware parsing, which, is more closer to a human-readable form. We list some notable syntax changes below.

Indexed family has the form

```
<dtype-decl>    ::= inductive <dtype-name> <dtype-ibinder>* : <dtype-univ>
                    "where"? ("|" <dtype-branch>)*
<dtype-branch>  ::= <dtype-ctor> : sepBy1("->", <dtype-binder>)
<dtype-ctor>    ::= <upper-ident>
<dtype-name>    ::= <upper-ident>
<dtype-binder>  ::= <dtype-tbinder> | <dtype-ibinder>
<dtype-tbinder> ::= (<ident> : <term>)
<dtype-ibinder> ::= {<ident> : <term>}
<dtype-univ>    ::= Type
```

An concrete index family example in *Tigrisd* is

```
inductive Vector (α : Type) (n : Nat) : Type where
  | Nil  : Vector α 0
  | Cons : {m : Nat} -> α -> Vector α (Succ m)
```

With type irrelevant binders wrapped in curly brackets {} i.e. not used in programming. A constructor's result type is checked against the type that is being defined to see if their *head* matches, in the case, they must be an application of `Vector`. Implicitly, two constraints are produced by this inductive family, identified by the variant[100]:

[98] https://github.com/sweirich/pi-forall

[99] as we've sort of did so in *Tigris*, but it is all inlined and not as a standalone package, which, can be difficult to use here.

[100] The word *variant* here is used literally, which, has no connection to variant values.

- in the `Nil` branch, n must equate to `Zero`;
- in the `Cons` branch, n must equate to `Succ m` where `m` is bound to the first field of `Cons`.

This example defines a List that encodes its length in the type, similar to the `Vector` in Lean. We can verify this with unnamed bindings, defined with **example**:

```
example: {α : Type} -> Vector α 0 = Nil
example: Vector Unit 2 = Cons 1 () (Cons 0 () Nil)
```

function declaration now takes the form of:

```
<tm-lam> ::= fun <dtype-binder>+ ⇒ <tm>
<tm-let> ::= let <ident> <dtype-binder>* : <tm> ⇒ <tm>
```

The let-binding syntactic sugar described in *Tigris* applies here:

```
let map {α : Type} {β : Type} {n : Nat} (f : α -> β) (as : Vector α n) : Vector β n =
  match as with
  | Nil         ⇒ Nil
  | Cons m x xs ⇒ Cons m (f x) (map α β m f xs)
```

Note that currently *Tigrisd* does not perform implicit inference or implicit synthesis, which requires users to provide every relevant binders. *Tigrisd* implements a simple bidirectional typechecker with limited inference ability.

Readers should now pay great attention to this new property:

function takes types just the same as taking values, since types are terms as well.

Which, yields some interesting rules that previously, as in *Tigris*, requires extensions, such as skolem variables. Consider replacing the recursive call of `map` with `xs`, i.e. `Cons m (f x) xs`, which, immediately yields a type error because α and β are distinctive parameters. The typechecker cannot establish that `Vector` α n equals `Vector` β n, unless the equality $h : \alpha = \beta$[101] is proven, only after which said types can be proved equal through α-equivalence; In this case, it is not and it can't be, as α and β are arbitrary types and there is not any assumption that entails h.

pattern matching and product elimination has been reduced to the primitive form

- `match e with ..` only accepts the variant pattern;
- `let (x, y) = a in b` is exclusively used for product (pair) elimination.

Tigrisd implements a lightweight version of discriminant refinement as mentioned above, we shall describe this later.

Type/Proof Irrelevance refers to the absent of proofs and types as they have no runtime representation and thus are erased. Therefore, it is crucial for the typechecker to make sure there is no unexpected erasure of meaningful values. *Irrelevance* is the general word that describe this rule, as it may apply to not just types or proofs (see below).

[101] Equality itself is a big topic. In this case, we are talking about definitional equality. Detailed information on this is described later.

modules are fancy names for files. A *Tigrisd* source file automatically constitutes of a module, with the module name being its filename. Modules $M_1, \ldots, M_i$ can be imported using the directive `import M_1,..,M_i;` which may only appear at the top of a file.

Builtin commands are either proofs or commands used in interactive sessions (e.g. REPL), they are

- `sorry` (proof): can prove anything, which is obivously unsound. In practice, it is used to stub out the proofs that is to be typed out later. Any use of this command results a warning from the typechecker, with every definition depending on it being marked irrelevant.
- `show` (cmd): is used to pretty print the typing context.
- `rfl` (proof): is short for reflexivity, or $a = a$, for some term a. Some might say this is the root of all equality proofs as one is essentially substituting one side of the equality that eventually becomes the other side, which then can be *discharged* with `rfl`. `rfl` is more nuanced than it seems: It respects α-equivalence and β-equivalence.
- `exfalso tm` (proof): or *ex falso quodlibet* (EFQ), is the famous principle of explosion. `exfalso` expects a term proved `false`, which then can be used to prove anything, or "construct" a value of any type. The system becomes unsound if some term can be proved false without assumptions.

Universe , or level, is 1 in *Tigrisd*, which is also *impredicative*, in the sense that `Type : Type`, instead of `Type 1`, reading `Type` lives in itself, or the `Type` of types is still `Type`. Although 1-level MLTT is unsound, as proven by Girard[102] , we adopt it nonetheless for its simplicity. Similarly, bounded recursion , termination checking or totality checking is nonexistent in *Tigrisd*.

6.2 A formal description of *Tigrisd*

Unlike *Tigris*, which can be considered *stable*[103], while *Tigrisd* already have a working prototype with desired features, huge refactors and design changes may still happen, such as rewriting it in Lean, implementing *implicit argument synthesis* to reduce the verbosity of the proofs, and more. Because of this, we omit the implementation notes here and instead jump straight to the formal description of *Tigrisd*, where the core calculus and the type system is most unlikely to change.

Core calculus & context is given by

[102] We do not show why it is unsound, as that is mathematicians' job. Readers only have to know that the reason is similar to why ZF sets need an axiom of regularity; Otherwise there could be a Russell-style paradox.

[103] Readers must acknowledge that "stable" here means that the chance to have a code refactoring, or any major change to the design of the language is quite unlikely. Not that *Tigris* self is stable: it is absolutely not and is not suitable for production environment.

tm, *a*, *b*, *A*, *B*	::=		terms
		Type	universe
	|	x	
	|	$\lambda x.\ a$	lambda abstraction
	|	$a\ b$	application
	|	$(x{:}A) \to B_x$	Π/product
	|	$(a : A)$	ascription
	|	**sorry**	stub
	|	**show**	show ctx
	|	$(x : A) \times B_x$	Σ/coproduct
	|	(a, b)	products
	|	$\mathbf{let}\,(x, y) = a\,\mathbf{in}\,b$	pair elim
	|	$a = b$	Eq
	|	**rfl**	reflexivity
	|	$b \triangleright a$	cast
	|	**exfalso** a	ex falso
	|	$\lambda\{x\}.\ a$	irr lambda
	|	$a\ \{b\}$	irr app
	|	$\{x{:}A\} \to B_x$	irr *Pi*
	|	$\mathbf{let}\,x = a\,\mathbf{in}\,b$	let

context, Γ	::=		contexts
	|	$\varnothing$	empty
	|	$\Gamma, x{:}A$	cons
	|	$x{:}A$	singleton
	|	$\Gamma, x{:}^{\epsilon} A$	irr cons
	|	$x{:}^{\epsilon} A$	irr single
	|	Γ^{ϵ}	demote
	|	$\Gamma, x = a$	cons eq hypo

where the construction of context and typing is given by

$\boxed{\Gamma}$ *(Context construction)*

$$\textsc{G-NIL}\ \frac{}{\vdash \varnothing} \qquad \textsc{G-CONS}\ \frac{\Gamma \vdash A : \mathsf{Type} \qquad \vdash \Gamma}{\vdash \Gamma, x{:}A}$$

$\boxed{\Gamma \vdash a : A}$ *(Typing)*

$$\textsc{T-VAR}\ \frac{x : A \in \Gamma}{\Gamma \vdash x : A} \qquad \textsc{T-LAMBDA}\ \frac{\Gamma, x{:}A \vdash a : B[x' \mapsto x] \qquad \Gamma \vdash A : \mathsf{Type}}{\Gamma \vdash \lambda x.\ a : (x' : A) \to B_{x'}}$$

$$\textsc{T-IMPRED-UNIV}\ \frac{}{\Gamma \vdash \mathsf{Type} : \mathsf{Type}} \qquad \textsc{T-APP}\ \frac{\Gamma \vdash a : (x{:}A) \to B_x \qquad \Gamma \vdash b : A}{\Gamma \vdash a\ b : B[x \mapsto b]}$$

$$\textsc{T-PI}\ \frac{\Gamma \vdash A : \mathsf{Type} \qquad \Gamma, x{:}A \vdash B : \mathsf{Type}}{\Gamma \vdash (x{:}A) \to B_x : \mathsf{Type}} \qquad \textsc{T-SIGMA}\ \frac{\Gamma \vdash A : \mathsf{Type} \qquad \Gamma, x{:}A \vdash B : \mathsf{Type}}{\Gamma \vdash (x : A) \times B_x : \mathsf{Type}}$$

$$\textsc{T-CAST}\ \frac{\Gamma \vdash a : A \qquad \Gamma \vdash A = B}{\Gamma \vdash a : B}$$

T-PAIR
$$\frac{\Gamma \vdash a : A \qquad \Gamma \vdash b : B[x \mapsto a]}{\Gamma \vdash (a, b) : (x : A) \times B_x}$$

T-LETPAIR
$$\frac{\Gamma \vdash a : (x : A_1) \times A_{2x} \qquad \Gamma, x{:}\, A_1, y{:}\, A_2 \vdash b : B \qquad \Gamma \vdash B : \mathsf{Type}}{\Gamma \vdash \mathbf{let}\,(x, y) = a\,\mathbf{in}\,b : B}$$

T-LET
$$\frac{\Gamma \vdash a : A \qquad \Gamma, x{:}\, A, x = a \vdash b : B \qquad \Gamma \vdash B : \mathsf{Type}}{\Gamma \vdash \mathbf{let}\,x = a\,\mathbf{in}\,b : B}$$

T-EQ
$$\frac{\Gamma \vdash a : A \qquad \Gamma \vdash b : A}{\Gamma \vdash a = b : \mathsf{Type}}$$

T-REFL
$$\frac{\Gamma \vdash a = b}{\Gamma \vdash a = b : \mathsf{Type}} \qquad \frac{\Gamma \vdash a = b}{\Gamma \vdash \mathbf{rfl} : a = b}$$

T-SUBST
$$\frac{\Gamma \vdash a : A[x \mapsto a_1][y \mapsto \mathbf{rfl}] \qquad \Gamma \vdash b : a_1 = a_2}{\Gamma \vdash b : A[x \mapsto a_2][y \mapsto b] \triangleright a}$$

T-EXFALSO
$$\frac{\Gamma \vdash A : \mathsf{Type} \qquad \Gamma \vdash a : True = False}{\Gamma \vdash \mathbf{exfalso}\,a : A}$$

Readers should note that some concrete types such as Bool and Unit, is omitted. These types basically the same rules as in *Tigris*. With irrelevant being the same, in terms of typing:

$\boxed{\Gamma \vdash a : A}$ *(Irrelevant Typing)*

T-EVAR
$$\frac{x :^{\mathsf{Rel}} A \in \Gamma}{\Gamma \vdash x : A}$$

T-ELAMBDA
$$\frac{\Gamma, x :^{\mathsf{Irr}} A \vdash a : B \qquad \Gamma^{\mathsf{Rel}} \vdash A : \mathsf{Type}}{\Gamma \vdash \lambda\{x\}.\ a : \{x{:}\, A\} \to B_x}$$

T-EAPP
$$\frac{\Gamma \vdash a : \{x{:}\, A\} \to B_x \qquad \Gamma^{\mathsf{Rel}} \vdash b : A}{\Gamma \vdash a\ \{b\} : B[x \mapsto b]}$$

T-EPI
$$\frac{\Gamma \vdash A : \mathsf{Type} \qquad \Gamma, x :^{\mathsf{Rel}} A \vdash B : \mathsf{Type}}{\Gamma \vdash \{x{:}\, A\} \to B_x : \mathsf{Type}}$$

The bidirectional type system *Tigrisd* implementation is (check/infer)

$\boxed{\Gamma \vdash a \Leftarrow B}$ *(type checking)*

C-LAMBDA
$$\frac{\Gamma, x{:}\, A \vdash a \Leftarrow B}{\Gamma \vdash \lambda x.\ a \Leftarrow (x{:}\, A) \to B_x}$$

C-SWITCH-MODE
$$\frac{a \text{ not applicable} \qquad \Gamma \vdash a \Rightarrow A \qquad \Gamma \vdash A \overset{\mathsf{impl}}{=\!=} B}{\Gamma \vdash a \Leftarrow B}$$

C-WHNF
$$\frac{A \notin \mathbf{WHNF} \qquad \Gamma \vdash a \Leftarrow nf \qquad \Gamma \vdash \mathbf{WHNF}\ A \rightsquigarrow nf}{\Gamma \vdash a \Leftarrow A}$$

C-PAIR
$$\frac{\Gamma \vdash a \Leftarrow A \qquad \Gamma \vdash b \Leftarrow B[x \mapsto a]}{\Gamma \vdash (a, b) \Leftarrow (x : A) \times B_x}$$

C-LETPAIR-RL
$$\frac{\Gamma \vdash z \Rightarrow (x : A_1) \times A_{2x} \qquad \Gamma, x{:}\, A_1, y{:}\, A_2 \vdash b \Leftarrow B[z \mapsto (x, y)]}{\Gamma \vdash \mathbf{let}\,(x, y) = z\,\mathbf{in}\,b \Leftarrow B}$$

C-LETPAIR-LR
$$\frac{\Gamma \vdash z \Rightarrow (x : A_1) \times A_{2x} \qquad \Gamma, x{:}\, A_1, y{:}\, A_2, z = (x, y) \vdash b \Leftarrow B}{\Gamma \vdash \mathbf{let}\,(x, y) = z\,\mathbf{in}\,b \Leftarrow B}$$

C-LET
$$\frac{\Gamma \vdash a \Rightarrow A \qquad \Gamma, x{:}\, A, x = a \vdash b \Leftarrow B}{\Gamma \vdash \mathbf{let}\,x = a\,\mathbf{in}\,b \Leftarrow B}$$

C-REFL
$$\frac{\Gamma \vdash a \overset{\mathsf{impl}}{=\!=} b}{\Gamma \vdash \mathbf{rfl} \Leftarrow a = b}$$

C-SUBST-L
$$\frac{\Gamma \vdash y \Rightarrow B \qquad \Gamma \vdash \mathbf{WHNF}\ B \rightsquigarrow (x = a_2) \qquad \Gamma \vdash a \Leftarrow A[x \mapsto a_2][y \mapsto \mathbf{rfl}]}{\Gamma \vdash y \Leftarrow A \triangleright a}$$

C-SUBST-R
$$\frac{\Gamma \vdash y \Rightarrow B \qquad \Gamma \vdash \mathbf{WHNF}\ B \rightsquigarrow (a_1 = x) \qquad \Gamma \vdash a \Leftarrow A[x \mapsto a_1][y \mapsto \mathbf{rfl}]}{\Gamma \vdash y \Leftarrow A \triangleright a}$$

C-EXFALSO
$$\frac{\Gamma \vdash a \Rightarrow A \qquad \Gamma \vdash \mathbf{WHNF}\ A \rightsquigarrow (a_1 = a_2) \qquad \Gamma \vdash \mathbf{WHNF}\ a_1 \rightsquigarrow True \qquad \Gamma \vdash \mathbf{WHNF}\ a_2 \rightsquigarrow False}{\Gamma \vdash \mathbf{exfalso}\,a \Leftarrow B}$$

$\boxed{\Gamma \vdash a \Rightarrow A}$ *(type inference)*

I-VAR
$$\frac{x : A \in \Gamma}{\Gamma \vdash x \Rightarrow A}$$

I-APP-NONWHNF
$$\frac{\Gamma \vdash a \Rightarrow (x{:}A) \to B_x \qquad \Gamma \vdash b \Leftarrow A}{\Gamma \vdash a\ b \Rightarrow B[x \mapsto b]}$$

I-APP
$$\frac{\Gamma \vdash a \Rightarrow A \qquad \Gamma \vdash \textbf{WHNF}\ A \rightsquigarrow (x{:}A_1) \to B_x \qquad \Gamma \vdash b \Leftarrow A_1}{\Gamma \vdash a\ b \Rightarrow B[x \mapsto b]}$$

I-PI
$$\frac{\Gamma \vdash A \Leftarrow \mathsf{Type} \qquad \Gamma, x{:}A \vdash B \Leftarrow \mathsf{Type}}{\Gamma \vdash (x{:}A) \to B_x \Rightarrow \mathsf{Type}}$$

I-IMPRED-UNIV
$$\frac{}{\Gamma \vdash \mathsf{Type} \Rightarrow \mathsf{Type}}$$

I-ASCRIBE
$$\frac{\Gamma \vdash A \Leftarrow \mathsf{Type} \qquad \Gamma \vdash a \Leftarrow A}{\Gamma \vdash (a : A) \Rightarrow A}$$

I-SIGMA
$$\frac{\Gamma \vdash A \Leftarrow \mathsf{Type} \qquad \Gamma, x{:}A \vdash B \Leftarrow \mathsf{Type}}{\Gamma \vdash (x : A) \times B_x \Rightarrow \mathsf{Type}}$$

I-LET
$$\frac{\Gamma \vdash a \Rightarrow A \qquad \Gamma, x{:}A, x = a \vdash b \Rightarrow B}{\Gamma \vdash \textbf{let}\ x = a\ \textbf{in}\ b \Rightarrow B[x \mapsto a]}$$

I-EQ
$$\frac{\Gamma \vdash a \Rightarrow A \qquad \Gamma \vdash b \Rightarrow B \qquad \Gamma \vdash A \overset{\mathsf{impl}}{=\!=} B}{\Gamma \vdash (a = b) \Rightarrow \mathsf{Type}}$$

I-SUBST
$$\frac{\Gamma \vdash b \Rightarrow B \qquad \Gamma \vdash \textbf{WHNF}\ B \rightsquigarrow (a_1 = a_2) \qquad \Gamma \vdash a \Rightarrow A}{\Gamma \vdash b \Rightarrow A \triangleright a}$$

Readers paying attention will notice that some inference rules use defintional (also judgmental in MLTT, and typal in 1-level TT) equality, such as rules C-SWITCH-MODE, I-APP, and C-WHNF. Note that these rules only specifies *what it means to be "equal"*, They are not computational rules nor used for implementation purposes. We define the equality as follows:

$\boxed{\Gamma \vdash A = B}$ *(Defintional/Judgmental/Typal equality)*

E-REFL
$$\frac{}{\Gamma \vdash A = A}$$

E-SYM
$$\frac{\Gamma \vdash A = B}{\Gamma \vdash B = A}$$

E-TRANS
$$\frac{\Gamma \vdash A_1 = A_2 \qquad \Gamma \vdash A_2 = A_3}{\Gamma \vdash A_1 = A_3}$$

E-ASCRIBE
$$\frac{\Gamma \vdash a_1 = a_2}{\Gamma \vdash (a_1 : A) = a_2}$$

E-SUBST-CONGR
$$\frac{\Gamma \vdash a_1 = a_2}{\Gamma \vdash b[x \mapsto a_1] = b[x \mapsto a_2]}$$

E-BETA
$$\frac{}{\Gamma \vdash (\lambda x.\ a)\ b = a[x \mapsto b]}$$

E-PI-CONGR
$$\frac{\Gamma \vdash A_1 = A_2 \qquad \Gamma, x{:}A_1 \vdash B_1 = B_2}{\Gamma \vdash (x{:}A_1) \to B_{1x} = (x{:}A_2) \to B_{2x}}$$

E-SIGMA-CONGR
$$\frac{\Gamma \vdash A_1 = A_2 \qquad \Gamma, x{:}A_1 \vdash B_1 = B_2}{\Gamma \vdash (x : A_1) \times B_{1x} = (x : A_2) \times B_{2x}}$$

E-APP-CONGR
$$\frac{\Gamma \vdash a_1 = a_2 \qquad \Gamma \vdash b_1 = b_2}{\Gamma \vdash a_1\ b_1 = a_2\ b_2}$$

E-LAM-CONGR
$$\frac{\Gamma, x{:}A_1 \vdash b_1 = b_2}{\Gamma \vdash \lambda x.\ b_1 = \lambda x.\ b_2}$$

E-PAIR-ELIM
$$\frac{}{\Gamma \vdash \textbf{let}\ (x, y) = (a_1, a_2)\ \textbf{in}\ b = b[x \mapsto a_1][y \mapsto a_2]}$$

E-PAIR-CONGR
$$\frac{\Gamma \vdash a = a' \qquad \Gamma \vdash b = b'}{\Gamma \vdash \textbf{let}\ (x, y) = a\ \textbf{in}\ b = \textbf{let}\ (x, y) = a'\ \textbf{in}\ b'}$$

E-VAR
$$\frac{x = a \in \Gamma}{\Gamma \vdash x = a}$$

E-LET-BETA
$$\frac{}{\Gamma \vdash \textbf{let}\ x = a\ \textbf{in}\ b = b[x \mapsto a]}$$

E-LET
$$\frac{\Gamma \vdash a_1 = a_2 \qquad \Gamma, x{:}A, x = a_1 \vdash b_1 = b_2}{\Gamma \vdash \textbf{let}\ x = a_1\ \textbf{in}\ b_1 = \textbf{let}\ x = a_2\ \textbf{in}\ b_2}$$

E-TYPALSUBST-BETA
$$\frac{}{\Gamma \vdash \textbf{rfl} \triangleright a = a}$$

E-TYPALSUBST-CONGR
$$\frac{\Gamma \vdash a = a' \qquad \Gamma \vdash b = b'}{\Gamma \vdash b \triangleright a = b' \triangleright a'}$$

E-TYPALEQ-CONGR
$$\frac{\Gamma \vdash a_1 = b_1 \qquad \Gamma \vdash a_2 = b_2}{\Gamma \vdash (a_1 = a_2) = (b_1 = b_2)}$$

$$\text{E-EXFALSO-CONGR}\ \frac{\Gamma \vdash a = a'}{\Gamma \vdash \mathbf{exfalso}\ a = \mathbf{exfalso}\ a'} \qquad \text{E-EAPP-CONGR}\ \frac{\Gamma \vdash a_1 = a_2}{\Gamma \vdash a_1\ \{b_1\} = a_2\ \{b_2\}} \qquad \text{E-ELAM-CONGR}\ \frac{\Gamma, x{:}^{\mathsf{Irr}}\, A \vdash a_1 = a_2}{\Gamma \vdash \lambda\{x\}.\ a_1 = \lambda\{x\}.\ a_2}$$

$$\text{E-EPI-CONGR}\ \frac{\Gamma \vdash A_1 = A_2 \qquad \Gamma, x{:}^{\mathsf{Rel}}\, A_1 \vdash B_1 = B_2}{\Gamma \vdash \{x{:}\,A_1\} \to B_{1x} = \{x{:}\,A_2\} \to B_{2x}}$$

Weak Head Normal Form or the judgment $\Gamma \vdash \mathbf{WHNF}\ tm \rightsquigarrow nf$ appeared above reads: under context Γ, tm reduces to nf, which is in *weak head normal form*[104]. Before introducing WHNF, let us recall the definition of a *value*. As mentioned in *Tigris*, a value v is in normal form if $\langle v, \sigma\rangle \Downarrow \langle v\rangle$ where σ denotes a state (if any). In *Tigrisd*, v is defined as

v	::=		values
	|	Type	
	|	$\lambda x.\ a$	
	|	$(x{:}\,A) \to B_x$	
	|	$(x : A) \times B_x$	
	|	(a, b)	
	|	$a = b$	
	|	$\mathbf{rfl}$	

WHNF has similar form to v and contains v, but is generalized to quantify more forms. Readers should pay attention to the phrase "cannot be reduced further" as this is precisely what WHNF quantifies. This additional part that WHNF quantifies, is called *neutral form*. A neutral term mainly contains a sequence of applications, with its head being a variable. Although we knew that the arguments to this variable may be redexes[105] themselves, this application itself cannot be reduced further. Therefore, it belongs in WHNF. In this calculus, the neutral terms are given by

neutral, ne	::=		neutral
	|	x	
	|	$ne\ a$	
	|	$\mathbf{let}\,(x, y) = ne\,\mathbf{in}\,a$	
	|	$ne \triangleright a$	

For example, the term $(x : \mathsf{Type}) \times (\lambda x.\,\mathsf{Nat})\ \mathsf{Unit}$ is in WHNF, because, though its RHS is β-reducible, this term itself is not: It is a coproduct; Or the neutral term $\mathsf{add}\ 1\ 3$, which, judging from its apperance, should reduce to 4 but we are unsure of that yet as we did not *unfold* the definition of add, therefore, this term is also irreducible in WHNF. We introduce this concept because it concerns with reduction strategies, or normalization of forms (as mentioned in the footnote), which in turn, matters a *lot* when implementing equality as we want equality check to not only cover terms that *look exact*[106], but also several basic equivalence in the lambda calculus, which though possible by a naive evaluator that performs *full reduction*, is not good enough and can potentially be harmful. Therefore, we want to set *standards* for terms (i.e. terms that should not be eagerly reduced) in order to *control reduction strategies* of properly constructed evaluators that respect these standards. WHNF is exactly the standard we are looking for.

Specifically,

[104] this term supposedly came from Peyton Jones (SPJ), for its relations to reduction/evaluation strategies.
[105] reducible expressions
[106] as in syntactic equality (w/o α-renaming)

- WHNF allow us to *defer*[107] the unfolding of definitions until needed, similar to a call-by-need evaluation strategy. This, intuitively, mimics the *symbolic* computational thought process of a normal human; Consider the equation $f\ 3 = f\ (1+2)$ where f represents a *lengthy computation*. Both sides are in WHNF, where a naive evaluator may spend time reducing all the application of f, A WHNF-respecting evaluator can instead compare $3, (1+2)$ because both sides share the same *head* f. *Deciding* this equality is simple enough that we do not need to unfold f at all. In fact, this is just a special case of the *congruence* relation of functions.
- However, the equality relation in *Tigrisd* is *semidecidable*, as in sometimes it is possible for a computation to diverge because totality is not guaranteed. WHNF *alleviates* this problem by – as mentioned above – deferring unfolding as even if f diverges, the equality still holds, proven by *Tigrisd*. Though it does *not* entail that no unfolding is taken place – As to prove $1+1=2$, we must unfold $+$, and in the case $+$ diverges, the reduction loops no matter what, leading to the eventual crash of the system when it runs out of stack/heap spaces; Though, the REPL implementation has similar feature to that of *Tigris*, allowing users to interrupt with `<C-c>`.

Furthermore, readers with relatively little prover knowledge may draw intuition from Lean's `rfl` tactic/term and Coq's `reflexivity` tactic, both of which instructs the typechecker to perform normalization to decide equality, which is arguably much more stronger than *Tigrisd*'s, implemented in similar way, if not more complex, of making rules to avoid eager reductions. Moreover, they have some form of *normalization by evaluation* (NBE) which compiles expressions to forms suitable for running, such as (OCaml) bytecode, if using Coq. It is about the granularity of these rules and the performance of normalization that set us apart.

We now define the *reduction*[108] rules to WHNF and finally, the implementation of equality in *Tigrisd*:

$\boxed{\Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow nf}$ *(WHNF normalization)*

WHNF-VAR
$$\frac{x = a \in \Gamma \qquad \Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow nf}{\Gamma \vdash \mathbf{WHNF}\ x \rightsquigarrow nf}$$

WHNF-LAM
$$\frac{}{\Gamma \vdash \mathbf{WHNF}\ \lambda x.\ a \rightsquigarrow \lambda x.\ a}$$

WHNF-PI
$$\frac{}{\Gamma \vdash \mathbf{WHNF}\ (x{:}A) \to B_x \rightsquigarrow (x{:}A) \to B_x}$$

WHNF-TYPE
$$\frac{}{\Gamma \vdash \mathbf{WHNF}\ \mathsf{Type} \rightsquigarrow \mathsf{Type}}$$

WHNF-LAM-BETA
$$\frac{\Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow (\lambda x.\ a') \qquad \Gamma \vdash \mathbf{WHNF}\ (a'[x \mapsto b]) \rightsquigarrow nf}{\Gamma \vdash \mathbf{WHNF}\ a\ b \rightsquigarrow nf}$$

WHNF-LETPAIR
$$\frac{\Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow (a_1, a_2) \qquad \Gamma \vdash \mathbf{WHNF}\ (b[x \mapsto a_1][y \mapsto a_2]) \rightsquigarrow nf}{\Gamma \vdash \mathbf{WHNF}\ \mathbf{let}\,(x,y) = a\,\mathbf{in}\,b \rightsquigarrow nf}$$

WHNF-PROD-CONGR
$$\frac{\Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow ne}{\Gamma \vdash \mathbf{WHNF}\ \mathbf{let}\,(x,y) = a\,\mathbf{in}\,b \rightsquigarrow \mathbf{let}\,(x,y) = ne\,\mathbf{in}\,b}$$

WHNF-DEF
$$\frac{x = a \in \Gamma \qquad \Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow v}{\Gamma \vdash \mathbf{WHNF}\ x \rightsquigarrow v}$$

WHNF-LET-BETA
$$\frac{\Gamma \vdash \mathbf{WHNF}\ (b[x \mapsto a]) \rightsquigarrow v}{\Gamma \vdash \mathbf{WHNF}\ (\mathbf{let}\,x = a\,\mathbf{in}\,b) \rightsquigarrow v}$$

WHNF-TYPALEQ-REFL
$$\frac{}{\Gamma \vdash \mathbf{WHNF}\ (a = b) \rightsquigarrow (a = b)}$$

[107] do note the difference between *deferred* evaluation and *lazy* evaluation. Usually the latter encapsulate the former.
[108] we emphasize *reduction* because this transformation does not change the "meaning" of a term.

WHNF-REFL
$$\frac{}{\Gamma \vdash \mathbf{WHNF}\ \mathbf{rfl} \rightsquigarrow \mathbf{rfl}}$$

WHNF-TYPALSUBST
$$\frac{\Gamma \vdash \mathbf{WHNF}\ b \rightsquigarrow r\,f \qquad \Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow nf}{\Gamma \vdash \mathbf{WHNF}\ (b \triangleright a) \rightsquigarrow nf}$$

WHNF-TYPALSUBST-CONGR
$$\frac{\Gamma \vdash \mathbf{WHNF}\ b \rightsquigarrow ne}{\Gamma \vdash \mathbf{WHNF}\ (ne \triangleright a) \rightsquigarrow (ne \triangleright a)}$$

$\boxed{\Gamma \vdash a \overset{\mathsf{impl}}{=\!=} b}$ *(equality implementation)*

IMPLEQ-REFL
$$\frac{}{\Gamma \vdash a \overset{\mathsf{impl}}{=\!=} a}$$

IMPLEQ-LAM-CONGR
$$\frac{\Gamma \vdash a_1 \overset{\mathsf{impl}}{=\!=} a_2}{\Gamma \vdash \lambda x.\ a_1 \overset{\mathsf{impl}}{=\!=} \lambda x.\ a_2}$$

IMPLEQ-APP-CONGR
$$\frac{\Gamma \vdash ne_1 \overset{\mathsf{impl}}{=\!=} ne_2 \qquad \Gamma \vdash b_1 \overset{\mathsf{impl}}{=\!=} b_2}{\Gamma \vdash ne_1\ b_1 \overset{\mathsf{impl}}{=\!=} ne_2\ b_2}$$

IMPLEQ-PI-CONGR
$$\frac{\Gamma \vdash A_1 \overset{\mathsf{impl}}{=\!=} A_2 \qquad \Gamma \vdash B_1 \overset{\mathsf{impl}}{=\!=} B_2}{\Gamma \vdash (x{:}A_1) \to B_{1x} \overset{\mathsf{impl}}{=\!=} (x{:}A_2) \to B_{2x}}$$

IMPLEQ-WHNF-EXT
$$\frac{\Gamma \vdash \mathbf{WHNF}\ a \rightsquigarrow nf_1 \qquad \Gamma \vdash \mathbf{WHNF}\ b \rightsquigarrow nf_2 \qquad \Gamma \vdash nf_1 \overset{\mathsf{impl}}{=\!=} nf_2}{\Gamma \vdash a \overset{\mathsf{impl}}{=\!=} b}$$

Which, reveals an important property of WHNF:

Theorem 1 (WHNF equality invariant)**.** Two terms are equal if and only if their WHNF reductions are equal.

From left to right, this property is guaranteed by the invariant of reduction; From right to left, this property is shown by rule IMPLEQ-WHNF-EXT.

7 Summary

Regarding *Tigris*, the work is somewhat completed. Regarding *Tigrisd*, while the working prototype is available, it's implementation notes and rules of dependent pattern matching and usage examples are omitted in this report, but I'm working on delivering it in the final thesis.

此外, 格式部分并不符合要求[109] , 语言部分也较为粗糙, 不够精炼, 甚至可能有打错的部分. 但作为中期报告以及论文初稿的核心内容 (除去上面 *Tigrisd* 中的部分) 来说应当是充分的.

[109]之后会将本文内容誊抄到 GitHub 上师兄师姐制作的学校本科生论文模板中.